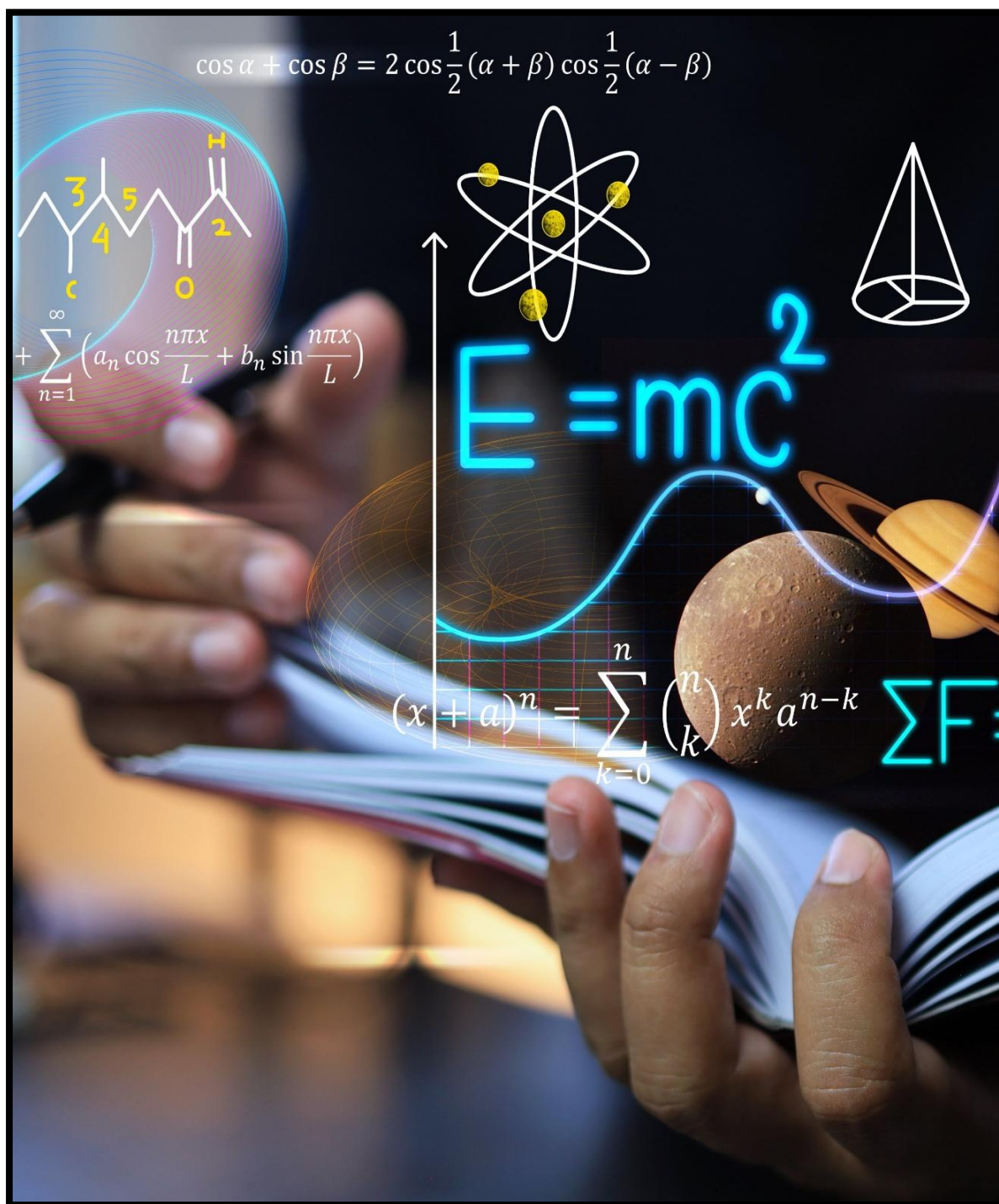


'O' LEVEL COMBINED SCIENCE

PHYSICS SECTION NOTES



❖ NEW CURRICULUM

❖ OVERVIEW OF THE BOOK



These notes were compiled by SIR SILLO, a facilitator and founder of Tinofamba Ne Vanofamba Study Club located in Kuwadzana 1, Harare. The prime objective of this book is to enable 'O' LEVEL learners to have access to simplified notes and a one stop study pack. This book is a quick study instrument and is suitable for use by learners who are about to write their final exams. This book provides all the necessary information and topics to be studied in preparation for 'O' LEVEL COMBINED SCIENCE EXAMS.

DATA PRESENTATION

- **Data** refers to the information collected in form of numbers.
- Data collected by scientists and researchers can be presented in different ways so that it is easier to understand, interpret and analyze the information.

Pie chart

- A **pie chart**, also known as a circle chart, is a circular statistical graphic that is divided into slices (sectors) representing proportions of a whole.
- Each slice (sector) visually represents the percentage of the total represented by a specific category.
- Pie charts are useful for illustrating how different parts relate to the whole, especially when there are a small number of categories.

Steps to create a pie chart from a table:

- **Calculate the Total frequency:** Add up all the values in the table to find the total quantity or frequency.
- **Calculate Percentages:** For each value in the table, divide it by the total quantity and multiply by 100 to get the percentage.
- **Calculate Angles:** Multiply each percentage by 360 degrees to get the angle of each slice(sector) in the pie chart.
- **Draw the Circle:** Use a compass to draw a circle representing the entire data.
- **Draw Slices:** Using a protractor, measure the calculated angles and draw lines from the center of the circle to the edge to create the slices(sectors). The end of sector is the beginning of each new sector. Moreover, Start with the largest angle and end with the sector with the smallest central angle.
- **Label or Key:** Clearly label each slice with the corresponding category or use a key to identify each slice.
- **Add a Title on top of the pie chart:** Give the pie chart a descriptive title to explain the data being shown.

Example

- A teacher asked students how they got to school. The table below shows the results of the obtained data. Construct a pie chart to represent the data.

Transport type	Number of learners(frequency)
Walk	8
Bicycle	2
Car	5
Bus	3

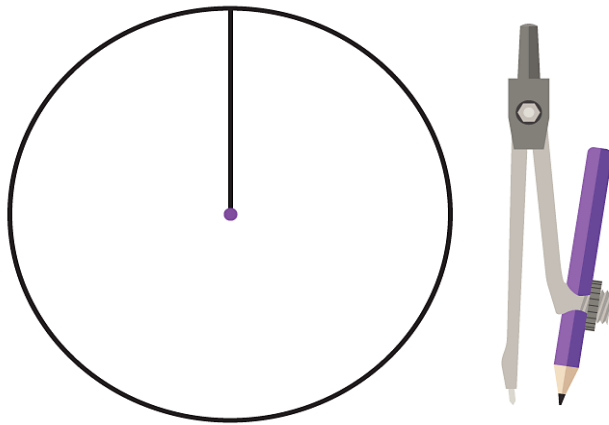
Solution

- To construct a pie chart the angles needed must be calculated, but firstly add the frequencies to obtain total frequency.

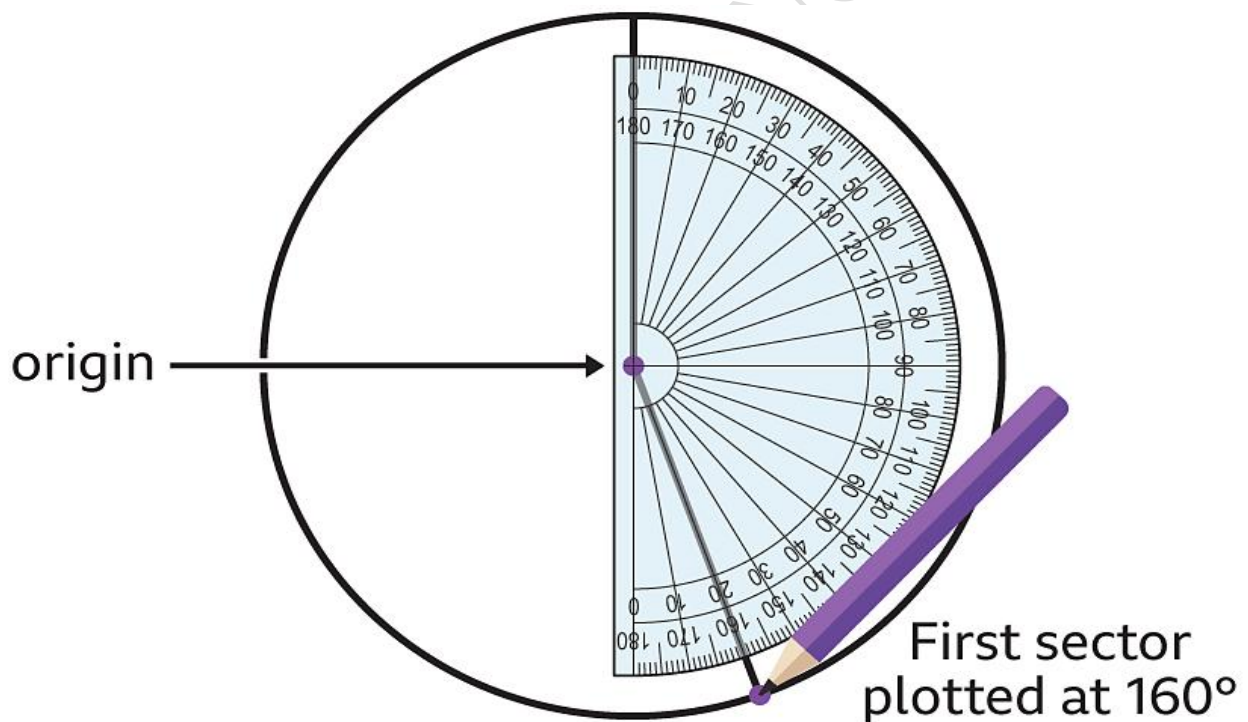
$$\begin{aligned}\text{Total frequency} &= 8+2+5+3 \\ &= 18\end{aligned}$$

Transport type	Number of learners	Size of angle (o)	% size of sector
Walk	8	$8/18 \times 360 = 160^{\circ}$	$8/18 \times 100 = 44 \%$
Bicycle	2	$2/18 \times 360 = 40^{\circ}$	$2/18 \times 100 = 11 \%$
Car	5	$5/18 \times 360 = 100^{\circ}$	$5/18 \times 100 = 28 \%$
Bus	3	$3/18 \times 360 = 60^{\circ}$	$3/18 \times 100 = 17 \%$
Total	18	360°	100%

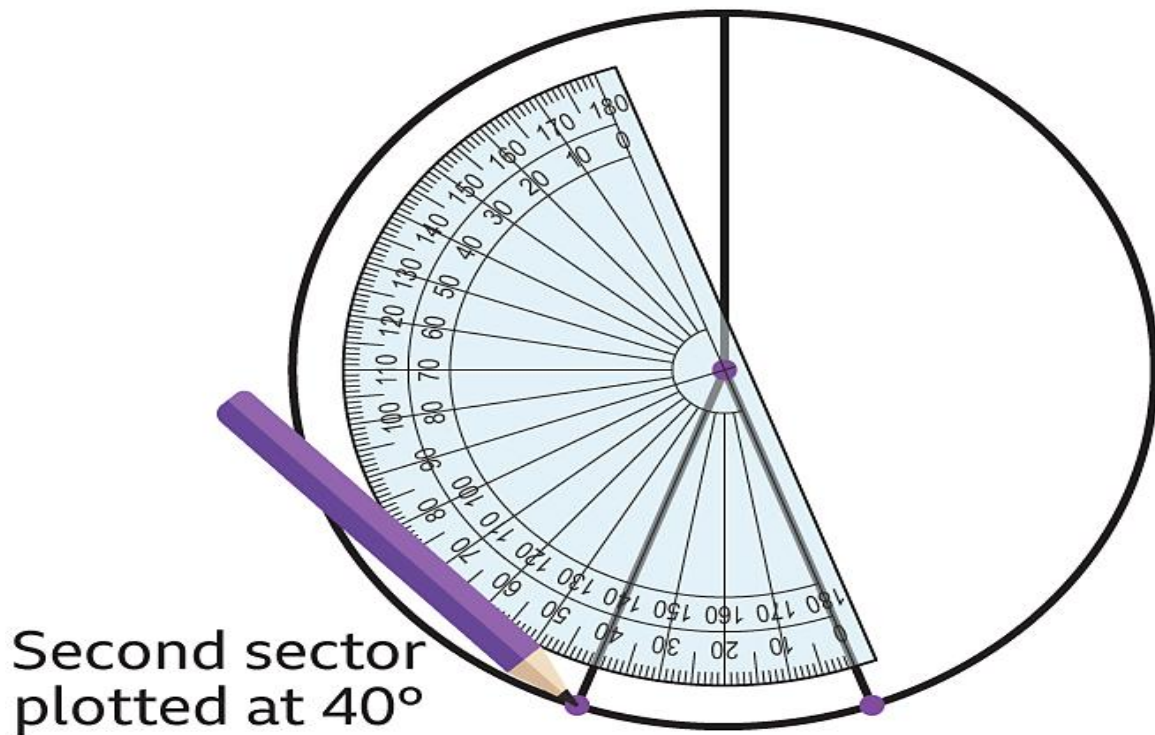
- **Construct a circle** using a pair of compasses. After that, add a vertical line from the centre to the top of the circumference.



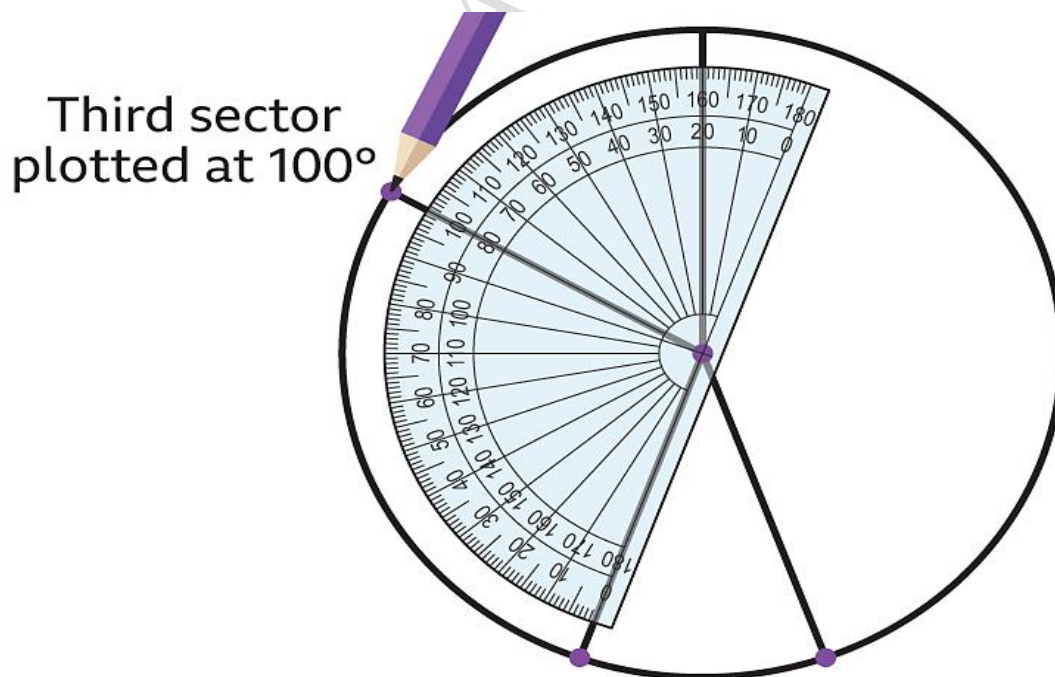
- The **first sector** to be drawn is to represent the friends who walk to school.
- This angle needs to be drawn at 160° .
- Position the protractor on top of the circle so that the origin is on the circle's center and the zero on the outer degree scale is aligned with the vertical line.
- Mark the angle at 160° . Remove the protractor and draw a straight line from the centre passing through this point.



- The **second sector** to be drawn is to represent the friends who cycle to school.
- This angle needs to be drawn at 40° .
- Rotate the protractor, keeping the origin on the centre of the circle, and with the zero on the outer degree scale now aligned with the previous line.
- Mark the angle at 40° .
- Remove the protractor and draw a straight line from the centre passing through this point.



- The **third sector** to be drawn is to represent the friends who use a car to travel to school.
- This angle needs to be drawn at 100° .
- Again, rotate the protractor, keeping the origin on the centre of the circle and aligning the zero on the outer degree scale with the previous line.
- Mark the angle at 100° .
- Remove the protractor and draw a straight line from the centre passing through this point.

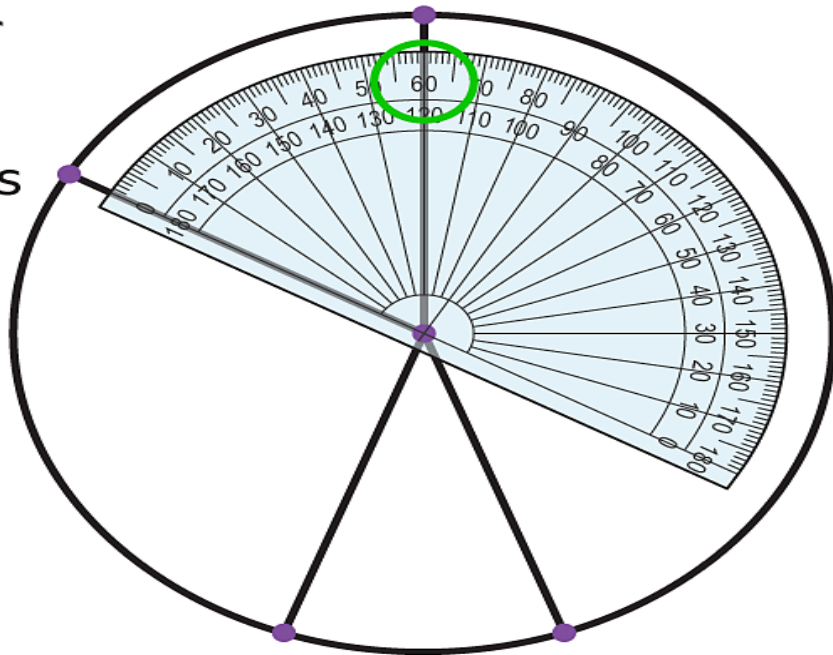


- The **last sector** is the angle that remains.
- This provides an opportunity to check how accurately the pie chart has been drawn.

- This angle should be 60°.
- This can be checked using a protractor.

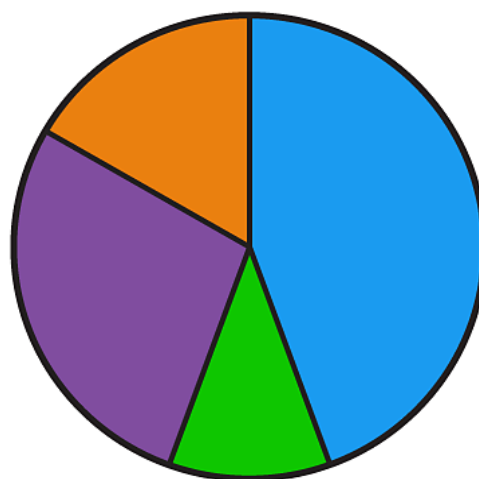
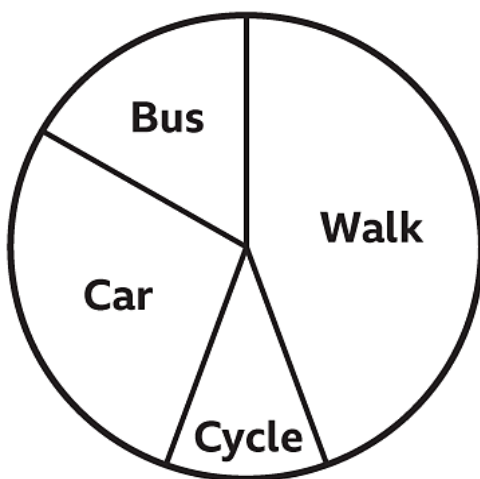
Final sector
checked

Angle equals
60° ✓



- Finally, we have constructed a pie chart.
- Add a key of label inside.
- Put the title on top of the pie chart.

A pie chart to show mode of transport to school



Practice work

- 50 learners were asked the type of mobile phones they use. The number of learners that use each mobile phone were recorded in the table below.
- Construct a pie chart from the given table.

Type of phone used	Number of learners
Samsung	9
Tecno	9

Iphone	5
Itel	20
Vivo	7

MEASUREMENTS

Physical Quantities

- Physical quantity is a property of an object or substance that can be measured using an appropriate measuring instrument.
- In the International System of Units (SI), there are **six major base units**. These are the fundamental units from which all other SI units are derived.

Physical quantity	SI Unit	Symbol
Length	metre	m
Mass	kilogram	kg
Electrical current	Ampere	A
Voltage	volt	V
Temperature	Degrees Celsius/Kelvin	$^{\circ}\text{C}$ / K
Time	Second	s
Quantity of a substance	mole	Mol

Measuring Time

- Measurement of time is important to find out the speed of chemical reactions.
- Time is measured using a stopwatch or stop-clock.
- The **SI unit for time is seconds (s)**. Other units used to measure time are minutes (min) and hours (h).



- Conversion of units:
 - 1 min = 60 s;
 - 1 h = 60 min.
- **Example 1:** Convert 100 s to min.

$$60 \text{ s} = 1 \text{ min}$$

$$100 \text{ s} = \frac{100 \text{ s}}{60 \text{ s}} \times 1 \text{ min}$$

$$= \mathbf{1.6 \text{ min}}$$

- **Example 2:** Convert 2 hrs to min.

$$60 \text{ min} = 1 \text{ hr}$$

$$x = 2 \text{ hr}$$

$$x = \frac{2 \text{ hr}}{1 \text{ hr}} \times 60 \text{ min}$$

$$= \mathbf{120 \text{ minutes}}$$

Measuring Distance

- Distance is referred to as the space between two points.
- A **tape measure** is used to measure distance, below is a diagram showing a tape measure.



- Distance can be measure in as little as nanometers (nm) up to kilometers (km) but the **SI unit for distance is meters (m)**.
- Conversion of units: $1\text{m} = 1000\text{mm} = 100\text{cm} = 0.001\text{km}$
- **Example 1:** Convert 10m to mm
- $$1\text{m} = 1000\text{mm}$$
- $$10\text{m} = \frac{10\text{m}}{1\text{m}} \times 1000\text{mm}$$
- $$= \mathbf{10000\text{mm}}$$
- **Example 2:** Convert 2.43km to m
- $$1\text{km} = 1000\text{m}$$
- $$2.43\text{km} = \frac{2.43\text{km}}{1\text{km}} \times 1000\text{m}$$
- $$= \mathbf{2430\text{m}}$$

Measuring Mass

- Mass refers to a measure of the amount of matter in a substance.
- A **top pan balance** is used to measure mass.



- A triple beam balance is less commonly used to measure mass these days.
- The **SI unit for mass is the kilogram (kg)**. In the Chemistry laboratory, a smaller unit of gram (g) is commonly used.
- Conversion of units: 1 kg = 1000 g.
- **Example 1:** Convert 120 g to kg.

$$1000 \text{ g} = 1 \text{ kg}$$

$$120 \text{ g} = \frac{120\text{g}}{1000\text{g}} \times 1 \text{ kg}$$

$$= 0.12 \text{ kg}$$

- **Example 2:** Convert 5 kg to grams.

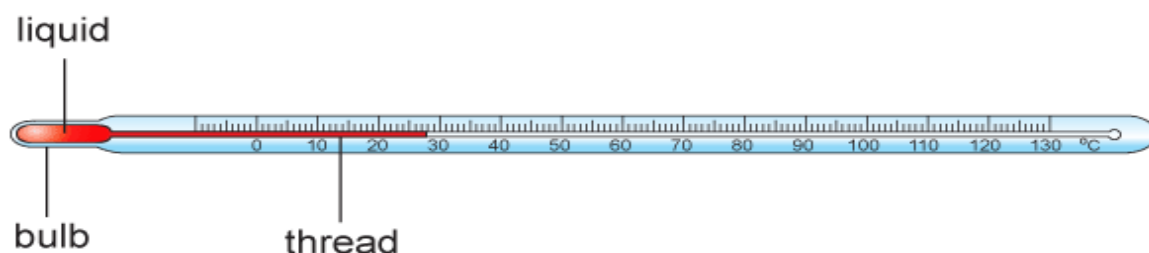
$$1 \text{ kg} = 1\,000 \text{ g}$$

$$5 \text{ kg} = \frac{5\text{kg}}{1\text{kg}} \times 1000\text{g}$$

$$= 5\,000 \text{ g}$$

Measuring Temperature

- A thermometer is used to measure the temperature of a substance.
- In Chemistry, some reactions have to take place at specific temperatures so a thermometer is important in finding out if the correct temperature has been achieved.
- A thermometer is also vital in determining reactions which result in an increase or decrease in temperature.
- The two types of liquids used in thermometers are alcohol and mercury in glass thermometers.
- The **SI unit for temperature is degrees Celsius (°C) or Kelvin (K)**.



Measuring electric current

- An electric current is a flow of charge in the electric circuit
- The symbol of current is **I**
- Ammeter measures the electric current in a circuit.
- Current is measured in **amps**.
- Ammeter is connected in a series.



Measuring voltage

- Voltage or electric potential difference is the amount of force /work done in moving charge (current) from one point to another in an electric circuit.
- Symbol for voltage/potential difference (p.d) is **V**.
- The SI unit for voltage is the **volt**.
- A voltmeter is used to measure voltage between two points in an electric circuit.
- Voltmeter is **always connected in parallel**



Derived units

- Derived units in the International System of Units (SI) are formed by combining one or more base units.
- They are used to measure quantities that cannot be expressed using only the six fundamental base units (meter, kilogram, second, etc.). Examples of derived units include speed (m/s), density (kg/m³), and area (m²).

- There are seven fundamental base units of measurements and all other S.I. units are derived by multiplying, dividing or powering the base units.

Quantity	Name of Unit	Abbreviation	Derivation
Area (A)	square meter	m ²	Length x width
Volume (V)	Cubic meter	m ³	Length x width x height
Density (D)	Kilograms per cubic meter	kg/m ³	Mass / volume
Molar mass (M)	Kilograms per mole	kg/mol	Mass / amt of substance
Concentration (c)	Moles per liter	M	Amt of substance / volume
Molar volume (V _m)	Cubic meters per mole	m ³ /mol	Volume / amt of substance
Energy (E)	joule	J	Force x length

Deriving the unit of area



- ❖ To find the area (A) of this rectangle you multiple its length (L) with its width (W).

$$\begin{aligned}
 A &= L \times W \\
 &= 5 \text{ m} \times 3 \text{ m} \\
 &= 15 \text{ m}^2
 \end{aligned}$$

- ❖ The units are also multiplied together to give m²

Deriving unit for Volume

- Volume refers to the space that a substance occupies.
 ➤ The volume of liquids and gases are usually measured in the Chemistry laboratory.
 ➤ Volume is calculated as follows:

$$\begin{aligned}
 \text{Volume} &= \text{Length (m)} \times \text{Width(m)} \times \text{Height(m)} \\
 &= \text{m}^3
 \end{aligned}$$

- The SI unit for volume is the cubic metre (m³).
 ➤ If the length, width and height are all in centimetres, therefore you multiply the units together to get cm³.

$$V = \text{cm} \times \text{cm} \times \text{cm}$$

$$= \text{cm}^3$$

- In the laboratory, volume is also measured in cubic centimetre (cm^3) and Cubic decimetre (dm^3).
- Conversion of units: $1 \text{ m}^3 = 10^6 \text{ cm}^3$; $1 \text{ dm}^3 = 1\,000 \text{ cm}^3$.
- **Example 1:** Convert 0.4 m^3 to cm^3 .

$$1 \text{ m}^3 = 10^6 \text{ cm}^3$$

$$0.4 \text{ m}^3 = \frac{0.4 \text{ m}^3}{1 \text{ m}^3} \times 10^6 \text{ cm}^3$$

$$= 400\,000 \text{ cm}^3$$

- **Example 2:** Convert 300 cm^3 to dm^3 .

$$1\,000 \text{ cm}^3 = 1 \text{ dm}^3$$

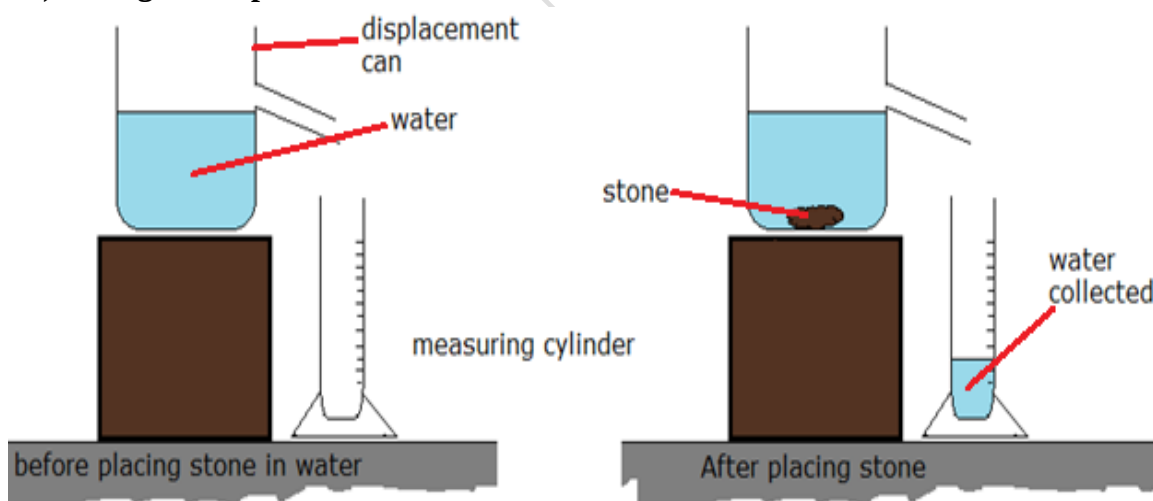
$$300 \text{ cm}^3 = \frac{300 \text{ cm}^3}{1000 \text{ cm}^3} \times 1 \text{ dm}^3$$

$$= 0.3 \text{ dm}^3$$

Experiment: finding the Volume of irregular shaped body

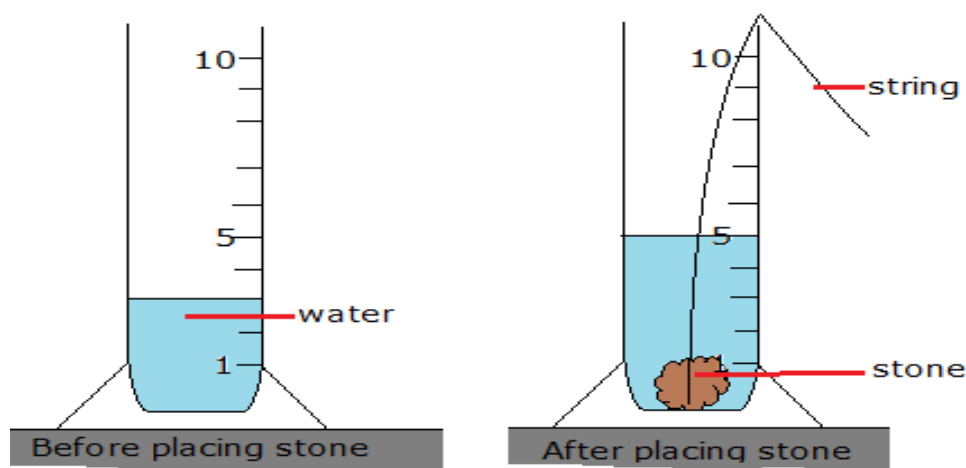
- The following methods can be used to find the volume of irregular shaped bodies.

1) Using the displacement can:



- Placing the stone in water as shown above, will cause the level of water to rise in the displacement can.
- The water will spill over the spout into the measuring cylinder.
- The water collected in the measuring cylinder (displaced by the stone), is equal in volume to that of the stone.

2) Placing the object in a measuring cylinder:



- Placing the stone in water will cause the level of the water in the measuring cylinder to rise as shown above.
- The difference between the initial reading (before placing stone in water) and final reading (after placing the stone in water) will give you the volume of the stone.

Deriving the unit of force

- Force is a push or a pull and the **units for force are the newtons**.
- Force can also be described by its effect; it causes a *mass* of an object to change its *velocity (acceleration)* hence.

$$F = m \times a$$

- **F** is in N (a derived unit),
- **m** is in kg (a base unit),
- $a = \frac{\text{velocity}}{\text{time}}$

Deriving the unit of Density

Density of liquids

- Density is the mass per unit volume
- The units of density is **g/cm³ or kg/m³**

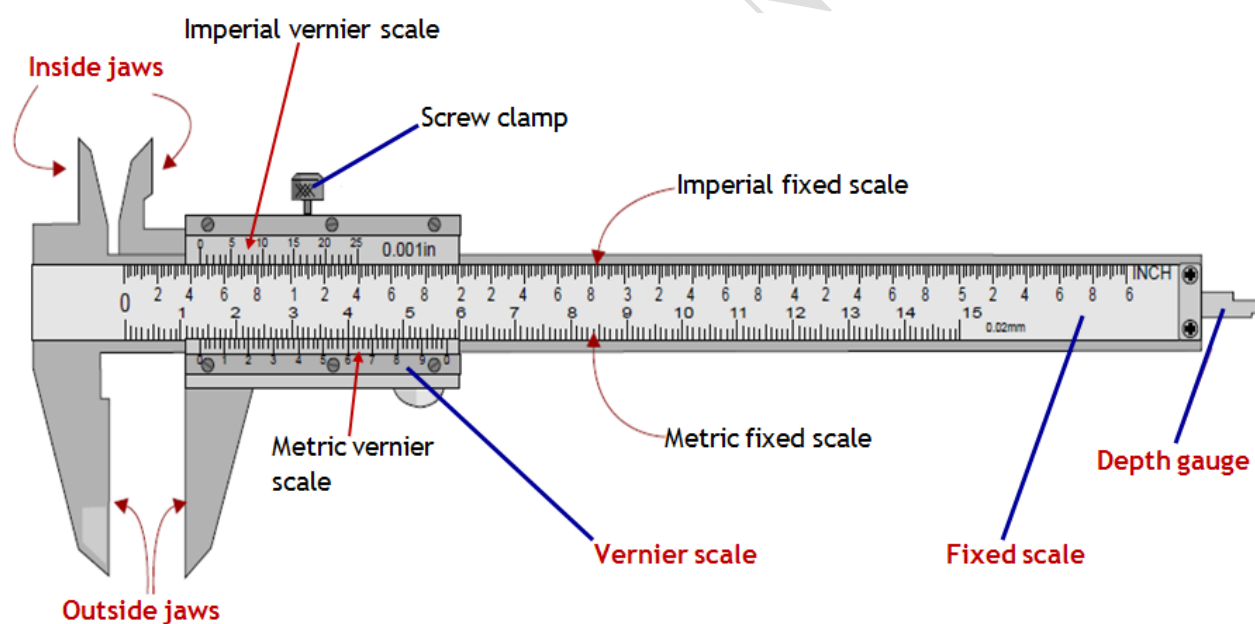
$$\text{Density} = \frac{\text{Mass (M)}}{\text{Volume (V)}}$$

- To determine density of a liquid, first determine both its mass and volume.
- Volume is found by pouring the liquid into a measuring cylinder.
- Mass is found by following these steps:
 - ✓ Weighing the beaker and record its mass in grams.
 - ✓ Pour the liquid into the beaker and measure the mass in grams.
 - ✓ Subtract the two measurements of mass, to find the mass of water.
 - ✓ Then use the mass and volume obtained to find its density

Measuring small objects

Vernier calipers

- A vernier caliper is a precision measuring instrument used to accurately measure lengths, depths, and diameters. It consists of a fixed main scale and a movable Vernier scale, allowing for precise readings beyond what's possible with a ruler alone. The main scale provides the primary reading, while the Vernier scale, which moves along the main scale, provides a finer, more precise reading.
- **The vernier calipers has various parts, including:**
 - ✓ three devices for measuring:
 1. **Outside jaws** are used to measure outer dimensions of objects, for example, the outer diameter of a pipe, test tube etc.
 2. **Inside jaws** are used to measure inner dimensions of objects, for example, the inner diameter of a pipe, test tube etc.
 3. **Stem or depth gauge** used to measure the depth of objects, for example, the depth of a small container such as a test tube.
 - ✓ two measurement scales:
 1. a **fixed scale** (metric and imperial).
 2. a **Vernier scale** a.k.a. moving/sliding scale (metric and imperial).



- A vernier calipers **measures short distances** ranging from **0 to 15cm** e.g. diameter of a test tube. It has an **accuracy of 0.01cm** i.e. it can accurately measure to the nearest tenth of a millimetre.
- **To measure**, gently grip the object with the straight edges of the outside or inside jaws.

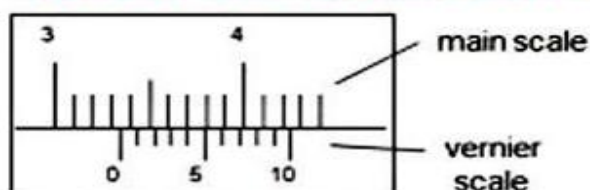
Reading vernier calipers

- Follow the following steps to read the vernier caliper:
 - 1) Read the value on the fixed scale that is located exactly at or just to the left of the zero on the Vernier (moving) scale. This is the fixed scale reading.
 - 2) Identify the next mark on the Vernier scale that aligns with a mark on the fixed scale. Read its value on the Vernier scale. This is the Vernier scale reading.

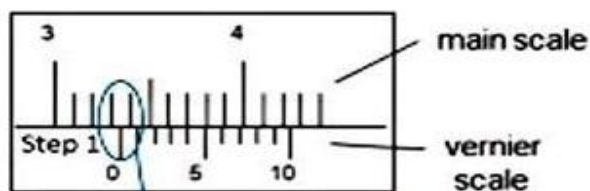
3) Add the two readings to get the actual or final reading.

Final reading = Fixed scale reading + Vernier scale reading.

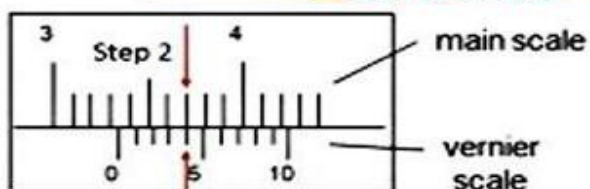
HOW TO READ A VERNIER CALIPER!



Reading: $\frac{\text{main scale}}{\text{vernier scale}}$ cm



The zero of Vernier scale is between 3.3 and 3.4
So, reading is 3.3__ cm.



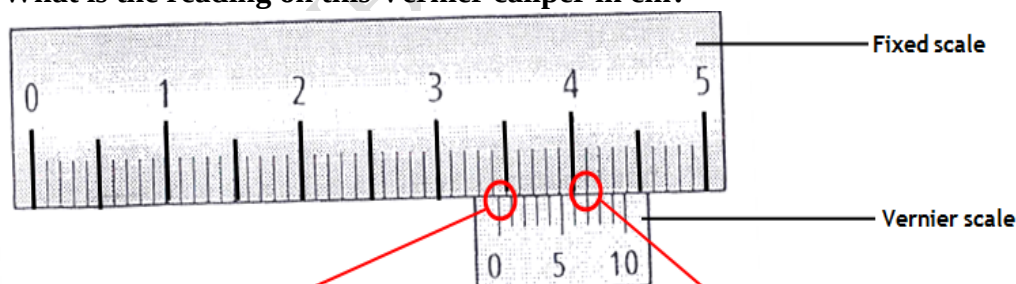
Next, look out for where the main scale and Vernier scale intercept. The markings appear as straight line from Vernier to main scale.

The point of intercept is at the 4th marking

Hence, Reading: 3.3cm + 0.04cm = 3.34cm

Example

➤ What is the reading on this Vernier caliper in cm?



1) Read the value on the fixed scale that is located exactly at or just to the left of the zero mark on the Vernier (moving) scale. This is the fixed scale reading.

Fixed scale reading = 3.40

2) Identify the next mark on the Vernier scale that aligns with a mark on the fixed scale. Read its value on the Vernier scale. This is the Vernier scale reading.

Vernier scale reading = 0.07

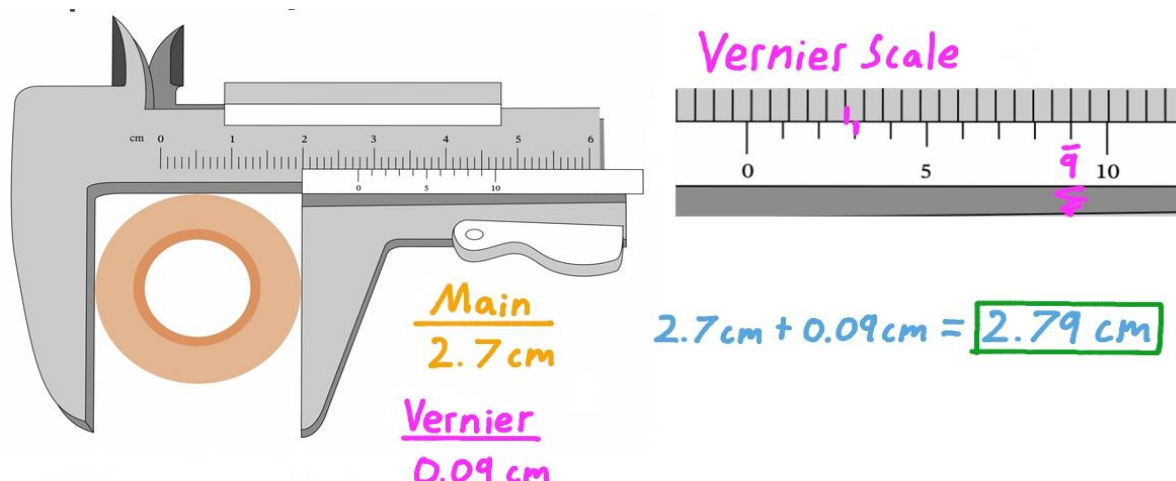
3) Add the two readings to get the final reading.

Final reading = Fixed scale reading + Vernier scale reading.

3.40
+ 0.07

3.47 cm

What is the length of the object being measured by the vernier caliper in the diagram?

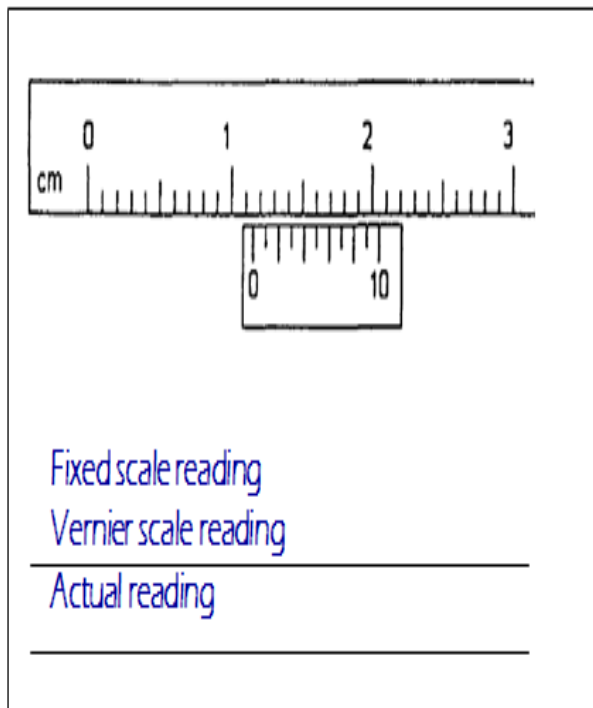
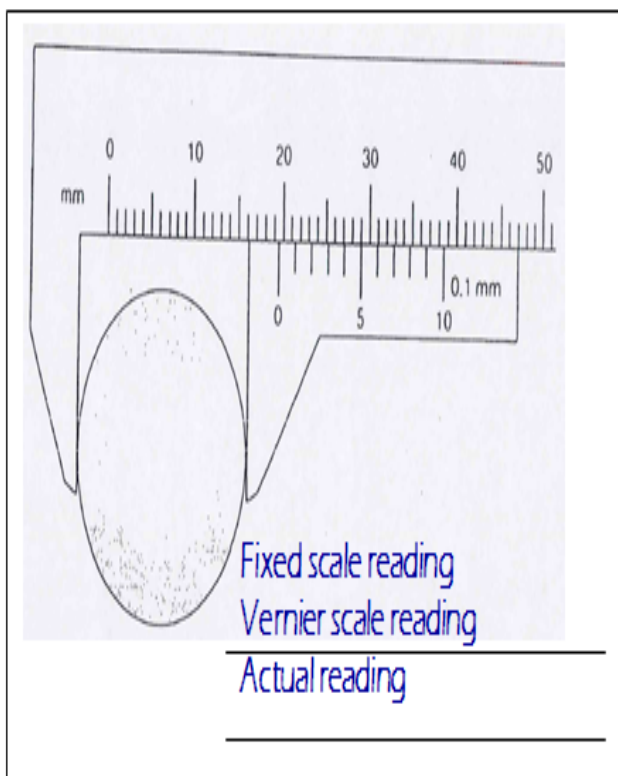
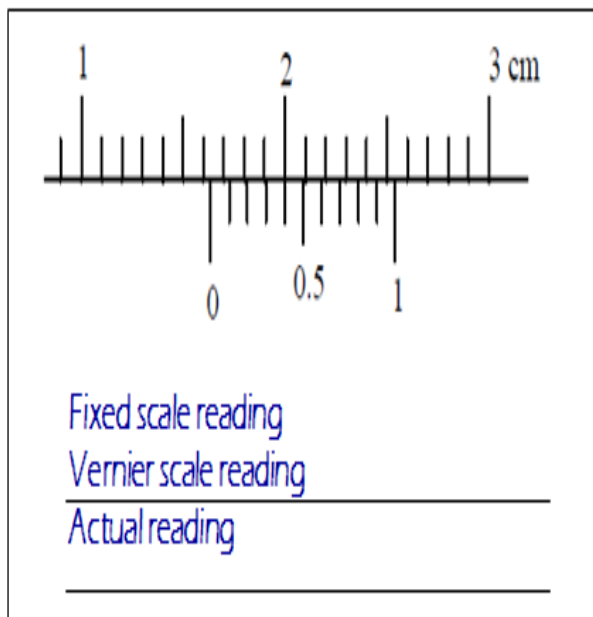
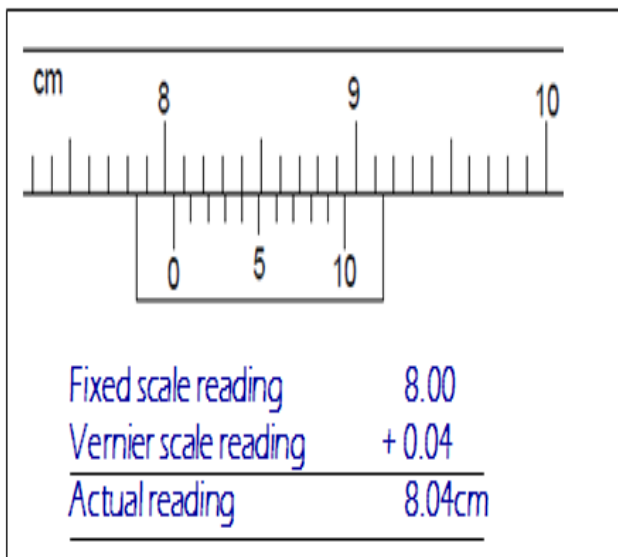


- We are measuring the length of this object when it is placed within the jaws of the caliper, which means we're measuring its outer dimensions.
- The first step to measuring an object within the jaws of a vernier caliper is to read its main scale, located on the side of the caliper itself.
- We take a measurement from this main scale and combine it with the vernier scale, which is located on the caliper's bottom jaw.
- When these scales are used together, they can measure lengths of up to one-tenth of a millimeter accurately. So, let's start measuring, starting with the main scale.
- In order to measure an object on the main scale when in the jaws of the caliper, we always start by measuring at the zero centimeters' mark. And we end our measuring at the zero mark, the zero on the vernier scale. So, let's get the measurement for our main scale.
- The largest labeled mark to the left of the endpoint is 2, indicating **2cm**. And then from there, we count the number of millimeter markings: 1, 2,3,4,5,6,7.
- We stop at the one that is to the left of the endpoint. This means the measurement is **2cm** and **7mm**, or **0.7 cm**. So, altogether, the main scale measurement is **2.7 cm**.
- Now that we have this, we can use the vernier scale to get an even more accurate measurement.
- The purpose of the vernier scale is to measure the distance here, in between the millimeter marking and the zero point, which means that it measures in between the millimeter markings. To see how we do this, we have a zoomed-in portion of the vernier scale, which is shown at the right part on the above image(diagram). We're not showing the numbers of the main scale above it because we don't need them this time, We just need their lines.
- In order to measure on the vernier scale, we have to find which of the lines on the vernier scale and the main scale match up the closest.
- Where the lines match up is where we'll find our value for the vernier scale.
- In this case, we see it is at **9** on the vernier scale because the lines match up there exactly.
- Now, each of these lines on the vernier scale, of which there are 10 total, represents one-tenth of a millimeter, or converted one one hundredth of a centimeter.
- This means that a nine on the vernier scale represents nine-tenths of a millimeter or nine hundredths of a centimeter.

- Now that we have our vernier scale measurement, we just have to combine it with our main scale measurement to find the total length of the object being measured.
- And to do this, we just add them together. **2.7 cm + 0.09 cm** gives us an answer of **2.79 cm**.
- The length of the object being measured by the vernier caliper in the diagram is **2.79 cm**.

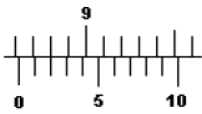
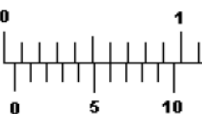
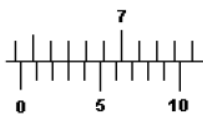
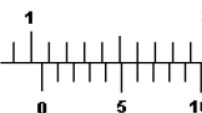
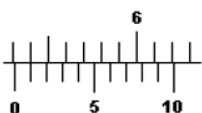
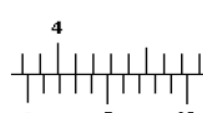
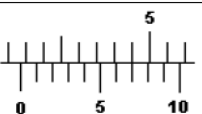
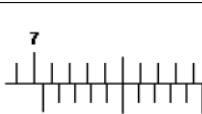
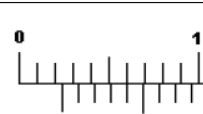
Vernier caliper readings exercise

- Try to read the following cm on the Vernier caliper.
- **N.B – the first one has already been done for you as an example.**



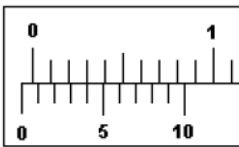
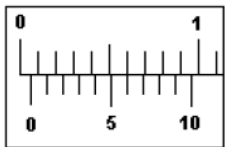
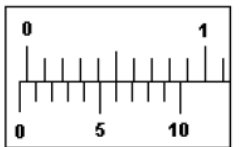
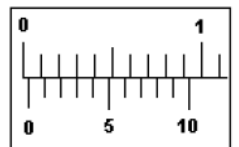
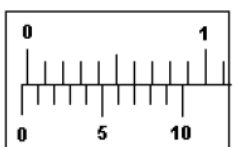
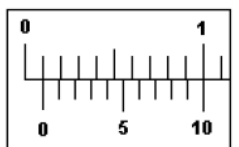
Exercise 1

Find the readings of the vernier calipers below.

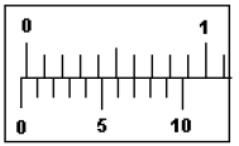
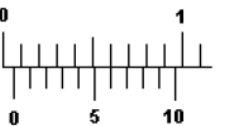
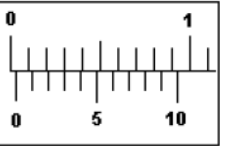
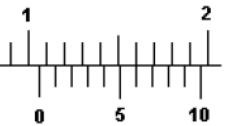
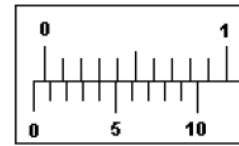
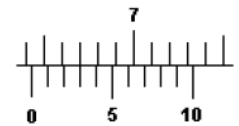
1. 	2. 	3. 
8.6 cm + 0.02cm = 8.62cm	0.0cm + 0.06cm = 0.06cm	6.4cm + 0.03cm = 6.43cm
4. 	5. 	6. 
1.0cm + 0.06cm = 1.06cm	5.3cm + 0.01cm = 5.31cm	3.8cm + 0.03cm = 3.83cm
7. 	8. 	9. 
4.3cm + 0.07cm = 4.37cm	7.0cm + 0.05cm = 7.05cm	0.2cm + 0.04cm = 0.24cm

Exercise 2

1. Assuming that the jaws of the calipers are tightly closed, find the zero error of the Vernier calipers below.

		
1. -0.06cm	2. +0.06cm	3. -0.04cm
		
4. +0.03cm	5. -0.03cm	6. +0.10cm

2. Find the zero error and the correct reading of the Vernier calipers below.

1.  Zero error = -0.03cm 	2.  Zero error = +0.03cm 	3.  Zero error = -0.04cm 
Correct reading: 0.06cm - (-0.03cm) = 0.09cm	Correct reading: 1.06cm - (+0.03cm) = 1.03cm	Correct reading: 6.43cm - (-0.04cm) = 6.47cm

Micrometer screw gauge

- A micrometer screw gauge is a precision measurement tool used to measure small lengths and diameters with a high degree of accuracy.
- It relies on a calibrated screw to make measurements, often used in fields like mechanical engineering, machining, and metrology.
- To read a micrometer screw gauge, first locate the main scale reading on the sleeve. Then, identify the coinciding mark on the thimble scale. Add these two readings together to get the final measurement.

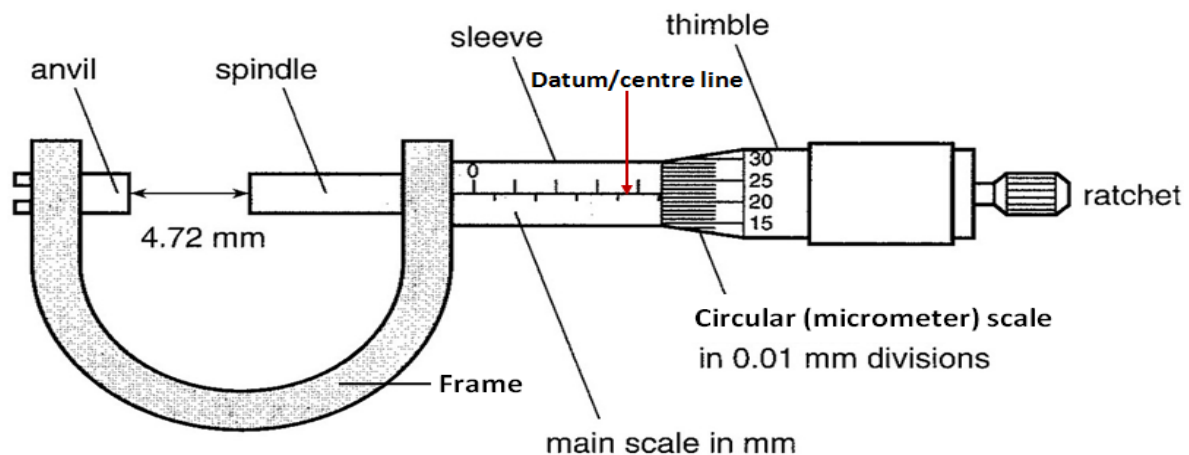


DIAGRAM: Micrometer screw gauge

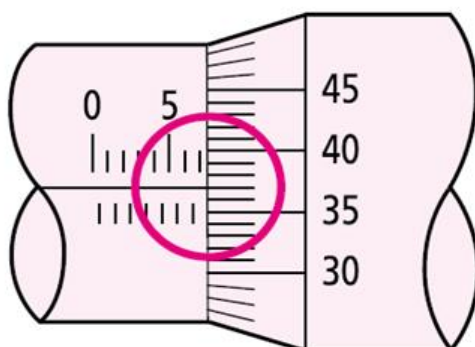
- A micrometer screw gauge has two scales: the main scale on the sleeve and the circular scale on the thimble.
- Each division on the thimble represents a distance of 0.01mm.
- A micrometer screw gauge **measures very short distances** ranging from **0 to 2.5** e.g. diameter of a metal wire. It has an **accuracy of 0.01mm**.

How to read a micrometer screw gauge

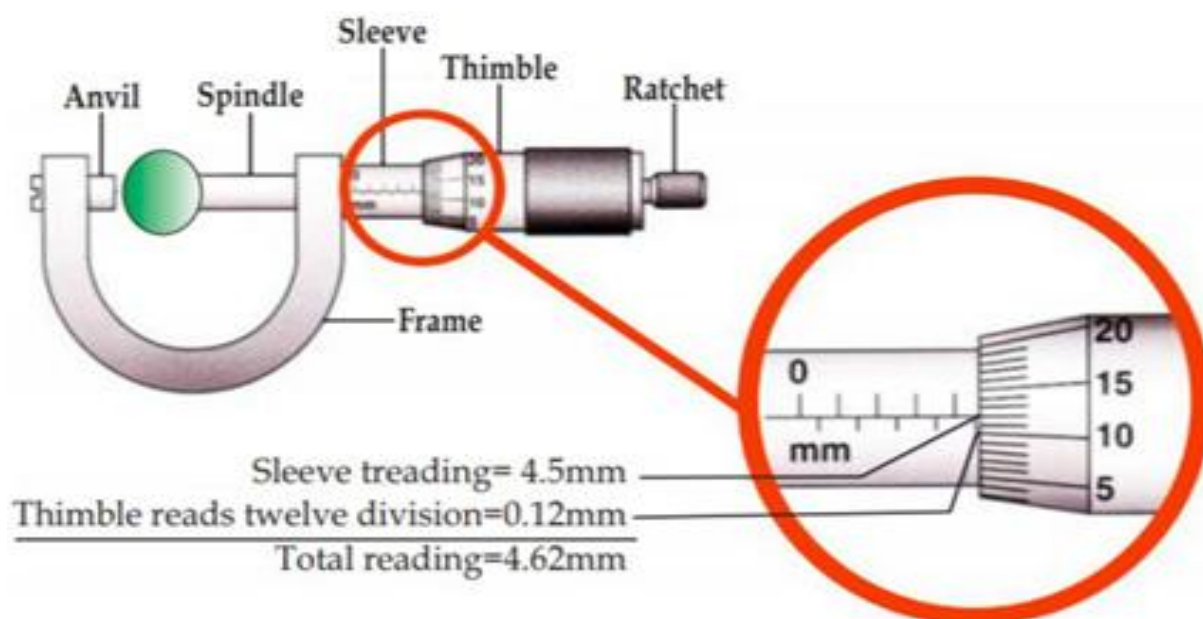
- Thimble is turned until the object is gripped gently.
 1. Read the main scale on the sleeve. This is the sleeve reading.
 2. Find a mark on the circular scale that aligns with the horizontal mark of the main scale. This is the thimble reading.
 3. Actual reading = Sleeve reading + thimble reading.

Example

What is the reading on this Micrometer screw gauge?

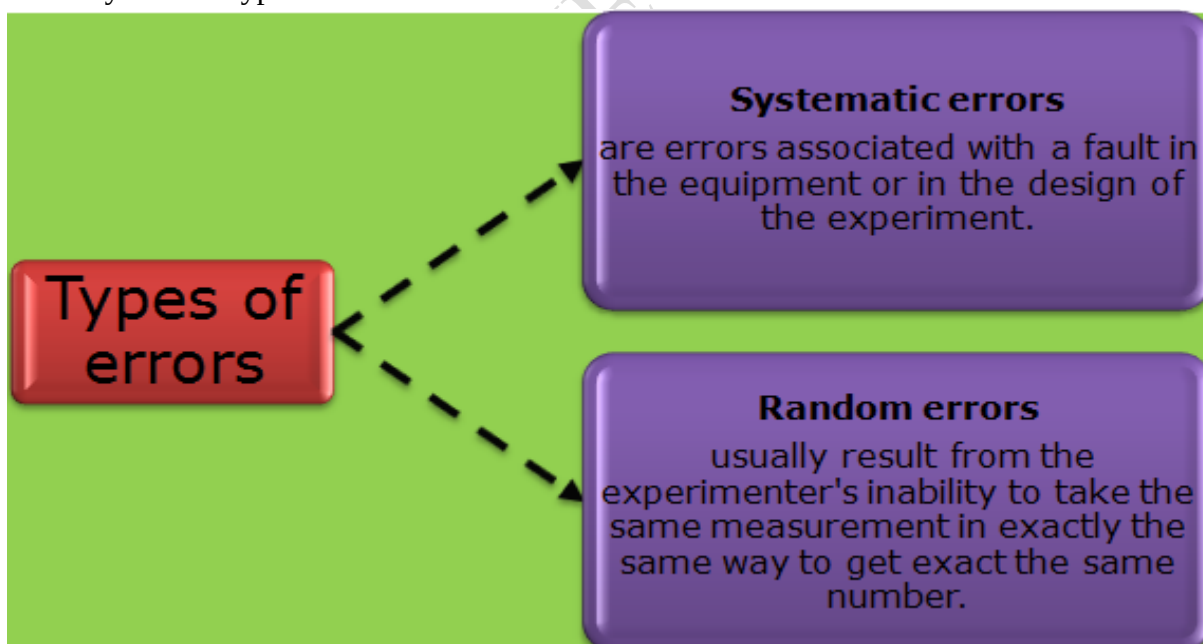


Sleeve reading	7.00 mm
Thimble reading	+ 0.37 mm
Actual reading	7.37 mm



✚ Measurements and Errors in experiments

- In physics experiments (and other sciences), we aim to obtain results that are both accurate and precise, but in all physical measurements errors occur.
- When a measurement is used in a calculation, the error in the measurement is carried through into the result.
- They are two types of errors:

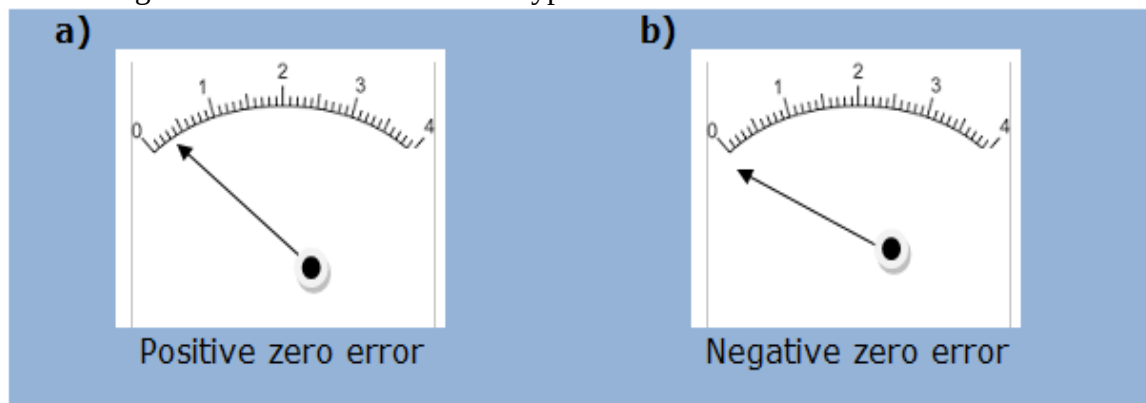


- Example of a **systematic errors** can be due to:
 - a) **Zero error** - occurs when a measuring instrument fails to return or read the zero mark.
 - b) Inaccurately calibrated instruments.
- Example of **random errors** can be due to:
 - a) **Parallax error** - not reading from the appropriate eye level.

- b) Careless in making reading or recording.
- c) Wrong techniques.

Zero error

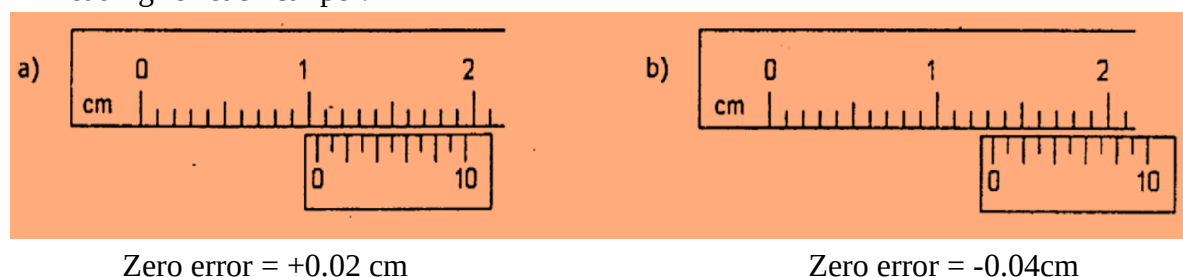
- Consider the following voltmeters that are **not** connected to a circuit. They should be reading zero but both have different types of zero error.



- For such an error the experimenter needs to consider the value of the error in their final reading.
- For the positive zero error, the error margin is subtracted from the final reading.
- For the negative zero error, the error margin is added to the final reading.

Example

- The following calipers have zero errors as indicated in each case. Find the actual reading for each caliper.



Answers:

- a) Actual reading = caliper reading - positive zero error
 $= (1.0 \text{ cm} + 0.5 \text{ cm}) - 0.02 \text{ cm}$
 $= \mathbf{1.03 \text{ cm}}$
- b) Actual reading = caliper reading + negative zero error
 $= (1.3 \text{ cm} + 0.03 \text{ cm}) + 0.04$
 $= \mathbf{1.37 \text{ cm}}$

FORCE

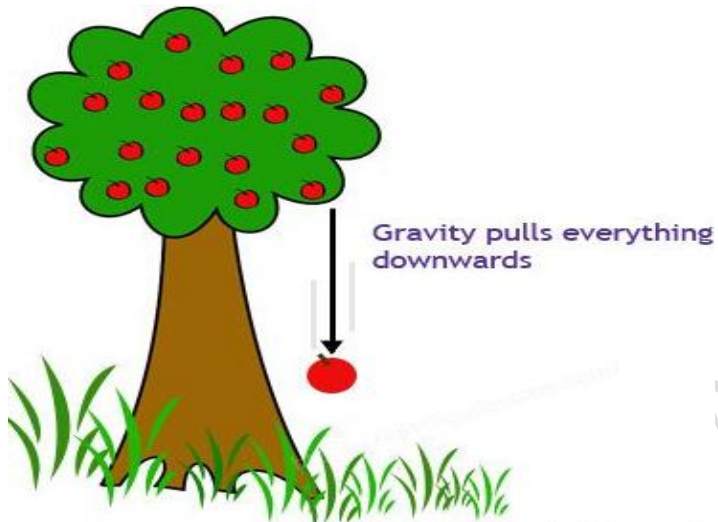
- **Force** is a push or pull that causes an object to change its motion or shape.

- It **changes an object's shape, speed or size.**
- Force is measured using a **spring balance.**
- The S.I unit of force is **Newton (N)**

Types of force

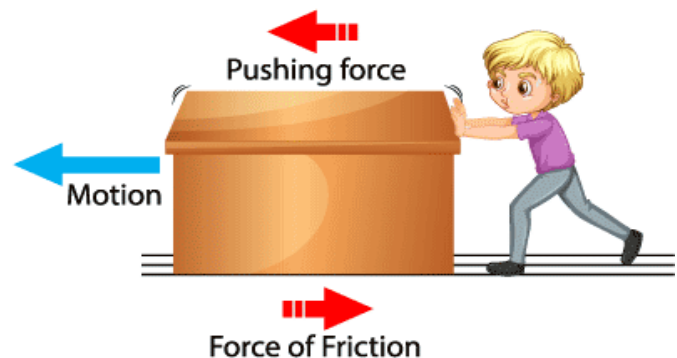
Force of gravity

- It is the force that brings objects downwards.
- It can also be defined as the pull of the earth acting on an object.



Force of friction

- It is the contact force which **opposes motion** of a body due to the roughness of the surface.



Walking



Driving a Car



Writing



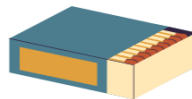
Braking a Bicycle



Rubbing Hands



Climbing a Hill



Lighting a Match



Playing Sports



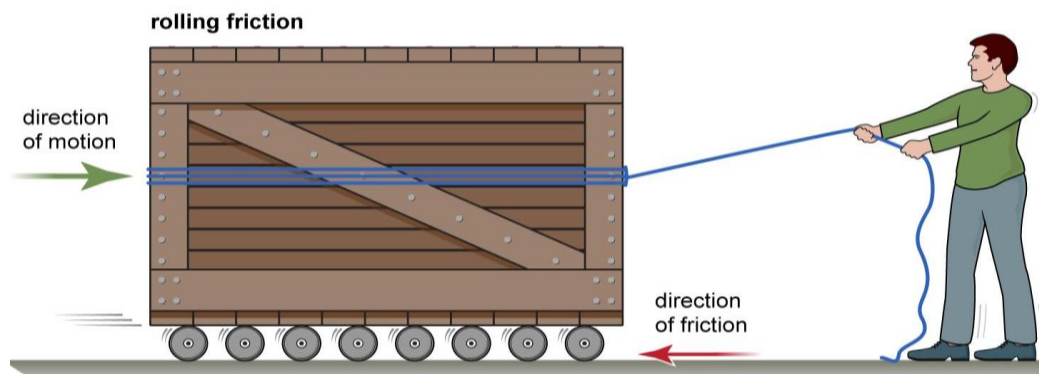
Skiing



Opening a Jar

Ways of reducing friction:

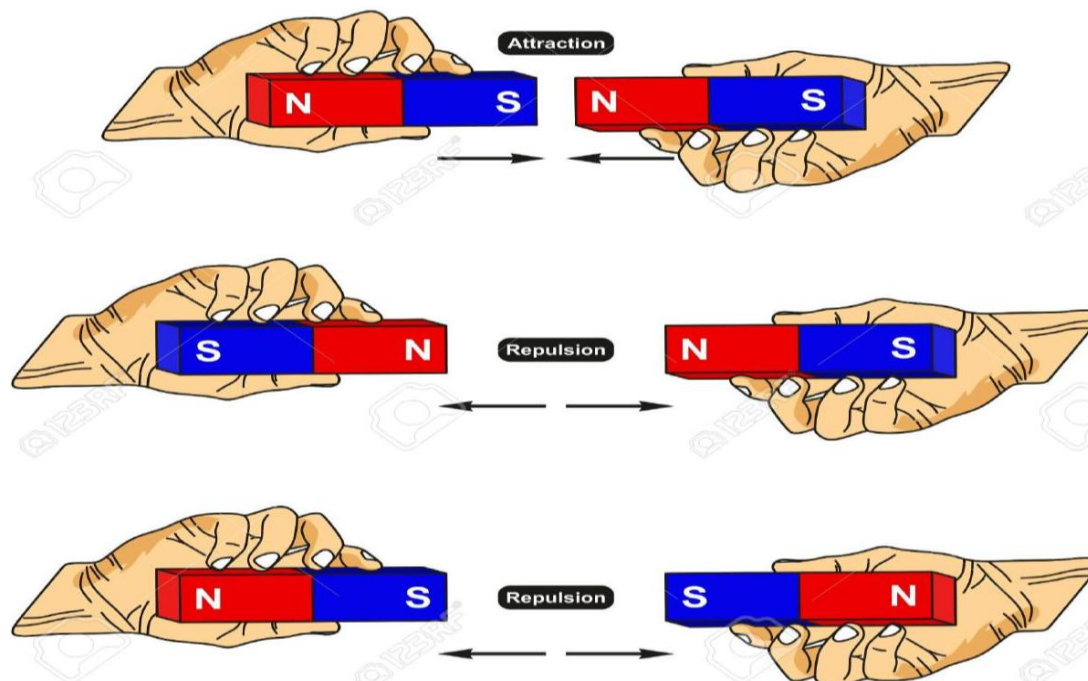
- 1) **Lubrication** - Applying lubricants like oil, grease, or wax to reduce friction between surfaces.
- 2) **Polishing surfaces** - Smoothing surfaces to reduce roughness and friction.
- 3) **Proper alignment of gear teeth.**
- 4) **Use of more pulleys.**
- 5) **Use of light machines.**
- 6) **Streamlining** – fast cars, boats, planes etc. have a streamlined body to allow air (or water in the case of boats) to easily flow by, without offering much resistance.
- 7) **Using materials with low friction coefficients** - Materials like Teflon or silicone can reduce friction.
- 8) **Using ball bearings or rollers** - Allowing objects to roll instead of slide, reducing friction.

**Magnetic force**

- It is the push or pull (force) between magnets.

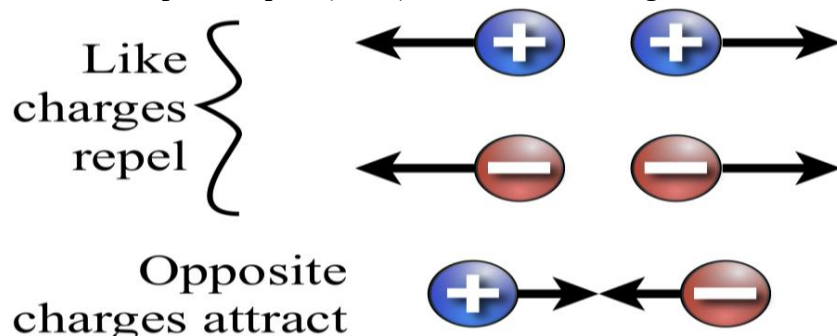
The Law of Magnets

Like Poles Repel, Unlike Poles Attract.



Electrical force

- It is the push or pull (force) between two charges.



Balanced vs. Unbalanced

- **Balanced forces**- When two or more forces acting on an object are equal in all directions.
 - Results in no movement
- **Unbalanced forces**- When two or more unequal forces act on an object.
 - Moves in the direction with more force.
 - Net force is the difference between these forces.



Balanced Forces

1. Two equal sized forces acting on the object in opposite direction
2. Does not produce acceleration
3. Does not cause any change in motion
4. Magnitude is always equal
5. Pushing a wall, a cup on a tray etc.



Unbalanced Forces

1. Two unequal sized forces acting on the object either in same or opposite direction
2. Produces acceleration
3. Makes change in the state of motion
4. Magnitude is not equal
5. Hammering, kicking a ball etc.

Mass and Weight

- **Mass** is the amount of matter in an object.
- It represents the quantity of matter in an object measured in **kilograms(kg)**.
- It is a measure of an object's resistance to change in motion (inertia).
- **Weight** is the amount of gravitational force acting on an object.
- Weight is measured in **Newtons (N)** using a **spring balance or forcemeter**.

$$\text{Weight} = \text{mass} \times \text{gravity}$$

$$W = mg$$

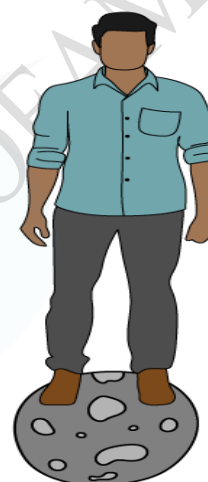
- Weight varies depending on the strength of gravity at a given location (e.g on earth, moon or another planet)
- **N.B** - On the moon there is low gravitational acceleration of 1.6 m/s^2 , while on earth there is high gravitational acceleration of 10 m/s^2 .
- A person with a **mass of 70kg** would weigh **687 N on earth** and **114 N on the moon**.

MASS = 70 kg
 $g = 9.81 \text{ N/kg}$
 WEIGHT = $70 \text{ kg} \times 9.81 \text{ N/kg}$
 WEIGHT = 687 N

MASS = 70 kg
 $g = 1.63 \text{ N/kg}$
 WEIGHT = $70 \text{ kg} \times 1.63 \text{ N/kg}$
 WEIGHT = 114 N



EARTH



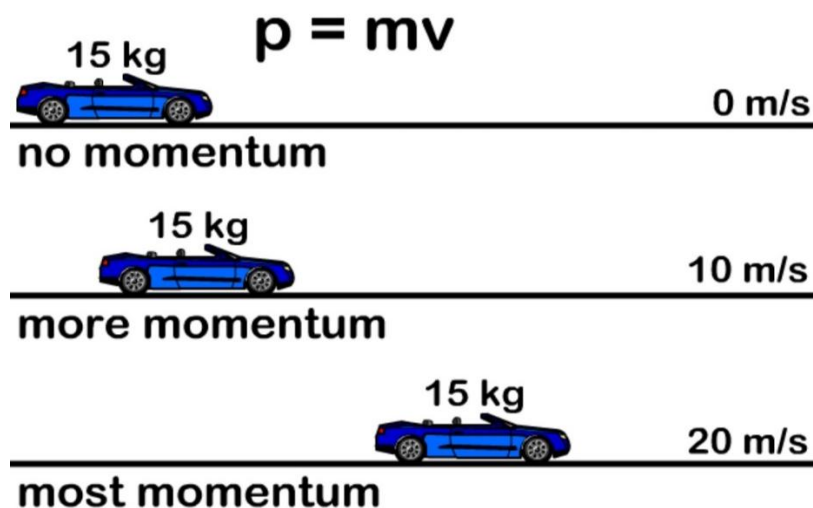
MOON

Sl. No.	Mass	Weight
1.	The mass is a scalar quantity.	The weight is a vector quantity.
2.	Mass of a rigid body is regular everywhere in the universe.	The weight of a rigid body alters from place to place and inclines zero at the center of the earth.
3.	Mass can be resulted by a traditional balance.	Weight can be defined as spring balance
4.	The unit of mass is kg or g.	The unit of weight is Newton.
5.	Mass can never be zero.	Weight can be zero based on the gravity acting upon it.
6.	Mass does not change based on location.	Weight changes based on location, depending on the gravity it experiences.
7.	Mass is measured using an ordinary weighing scale.	Weight is measured using spring balance.

Momentum

- **Momentum** is defined as the product of mass and velocity.
- It is a measure of an object's resistance to stop.
- The unit of momentum is kgms^{-1}
- It is calculated as follows:

$$\text{Momentum} = \text{Mass} \times \text{Velocity}$$

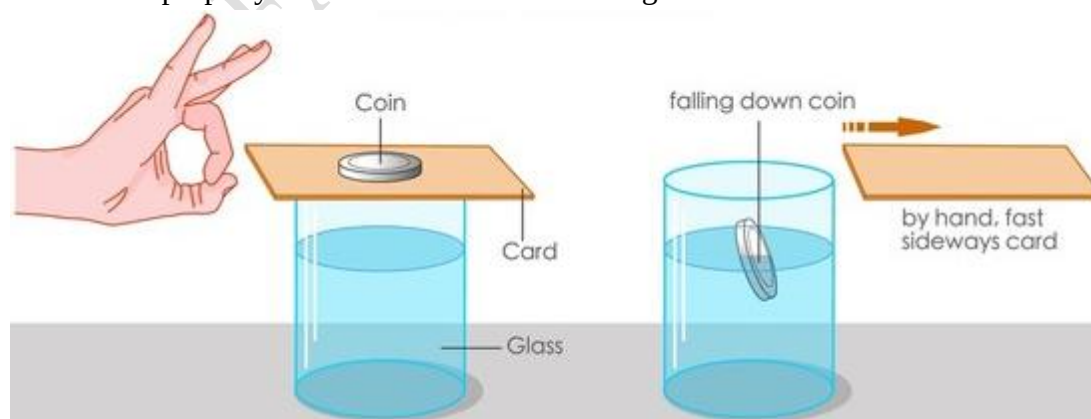


Example

- A body of mass 2 kg travelling 3 ms^{-1} has a momentum of 6 kgms^{-1} .

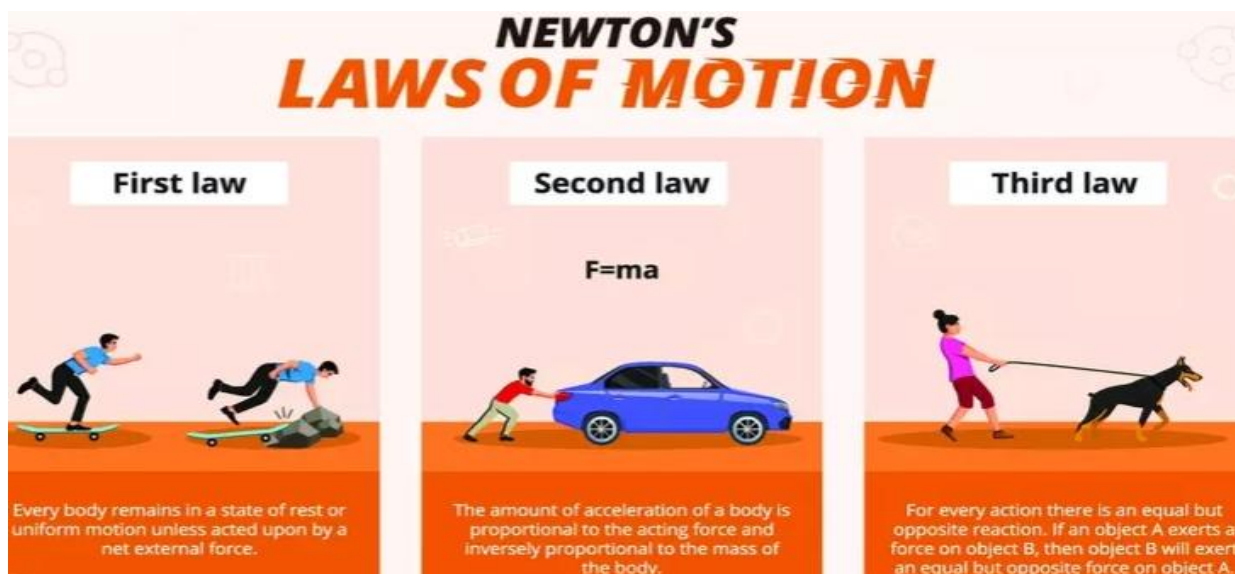
Inertia

- The inertia refers to the reluctance or resistance of the object to start moving if it is stationary or reluctance of the object to stop moving if it is moving in the first instance.
- An object that has inertia will try to resist a change in its state of motion or rest
- Objects with a greater mass have more inertia.
- It is the property of a mass which resists change from its state rest or uniform motion.



Newton's Laws of motion

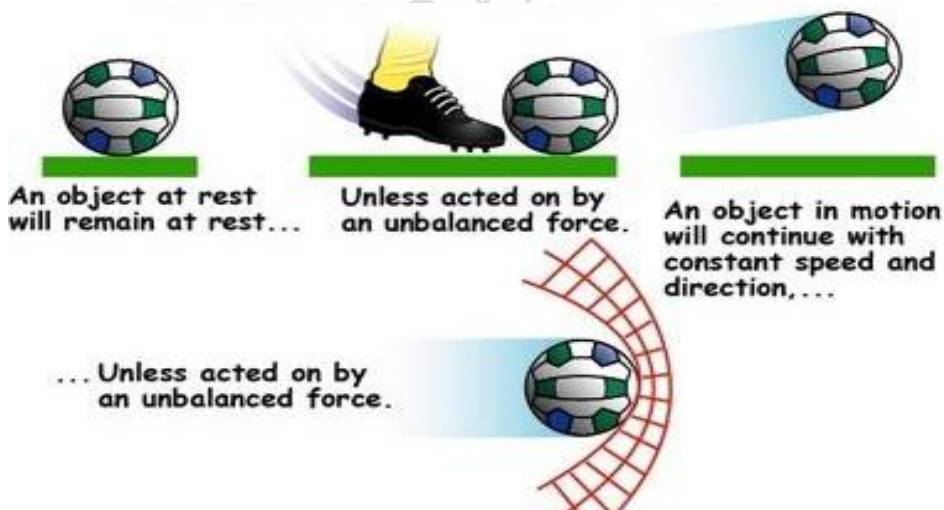
- Newton's laws of motion are three physical laws that **describe the relationship between the motion of an object and the forces acting on it.**



Newton's first law of motion – (law of inertia)

A body will remain at rest, or to move with a constant velocity, unless acted upon by an external force

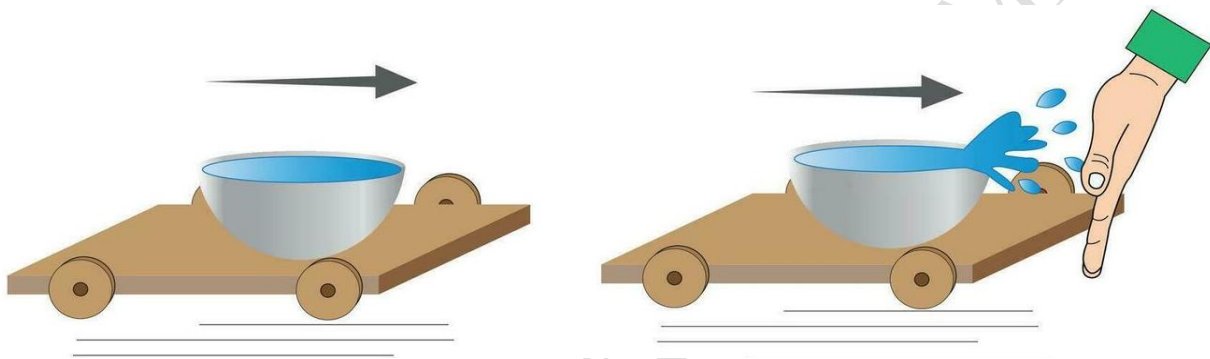
- This law tells us two things:
 - 1) If there is no resultant force acting on an object at rest, the object will remain at rest.
 - 2) If there is no resultant force acting on a moving object, the moving object will continue to move at a constant speed in a straight line.
- Application of this law includes the following:



- A football will remain stationary forever unless moved by some external force
- Drivers need to wear seat belts when driving, without the seat belt, if the driver suddenly applies his car brakes, he will continue to move forward and crash into the windscreen.
- Moreover, if there is an object on top of a car which is not tied – it will continue moving forward when the brakes are applied.



- Slide a glass full of water across a table to the right. The water in the glass will seem as though it is moving to the left of the glass.

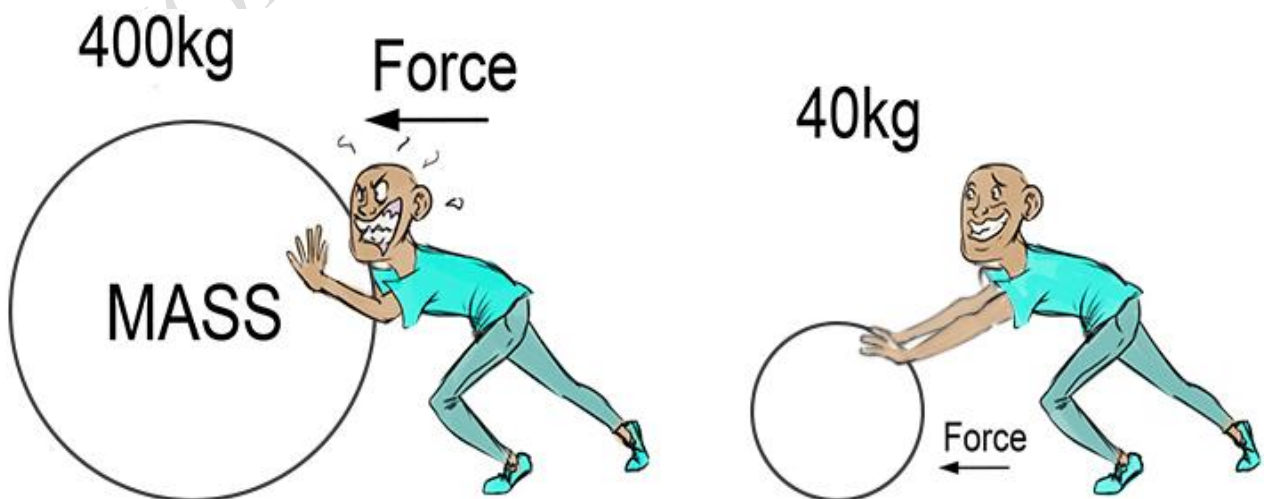


Newton's second law of motion – (law of force and acceleration)

The acceleration of mass is proportional to the force acted on it, provided the mass stays the same

- Newton's second law tells us that when there is a resultant force acting on an object, the object will either slow down (decelerate) or speed up (accelerate) in the direction of the net force.

$$F = ma$$

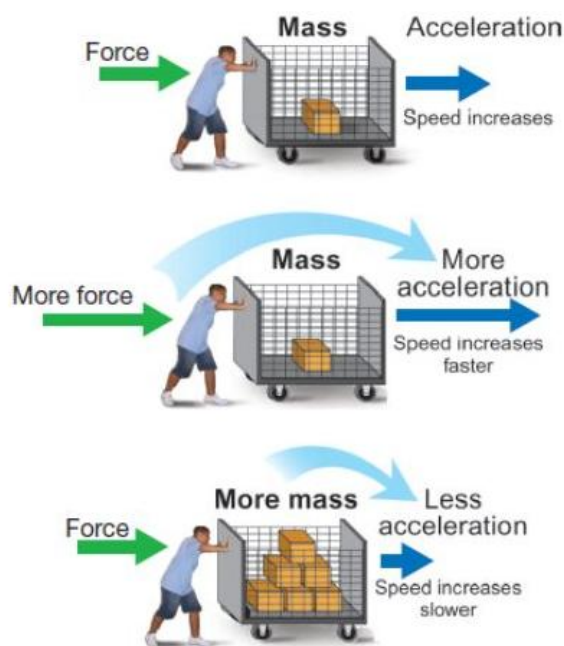


$$a = \frac{F}{m}$$

Acceleration (m/s^2)

Force (N)

Mass (kg)



Questions

- A boy pushed a box of mass 20kg with a force of 50N. What is the acceleration of the box?
- What is the force that acts on 2kg mass when a constant acceleration of 3m/s^2 is expressed?

Newton's third law – (law of action and reaction)

Action and reaction are equal and opposite

- This law tells us that, if body A exerts a force on body B, then body B will exert a force on body A of equal magnitude but in opposite direction.
- Forces occur in pairs i.e. **action force** and **reaction force**.
- The action and reaction are equal in magnitude, act opposite to one another and act on different bodies.
- Application of this law includes the following:



Boxing Punch



Jumping in Basketball



Soccer Player Kicking the Ball



Swimming Strokes



Tennis Serve



Rowing



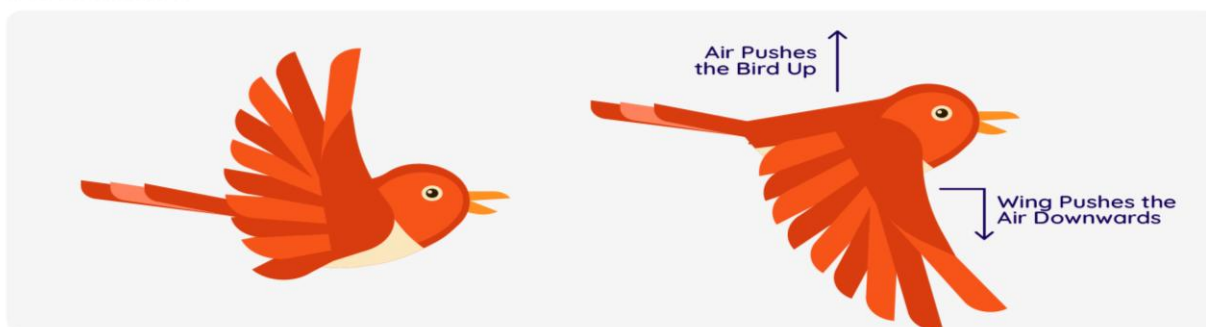
Jumping



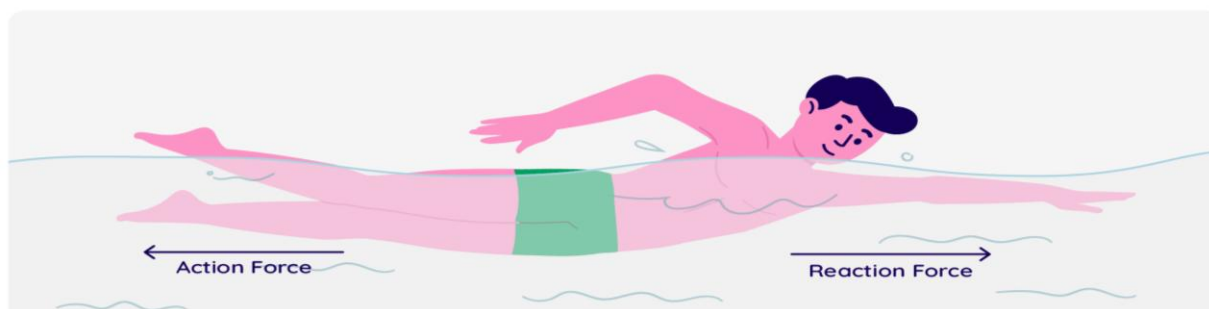
Bicycle Racing

- A high jump will exert an action on the ground which reacts by causing the jumper to spring upwards.
- As birds push down on the air with their wings, the air pushes their wings up and gives them lift thus making it possible for them to fly

Example 2: Birds also fly using the Third Law of Motion. The wings of the bird push the air, creating an upward force which helps them move forward. Airplanes and helicopters also use this technique.



Example 3: While swimming or moving in a rowboat, when you push the water backward, there is an equal and opposite force generated which causes forward motion.



MECHANICAL SYSTEMS

Machines

- A **machine** is a mechanical system designed to make hard work easier
- A machine enables less force to move an object.

➤ Examples of machines include:

- ✓ Inclined planes,
- ✓ gears,
- ✓ pulleys
- ✓ levers

Mechanical Advantage (MA)

- It is the measure of how much easier it is to complete a task using a simple machine.
- The greater the MA, the easier it is to do work
- It is calculated as follows:

$$MA = \frac{\text{Load (L)}}{\text{Effort (E)}}$$

- If MA is greater than 1 (> 1), it means less effort is needed to lift the load.
- If MA is equal to 1 (= 1), it means that the same amount of effort is needed to lift the load.
- If MA is lesser than 1 (< 1), it means more effort is needed to lift the load

Example:

Calculate the mechanical advantage.

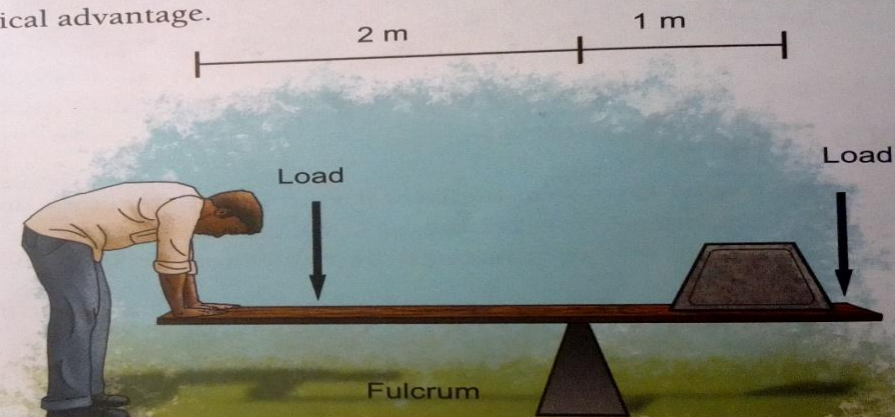


Figure 4.3

$$MA = \frac{\text{Effort arm distance}}{\text{Load arm distance}}$$

$$MA = \frac{2}{1}$$

$$MA = 2$$

Velocity Ratio (VR)

- Is the ratio of two distances i.e. distance moved by the effort and the distance moved by load.

$$VR = \frac{\text{Distance moved by effort}}{\text{Distance moved by load}}$$

Efficiency

- It refers to the ratio of work output (MA) and work input (VR).
- Efficiency shows how input energy (force applied) is converted to useful output energy (energy to lift the load).
- It can be expressed as a percentage.

$$\text{Efficiency} = \frac{MA}{VR} \times 100\%$$

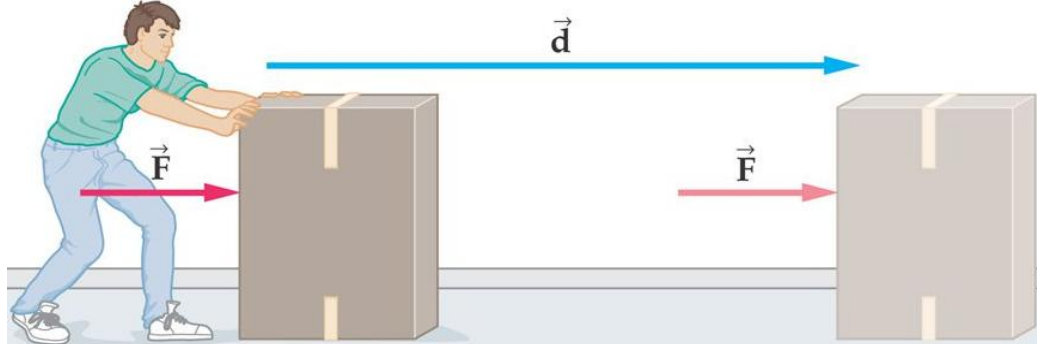
NB - Efficiency, VR and MA are all ratios hence they all have no units.

Work done

$$W = F \times d$$

$$\begin{array}{ccc} \text{Work} & = & \text{Force} \times \text{displacement} \\ (\text{J}) & & (\text{N}) \quad (\text{m}) \end{array}$$

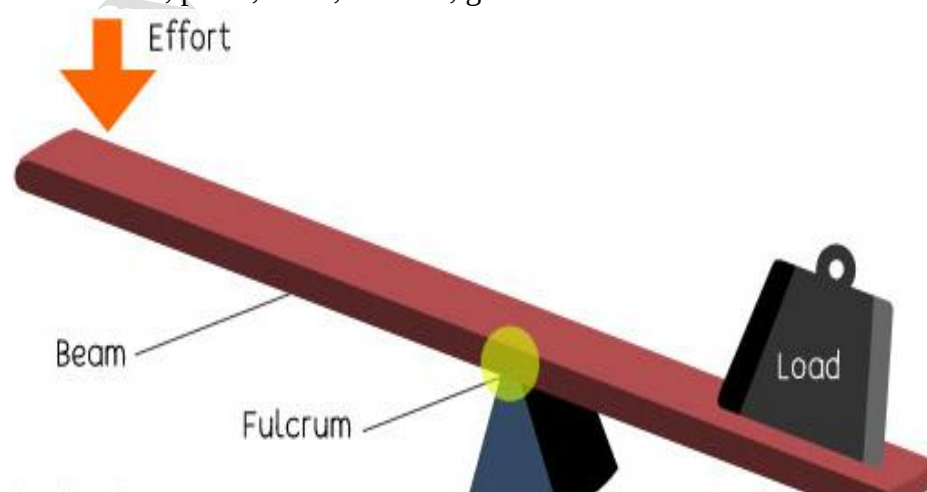
- Work is done when a constant force is applied on an object to move in the same direction the force was applied, thus the force is parallel to displacement.



Types of simple machines

Levers

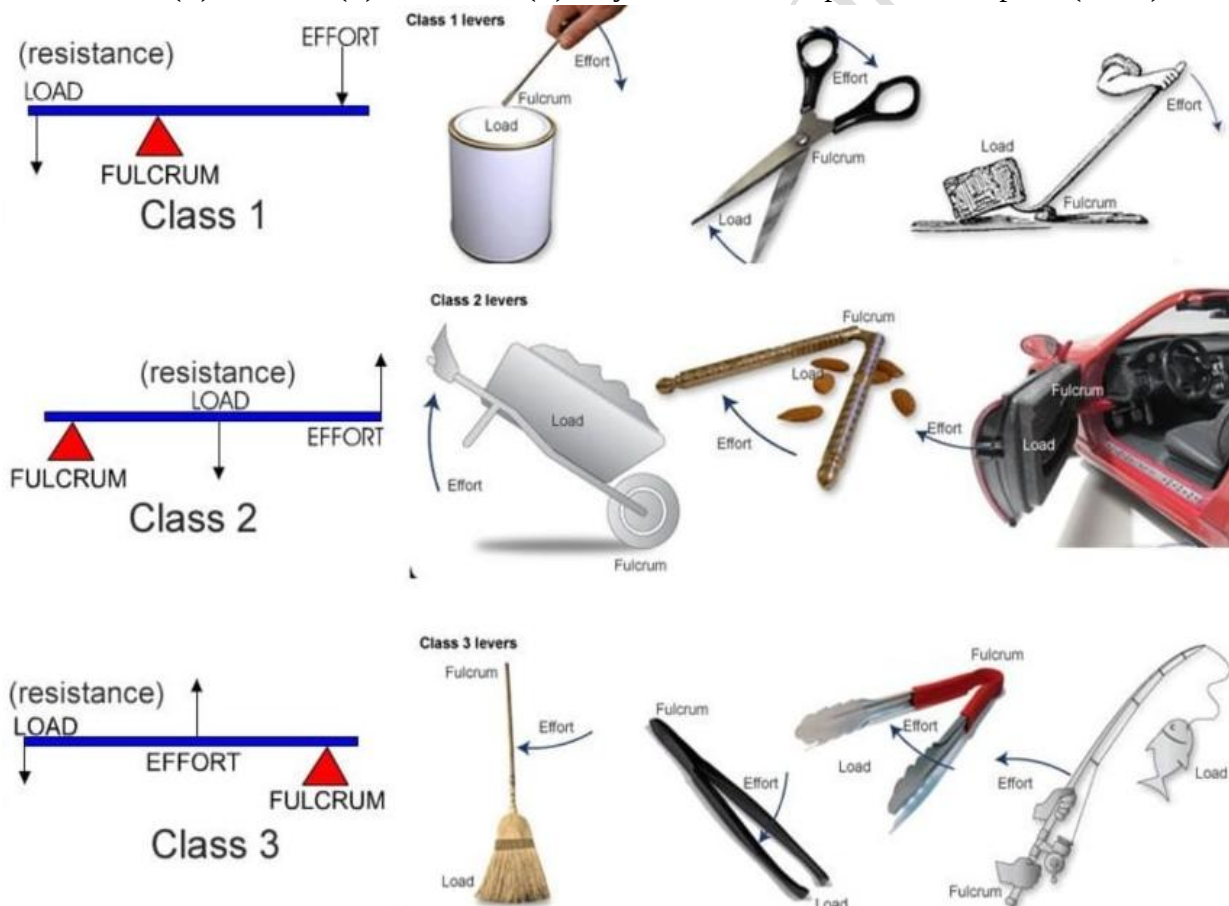
- A **lever** is a rigid bar that rotates around a fixed support called fulcrum to lift the load.
- It is perfect for lifting or moving heavy things.
- The lever is a machine which acts as a force multiplier i.e. it allows a smaller force (effort) to overcome a large force (load).
- When the effort distance is increased, MA and VR increases while efficiency remains the same.
- Efficiency is improved through lubrication as it reduces friction.
- Good examples of levers include the seesaw, crowbar, fishing line, wheelbarrows, Claw hammer, pliers, knife, scissors, garden shovel etc.



- Levers have four important parts:
 - ✓ bar or beam,
 - ✓ fulcrum (pivot or the turning point),
 - ✓ effort (or force),
 - ✓ and the load.
- A lever may be wood, metal, or any durable material.
- The beam rests on a fulcrum (a point on the bar creating a pivot).
- When you push down one end of a lever, you apply a force (input) to it.
- The lever pivots on the fulcrum and produces an output (lift a load) by exerting an output force on the load.
- A lever makes work easier by both increasing your input force and changing the direction of your input force.

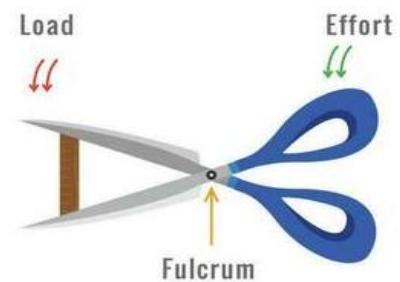
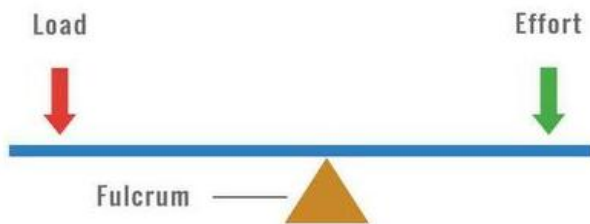
The three classes of levers

- The parts of the lever are not always in the same arrangement.
- The load (L), fulcrum (F), and effort (E) may be at different places on the plank (beam).

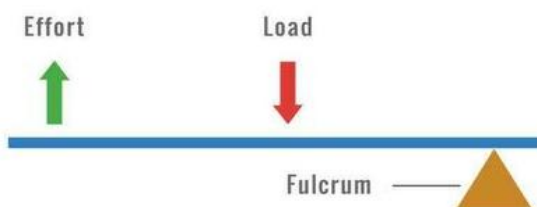


1) Class One Lever

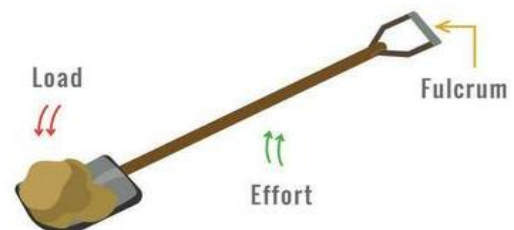
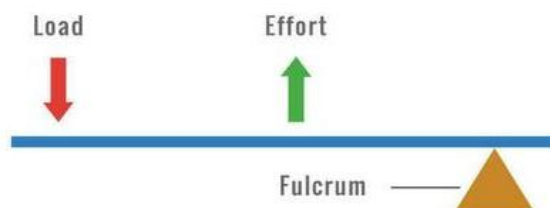
- In this class, the Fulcrum is between the Effort and the Load.
- The mechanical advantage is more if the Load is closer to the fulcrum.
- Examples of Class One Levers include seesaws, boat oars, scissors and crowbars.

First Class**2) Class Two Lever**

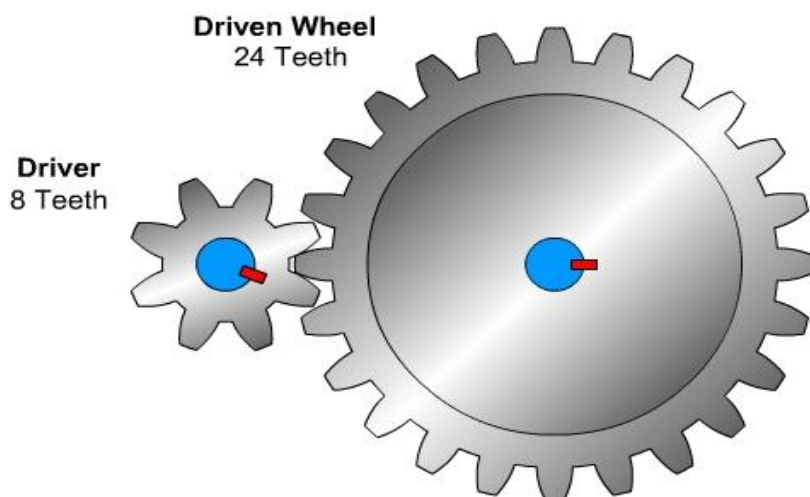
- In this class, the Load is between the Effort and the Fulcrum.
- The mechanical advantage is more if the load is closer to the fulcrum.
- Examples of Class Two Levers include wheelbarrows.

Second Class**3) Class Three Lever**

- In this class, the Effort is between the Load and the Fulcrum.
- The mechanical advantage is more if the effort is closer to the load.
- An example of a Class Three Lever is a garden shovel.

Third Class**Gears**

- A **gear** is a system of wheels with interlocking teeth so that one-wheel drives another by transferring force from one gear to another.
- There are two gears that will enable work to be done, namely effort gear (driving gear) and load gear (driven gear).
- The effort gear is smaller than the load gear.
- This means that the effort gear has to be turned several times for the load gear to turn.



- The diagram above shows a load gear (driven wheel) with 24 teeth and an effort gear (driver wheel) with 8 teeth. This means that the effort gear (with 8 teeth) makes three (3) complete revolutions in order to drive and rotate the load gear (with 24 teeth).

$$\text{MA} = \frac{\text{diameter of load (driven) gear}}{\text{diameter of effort (driver) gear}}$$

Calculating velocity ration of gears

- As gears are used to transmit motion and force from one shaft to another, it is important to be able to calculate the velocity of the driven gear from the velocity of the driver gear.
- When the velocity of the driver gear is known, the velocity of the driven gear may be calculated by using one of the two formulae below:

$$1) \text{ Velocity ratio (gear ratio)} = \frac{\text{distance moved by the effort (driver gear)}}{\text{distance moved by load (driven gear)}}$$

$$2) \text{ Velocity ratio (gear ratio)} = \frac{\text{number of teeth on the driven gear (load)}}{\text{number of teeth on the driver gear (effort)}}$$

a. Using method 1 to calculate VR

- In the diagram above, the driver gear has 8 teeth and the driven gear has 24 teeth.
- For 1 complete revolution of the driver the driven wheel moves 1/3 of a revolution, i.e. the driver revolves 3 times for every revolution of the driven gear.

$$\text{Velocity ratio} = \frac{\text{distance moved by the effort (driver gear)}}{\text{distance moved by load (driven gear)}}$$

$$= \frac{1}{1/3}$$

$$= \frac{3}{1}$$

$$= 3$$

- The distance moved by the driver gear divided by the distance moved by the driven gear may be shown as a ratio, e.g. gear ratio of 3:1

b. Using method 2 to calculate VR

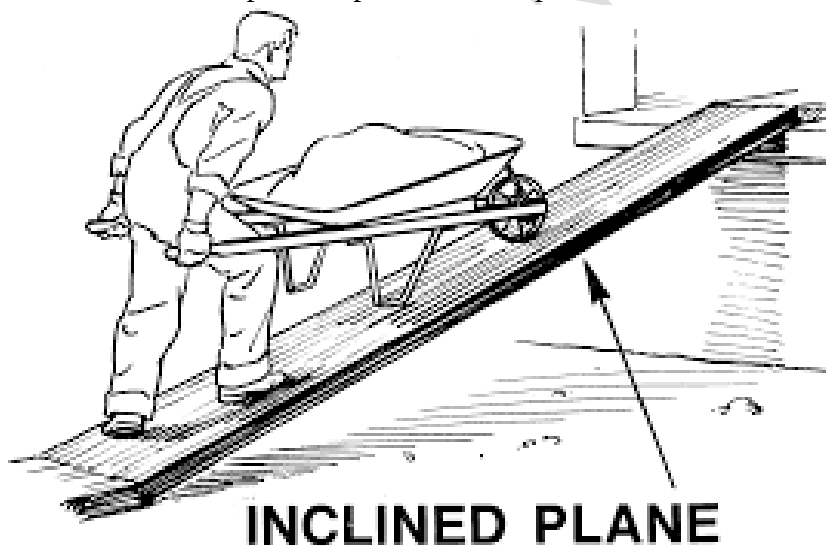
$$\begin{aligned}
 \text{Velocity ratio} &= \frac{\text{number of teeth on the driven gear (load)}}{\text{number of teeth on the driver gear (effort)}} \\
 &= \frac{24}{8} \\
 &= \frac{3}{1} \\
 &= 3
 \end{aligned}$$

Improving the efficiency of gears

- Lubrication
- Appropriate shaping of gear teeth
- Proper alignment of gear teeth to reduce friction

Inclined planes

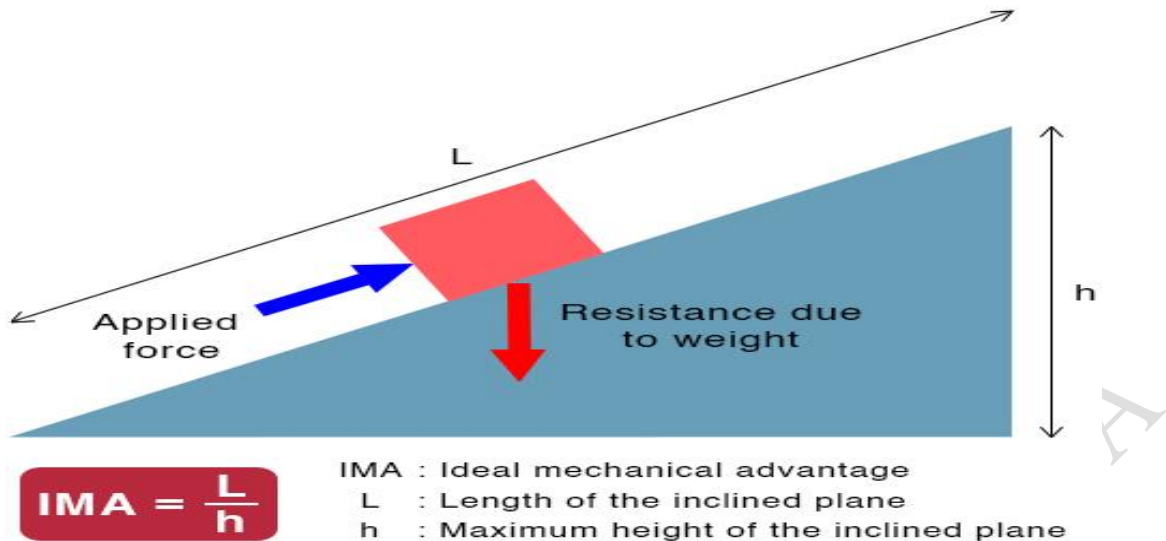
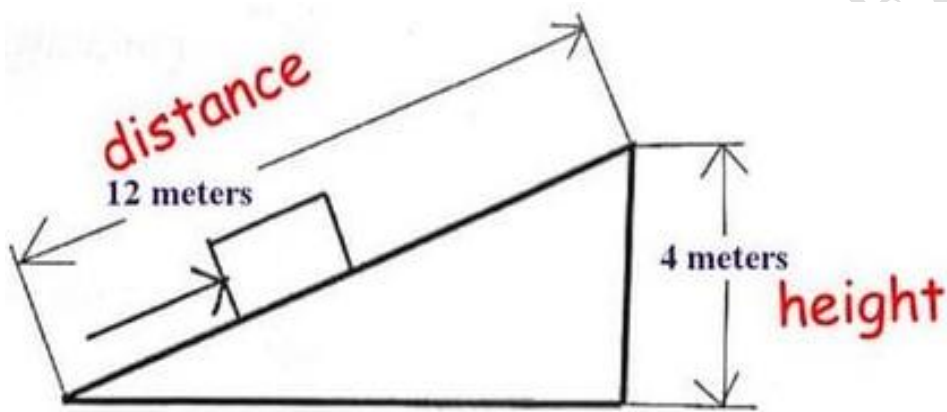
- An **inclined plane** is a form of simple machine that consists of a sloping surface with one end higher than the other.
- They are generally used for raising and lowering a load over a vertical obstacle.
- An example of an inclined plane is shown in the diagram below. Rather than lifting the wheelbarrow up the step, an inclined plane is used to move it up with less force.



- The mechanical advantage (MA) of an inclined plane is the ratio of the **length of the ramp** to its **height** ($MA = L / h$).

$$MA = \frac{\text{Length of Inclined plane}(L)}{\text{Height of Inclined plane}(H)}$$

- A higher MA means the force needed to move an object up the ramp is reduced, but the distance it must travel is increased.

**Example**

$$\begin{aligned}\text{MA} &= \frac{\text{Length/Distance}}{\text{Height}} \\ &= \frac{12}{4} \\ &= 3\end{aligned}$$

Take note:

- By using a longer, less steep ramp (increasing L), you can move an object to a higher height (h) with less force.
- The longer the ramp, the lower the angle, and the easier it is to push or pull the object up, but the longer the distance traveled.

Improving the efficiency of an inclined plane

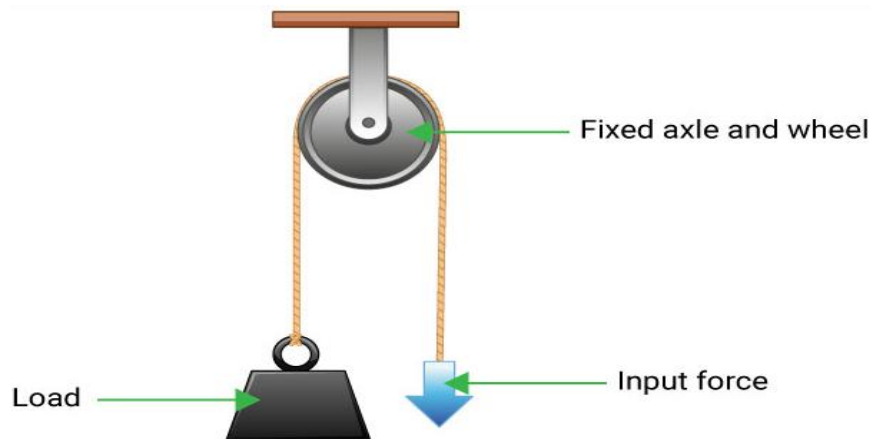
- Lubrication
- Increasing length of incline

Pulleys

- A **pulley** is a wheel with a groove through which a rope or belt passes.

- There are four types of pulleys, namely **single fixed pulley**, **single moveable pulley**, **compound pulley** and **block and tackle pulley**.

1) Single Fixed pulley



- A fixed pulley changes the direction of the input force but does not change the amount of force applied to it.
- Fixed pulleys are usually attached to a stationary object above the load, which means its axle and wheel will stay in one place.
- **N.B** - It is easier to lift a load by pulling down the rope than lifting the load.
- The **MA of a fixed pulley is equal to 1 (MA = 1)** because the values of the load and effort will be the same.
- The **VR of a fixed pulley is equal to 1 (VR = 1)** because distance moved by the load and effort must be the same.
- Examples of fixed pulley includes:
 - ✓ Flag pole
 - ✓ Water bucket of the well

Example of finding MA of the pulley

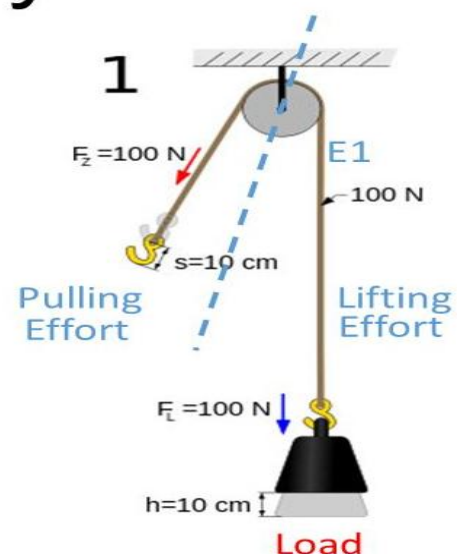
1 Pulley

$$\text{Mechanical Advantage (MA)} = \frac{\text{Load}}{\text{Pulling Effort}}$$

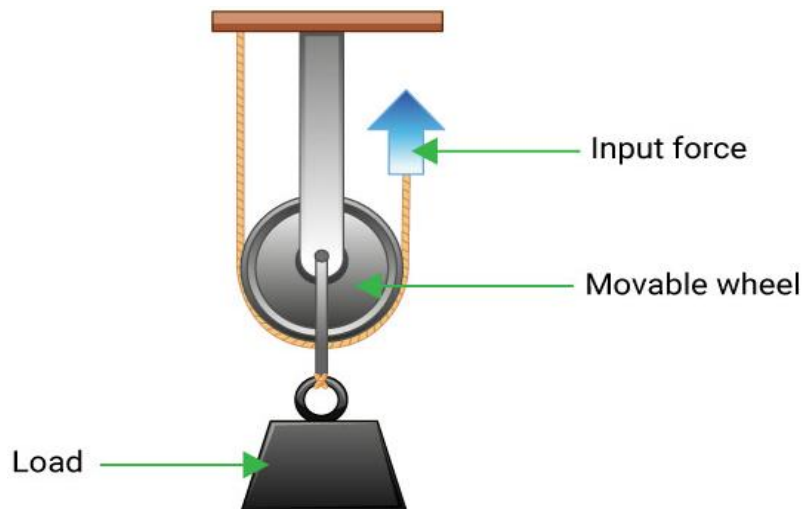
$$\text{Pulling Effort} = \text{Lifting Effort on E1}$$

$$\text{Mechanical Advantage} = \frac{100 \text{ N}}{100 \text{ N}}$$

$$\text{Mechanical Advantage} = 1 = \text{No Advantage}$$

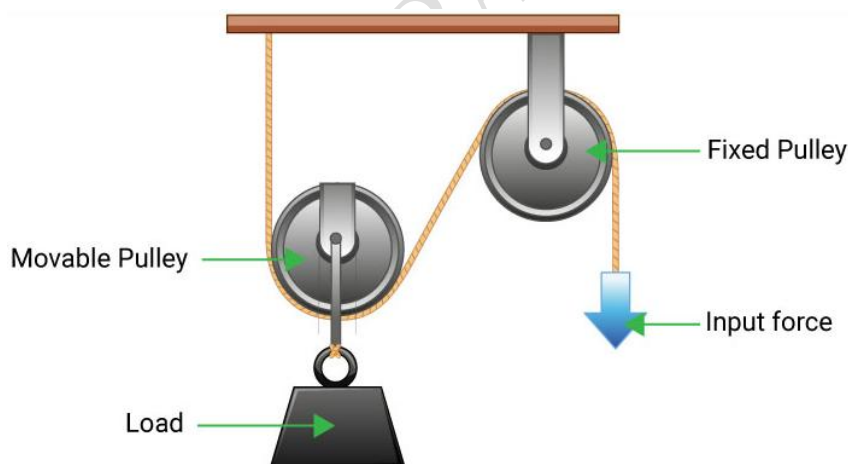


2) Single moveable pulley

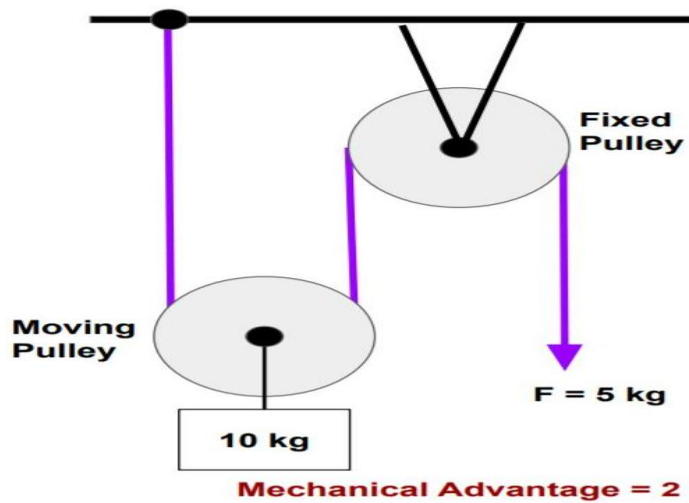
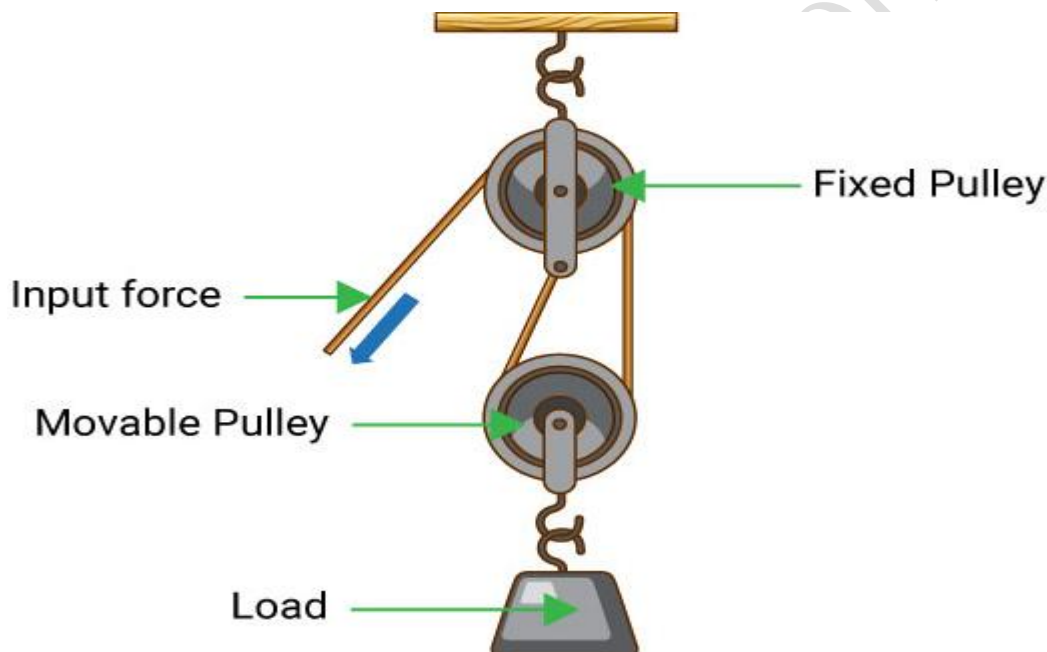


- A movable pulley does not change the direction of the force but increases the amount of force applied.
- Moving pulleys are attached to the loads, which means its axle is in a movable block.
- They make the load feel much lighter.
- The **MA = 2**, because the load is two times the effort.
- The **VR = 2**, because as the pulley moves up the load, the string moves twice the distance
- Examples of the movable pulleys include:
 - ✓ Elevators
 - ✓ Construction cranes

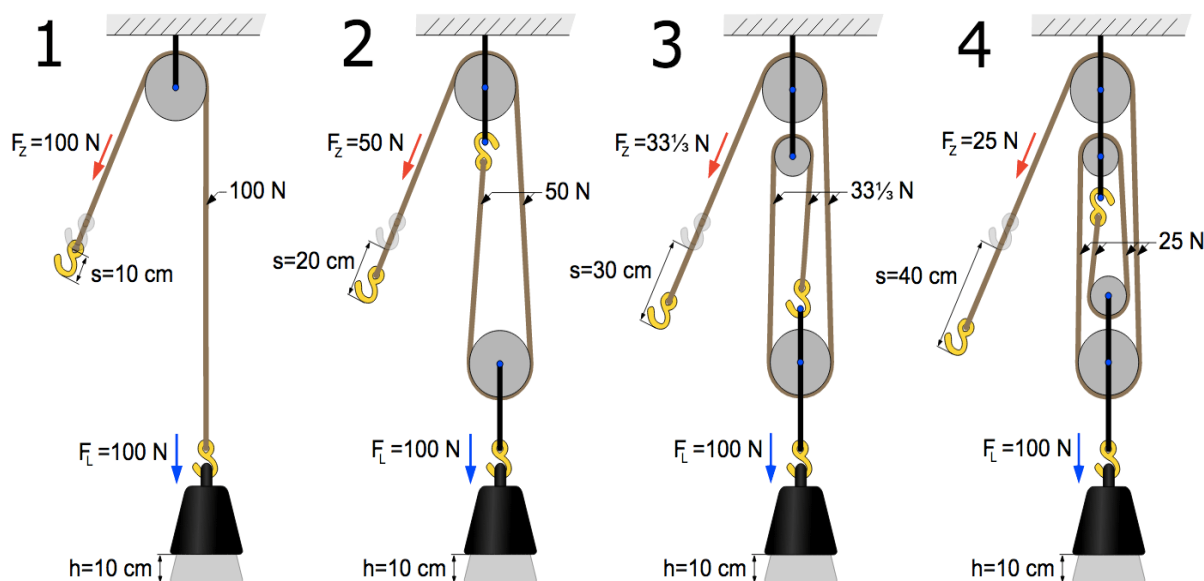
3) Compound pulley



- Compound pulleys are also known as combination pulleys, and they are made up of both fixed and movable pulleys.
- Together, they let you change the direction of force and make the weight feel lighter.
- The size of the wheel and the length of the rope, chain, and belt of this simple machine are designed to lessen the effort.
- An example of a compound pulleys:
 - ✓ Gym equipment such as chest exercises

Example of finding MA of the pulley4) Block and tackle pulley

- A block and tackle pulley comprise several movable pulleys, and fixed pulleys are arranged parallel.
- They are known as a specific form of compound pulleys where movable pulleys are aligned parallel with movables and fixed pulleys are fixed.
- When two or more pulleys are used, the machines become a force multiplier as well as direction changer.
- Using more pulleys/wheels reduce the effort you need to lift the load, thus increases the MA of the pulley.
- N.B - Overall, the block and tackle systems have two types of pulleys: fixed and moveable, and the mechanical advantage of all their examples varies.
- The **velocity ration (VR)** is equal to the **number of pulleys/ropes**.
- The following numbers (**1,2,3, and 4**) are showing the VR of the pulleys (you just have to count the number of ropes or pulleys)



Improving the efficiency of pulleys can be improved through

- Lubrication
- Using lighter pulleys
- Increasing number of pulleys

Examples of pulleys in everyday life

- A man uses almost all types of pulleys in a single day.
- Here are a few examples of pulleys we use in our everyday life:

1. Escalators & Elevators

- Elevators and escalators utilize pulley systems.
- Multiple pulleys are used to make one elevator or escalator, and they have made our life easy.

2. Gym or Exercise Equipment

- Pulley systems are used for the development of exercise machines.
- They are used to make gym or exercise equipment's weight lifting and movement easier, effortless, and safe. Without pulley systems, exercise equipment cannot operate.

3. Curtains & Blinds

- Pulleys are used in blinds and curtains to pull them down and up easily.
- Also, pulleys lift the curtains and drop them down in theatres.

4. Hanging Plant Pots

- Pulleys hang plant pots or flower baskets on the roof or walls as you can conveniently lower them to water or cleaning.

5. Well Bucket

- In some areas (especially rural areas), grooved wheel, moveable pulleys are used to pull the water bucket out of the well.

6. Kitchen Hangings

- Look in your kitchen, and you will find different applications of pulleys hanging around, which can be lowered for use, for example, hanging charts, dumbwaiters, or lights.

7. Sliding Doors

- We can easily slide the doors because of the types of pulleys used.
- Even the garage doors have pulleys to make opening and closing effortless.
- Sliding doors are mostly used by supermarkets.

8. Rock Climbing Equipment

- Rock climbers take the help of pulley systems to climb the rocks safely and easily.
- Flexible material, different pulleys, grooved wheels are used to balance their heavyweights and the direction of the force.

9. Construction Cranes

- Laborers use crane pulleys to move extremely heavy material or objects up and down the building.
- Moreover, some crane pulleys are used to collect the debris.

10. Fishing Rod

- Fishing couldn't be possible without pulleys.
- The enhancement of the grooved wheel in the fishing rod helps you catch the fish.

11. Washing Machine

- The motors used in the washing machines use belts connected by pulleys because they provide locomotion to move up and down.

Energy losses in machines

- Machines are used to do work, converting energy from one form to another.
- In practice machines are never 100% efficient.
- This means that total energy input is greater than the useful work output.
- Some of the energy input is used to overcome mass of moving parts of the machine hence there is need for mass reduction (use light machines).
- Some of the energy input is converted to unwanted forms such as heat and sound.

Ways of reducing friction or improving efficiency of machines

- **Lubrication**

- To overcome or minimize friction, the movable parts of the machine are lubricated
- **Using rollers and ball bearings.**
- Are made of hardened chrome steel
- Are used with lubrication
- **mass reduction**
- Some of the energy input is used to overcome mass of moving parts of the machine hence there is need for mass reduction by constructing a machine with light parts
- **Reducing contact area**
- Reducing the area over which friction occurs improves efficiency of machines.

PRESSURE

1. Pressure in solids

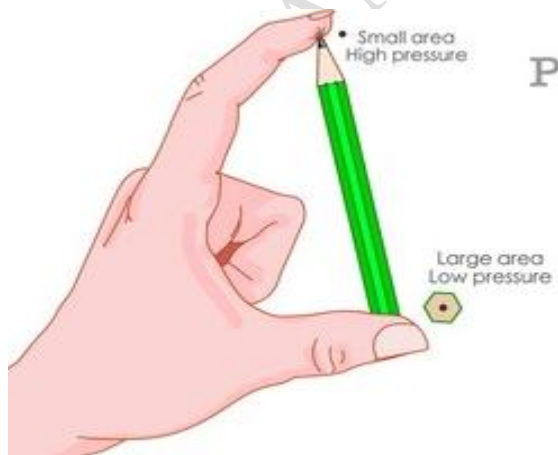
- Pressure is defined as the force exerted per unit area.
- It is calculated as follows:

$$P = \frac{F}{A}$$

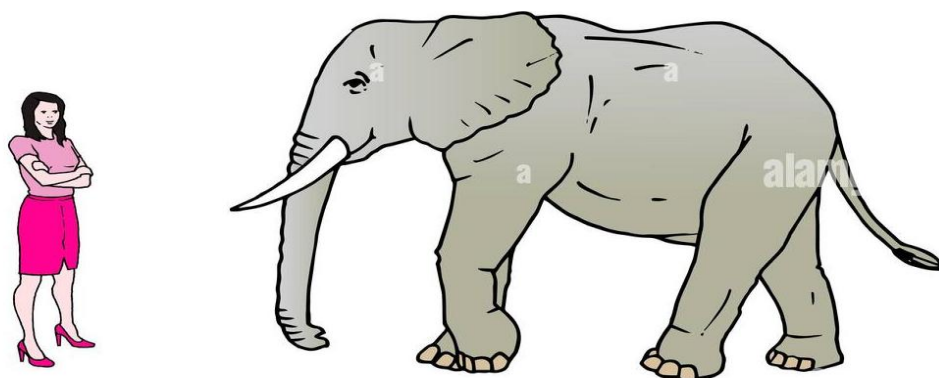
Pressure → P Force F (N) Area A (m^2)

$$1 \text{ Pa (Pascal)} = 1 \text{ N/m}^2$$

- Unit for Pressure is in $\frac{N}{m^2}$, which is known as **pascal (Pa)**.
- For high pressures, the pressure is expressed in kilopascals (kPa).
- **N.B** - $1 \text{ kPa} = 1000 \text{ Pa}$



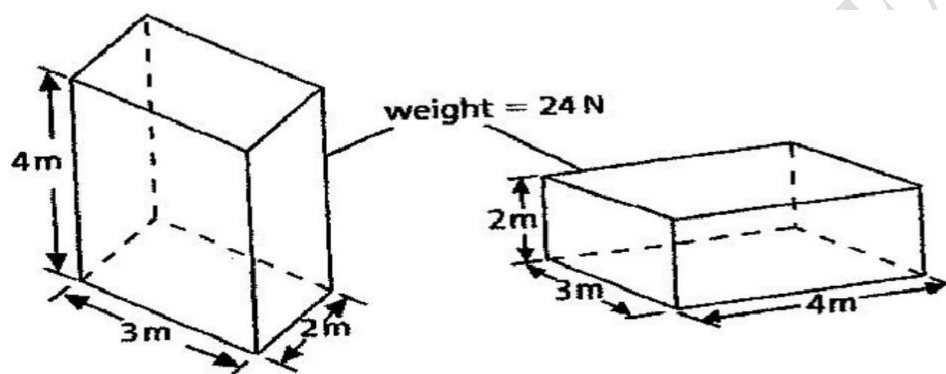
- **The greater the area** over which a force acts, **the lesser the pressure.**
- **The lesser the area** over which a force acts, **the greater the pressure.**
- **N.B** – the high heel can exert more than 15 times the pressure of an elephant's foot,



Higher pressure
Small area

Lower pressure
Large area

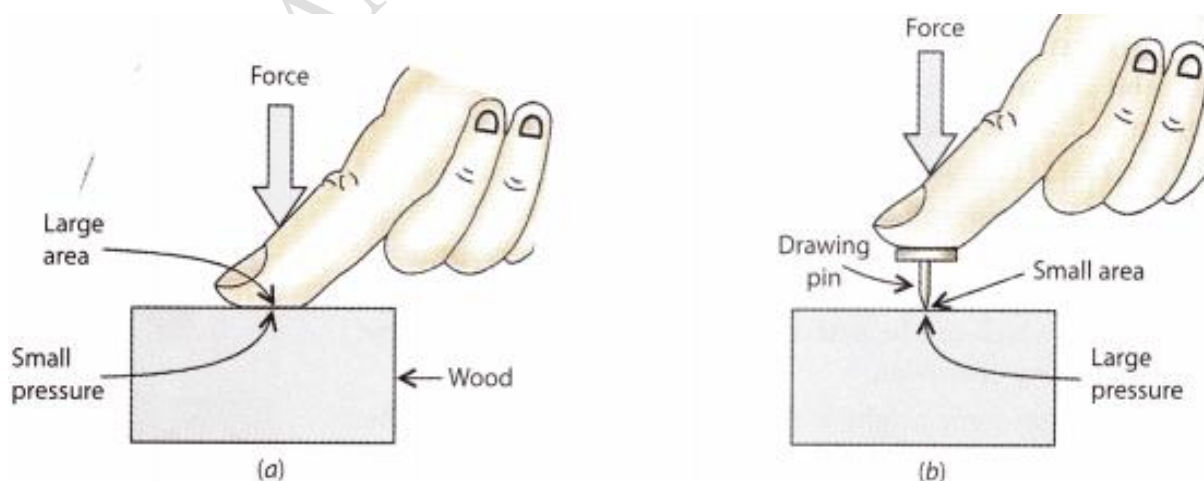
- The diagram below is showing a box placed on the ground on different sides.
- These sides have different areas, therefore resulting in different pressure.



a area = 6m^2
pressure = $\frac{24\text{ N}}{6\text{m}^2}$
 $= 4\text{ Pa}$

b area = 12m^2
pressure = $\frac{24\text{ N}}{12\text{m}^2}$
 $= 2\text{ Pa}$

- The pressure is large when the area is small and this why nails are made with sharp points.



Example

Q: A force of 10N acts over an area of 2m^2 . What pressure is created by the force?

Solution

$$\begin{aligned}\text{Pressure} &= \frac{\text{Force}}{\text{Area}} \\ &= \frac{10}{2} \\ &= 5 \text{ N/m}^2 \text{ or Pa}\end{aligned}$$

Q: A hammer is used to drive a nail into a wooden floor. The hammer is brought down with a force of 200N. The area of the top of the nail 0.5 cm^2 . What pressure is put upon the top of the nail by the hammer blow?

Solution

$$\begin{aligned}\text{Pressure} &= \frac{\text{Force}}{\text{Area}} \\ &= \frac{200 \text{ N}}{0.5 \text{ cm}^2} \\ &= 400 \text{ N/cm}^2\end{aligned}$$

Exercise : Pressure Calculations 1

1. What are the units of pressure, force and area?

Pressure measured in Pascals.

Force measured in Newtons.

Area measured in metres squared.

2. A force of 500 N acts over an area of 10 m^2 . What pressure is created by this force?

50 Pa.

3. A girl of weight 400 N has feet of area 100 cm^2 . What pressure does she put on the ground?

4 N/cm².

4. A car of weight 6 400 N has four wheels. Each wheel has an area of 80 cm^2 . What pressure does the car put on the ground?

20 N/cm².

Exercise : Pressure Calculations 2 – Pressure worksheet

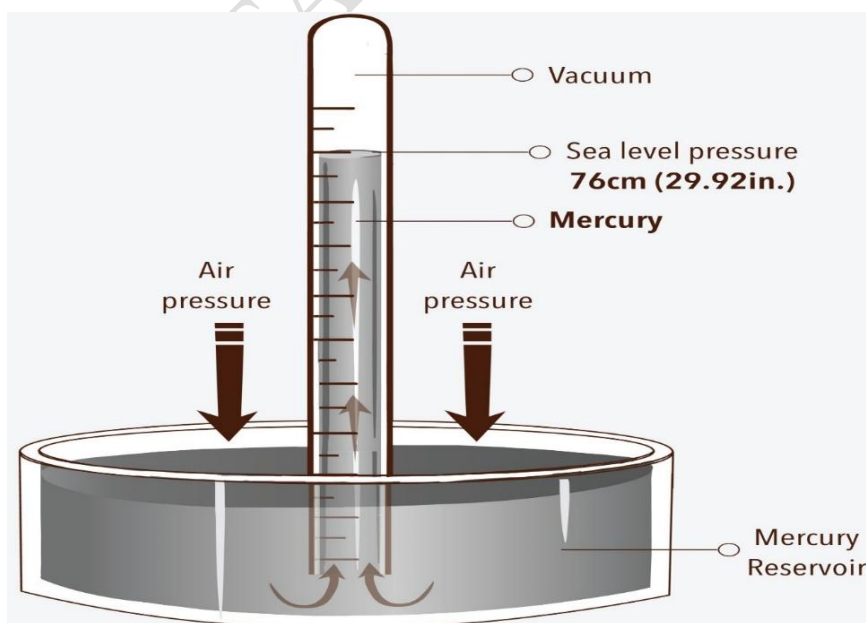
1. A force of 300 N creates a pressure of 4 N/m². Over what area is the force acting?
75 m².
2. A pressure of 200 Pa is created over an area of 5 m². What force acts to create this pressure?
1 000 N.
3. A tractor has tyres of area 3 m². It has a weight of 12 000 N. What pressure does the tractor put on the ground?
4 000 Pa.
4. A truck has tyres of area 5 m². It puts a pressure of 15 000 N/m² on the road. What is the weight of the truck?
75 000 N.

2. Atmospheric pressure

- Atmospheric pressure, the weight of the atmosphere pressing down on Earth.
- It is also defined as the pressure exerted by the air on earth.
- Pressure is exerted equally in all directions.
- Atmospheric pressure is low in uplands like mountains and high in low lying areas.
- Atmospheric pressure is measured using a **barometer**.
- Mercury barometers measure the pressure exerted by the atmosphere by balancing it with the weight of a liquid (like mercury) or a sealed compartment.

How a mercury barometer Work

- The height of the mercury column is directly proportional to the atmospheric pressure.
- A higher column means higher pressure, and a lower column means lower pressure.



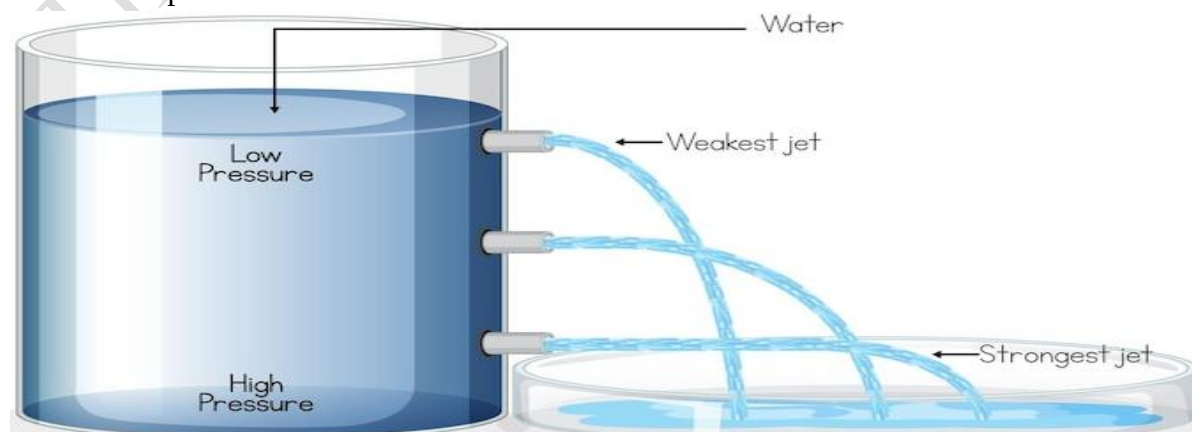
3. Liquid pressure

- A liquid exerts pressure on the sides of its container and on any object in the liquid.
- The **pressure exerted by the liquid increases as the depth increases.**
- Pressure also depends on the density of the liquid. The denser the liquid, the greater the pressure at any given depth.
- Liquid pressure is calculated as follows:

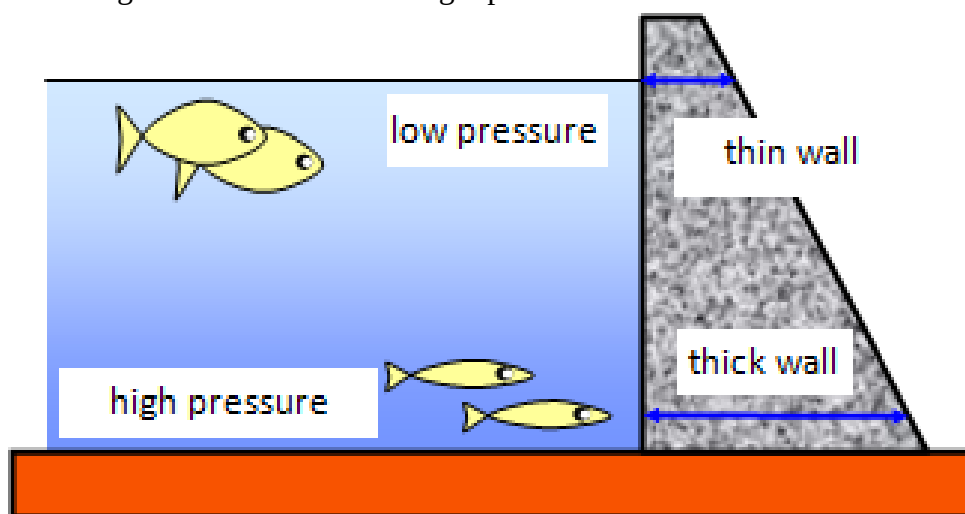
$$P = \rho gh$$

Where

- ✓ **ρ** is the density of the fluid ($\frac{\text{kg}}{\text{m}^3}$)
 - ✓ **g** is acceleration due to gravity ($10\frac{\text{m}}{\text{s}^2}$)
 - ✓ **h** is the depth of the liquid from the surface in meters (or height of the liquid)
- If we want to determine the absolute (total) pressure; we would have to add the atmospheric pressure
- $$P_{\text{total}} = \rho gh + P$$
- Pressure in a liquid increase with depth because the further down you go, the greater the weight of the liquid above.
 - At the same level, pressure is the same in all directions.
 - When a container with water is pierced to obtain many holes at different heights, Water escapes at different distances/jets.
 - Water escaping from the lower holes will travel further because it is at a high pressure.
 - Moreover, Water escaping from the upper holes will travel a shorter distance because it is at a low pressure.



- At the surface of the water in the dam; there is low pressure while at the bottom of the water in the dam there is high pressure
- This means that the dam wall needs to be thicker at the bottom than at the top so that it is stronger and can withstand larger pressure



EXAMPLE

- If the density of water is 1000kg/m^3 , what is the pressure due to the water at the bottom of a swimming pool 2m deep?

$$\rho = 1000\text{kg/m}^3 \quad g = 10 \text{ N/kg} \quad h = 2\text{m}$$

$$p = \rho g h$$

$$p = 1000\text{kg/m}^3 \times 10 \text{ N/kg} \times 2\text{m} = 20000 \text{ Pa}$$

$$\text{pressure} = 20 \text{ kPa}$$

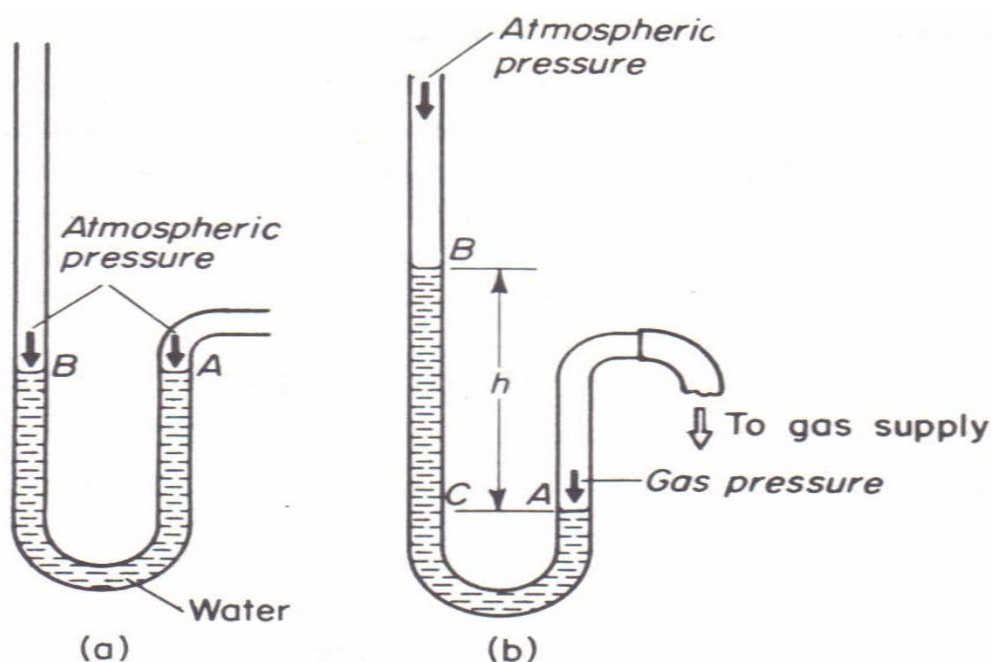
4.GAS PRESSURE

- Gas pressure is the force exerted by the gaseous molecules on the vessel that holds them.
- Gas particles move quickly and randomly in straight lines, colliding with both each other and the walls of their container.
- The greater the number of collisions that occur with the container walls, the higher the pressure of the gas.
- Gas pressure is measured using a manometer.

Manometer

- A manometer works by measuring pressure differences through a liquid column's displacement.
- It Consist of a U-tube of transparent material half filled with a liquid.

- One end of a manometer is connected to the gas pressure being measured, while the other is open to the atmosphere.
- The pressure difference causes the liquid to rise or fall in one column, which is then used to determine the pressure.
- When the manometer is not connected to any gas supply, only atmospheric pressure acts on both surfaces of the liquid.
- The liquid settle at same level since atmospheric pressure above the water surface at both ends is the same



- If one side is connected to a gas supply, the gas exerts a pressure on the surface A and level B rises until pressure of gas equals atmospheric pressure + pressure due to liquid column BC.
- The pressure of the liquid column BC therefore equals the amount by which the gas pressure exceeds atmospheric pressure
- The water level in that arm is depressed and the liquid in the other arm rises.
- A measurement of height (h) is recorded
- The pressure of source equals the difference in levels plus atmospheric pressure

$$\text{Gas pressure} = \text{Atmospheric pressure} + \rho gh$$

Practice 1

- Atmospheric pressure at sea level supports a column of 0.76m of mercury, density 13600kg/m³. If g is taken to be 10m/s², calculate pressure

Practice 2

- Water whose density is 1000kg/m³ fills a tank 5m high. Calculate the pressure exerted at the bottom of the tank. [g = 10ms⁻²]

Practice 3

- Given that acceleration due to gravity is 10ms^{-2} and the density of water is 1000kg/m^3 . Calculate the water pressure at a point 20cm deep.

Practice 4

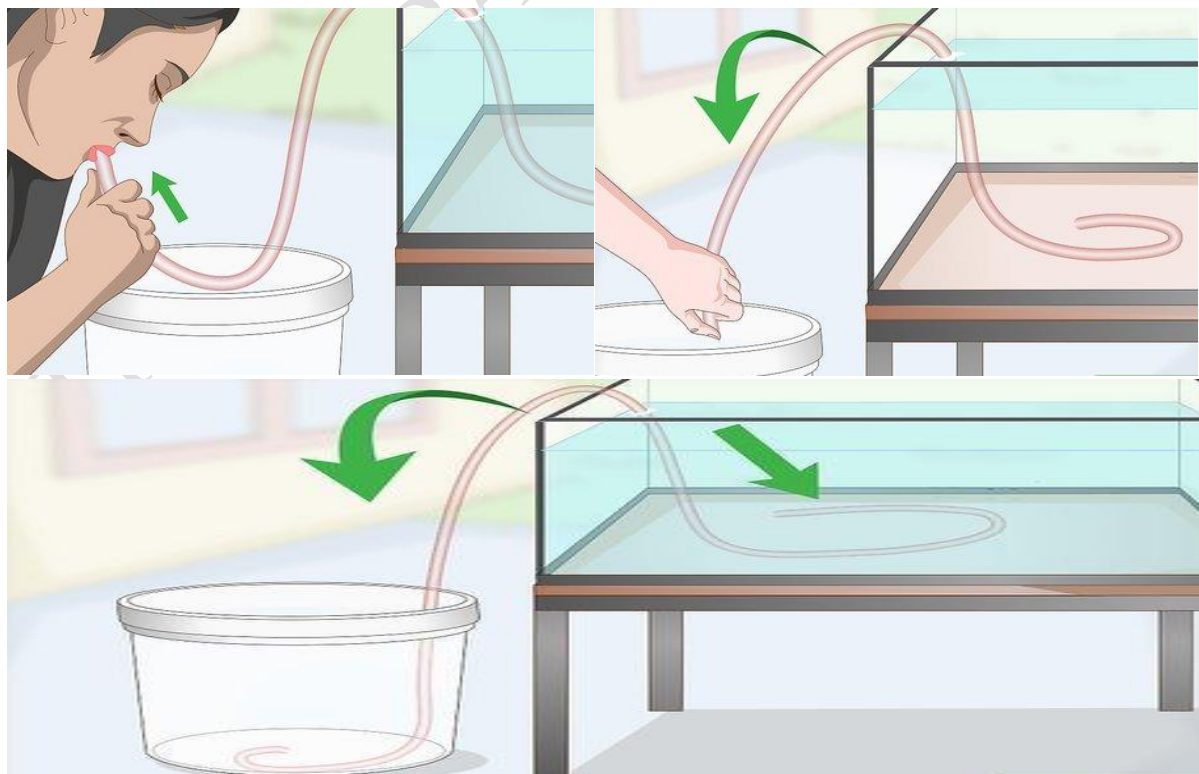
- In a reservoir, the height of water at the dam is 50m . Given that the density of water is 1000kg/m^3 and acceleration due to gravity is 10ms^{-2} . Calculate the pressure exerted at the bottom of the dam (ignore atmospheric pressure)

SIMPLE FLUID SYSTEMS

- Simple fluid systems help us to transport water from one place to another

Siphon

- A **siphon** is a flexible tube that transfers liquid from the source container (a higher container) to another container (a lower one) without a pump, relying on gravity and atmospheric pressure.
- The Source container must be at a higher level than the receptacle container for siphon to work.
- To start the siphon process, the tube must be filled with the liquid which is done by gently sucking air out and then releasing the liquid through the lower end.
- The liquid will then move up into the tube and move freely as long as the upper arm remains in the liquid in the tank.
- The atmospheric pressure will be exerting pressure on the liquid to continue flowing.
- **N.B** - The principle behind a siphon is that: the difference in height between the fluid level in the source container (reservoir) and the end of the siphon creates a pressure difference, allowing the fluid to flow.



Working Principle summary of siphon:**1. Initial Setup:**

- The siphon tube is initially filled with fluid and then one end is placed in the source container(reservoir).

2. Flow:

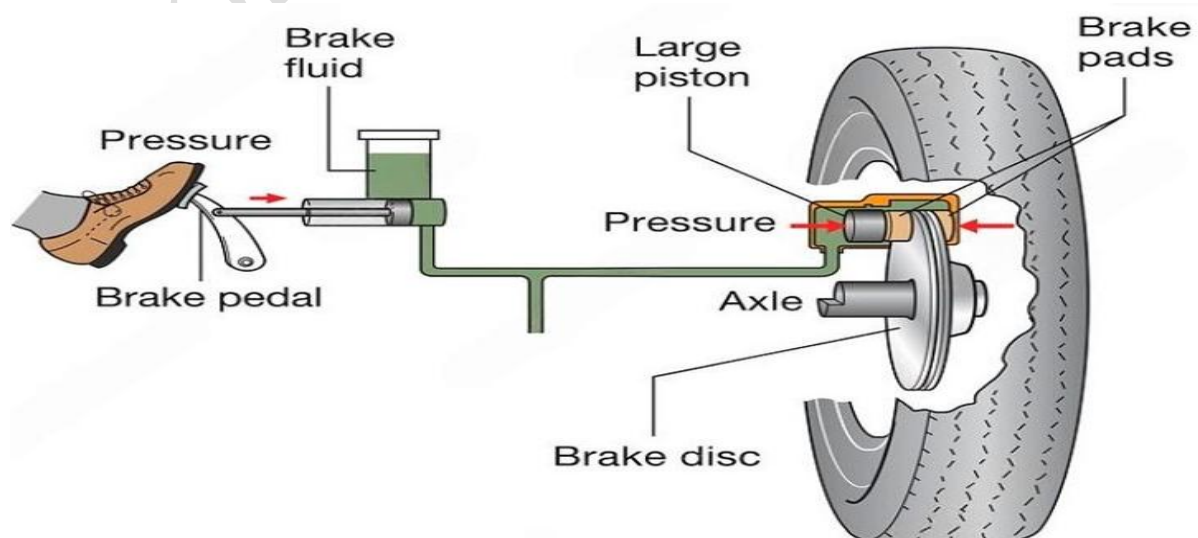
- The pressure difference between the source reservoir and the end of the siphon, along with gravity, causes the fluid to flow uphill through the inlet leg and then downhill through the outlet leg.

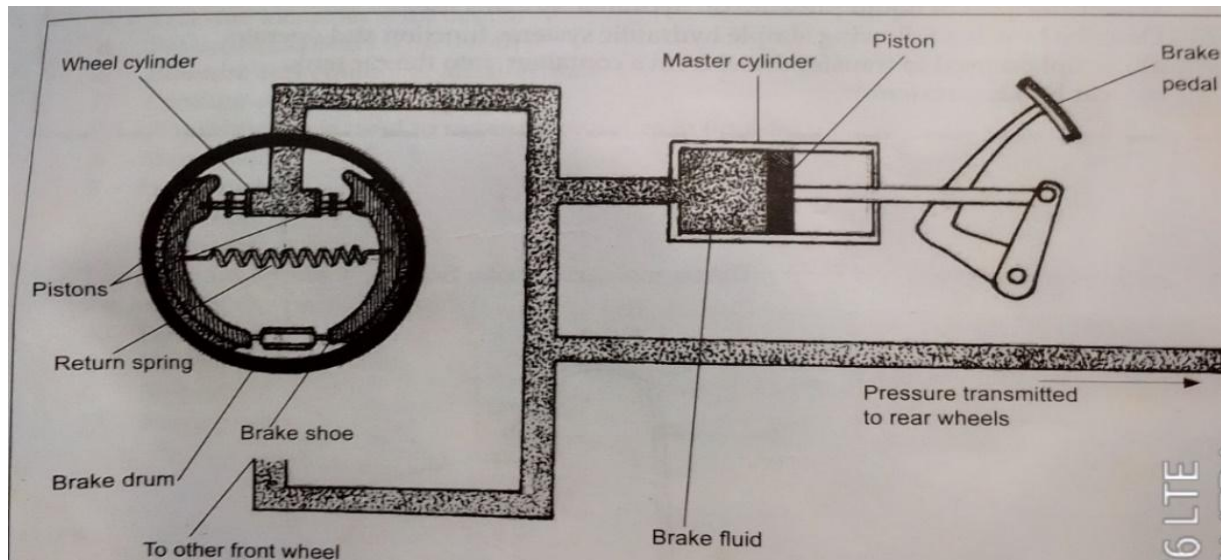
3. Continued Flow:

- As long as the inlet and outlet legs are submerged and there are no leaks, the fluid will continue to flow until the fluid levels in the source and receiving reservoirs equalize.
- Atmospheric pressure on the surface of the water pushes water out.
- The theory for the siphon action is based on the molecule's nature of matter and the fact that the molecules in the liquid have a force of attraction between them.
- Once the liquid is moving under these conditions, the molecular force of attraction in the remaining stream will allow the liquid to continue to flow.

HYDRAULIC SYSTEMS

- Are special systems that use fluids to transfer a force from one point to another
- Hydraulic systems use the following principle of liquids:
 - ✓ **Liquids transmit pressure because they cannot be compressed (are incompressible)**
 - ✓ **Pressure is the same throughout the liquid (transmit pressure equally in all directions)**
 - ✓ **Prevents bubble formation-presence of bubbles in the fluid means any transmitted force would compress the bubbles instead of passing on the pressure.**

Hydraulic car braking system

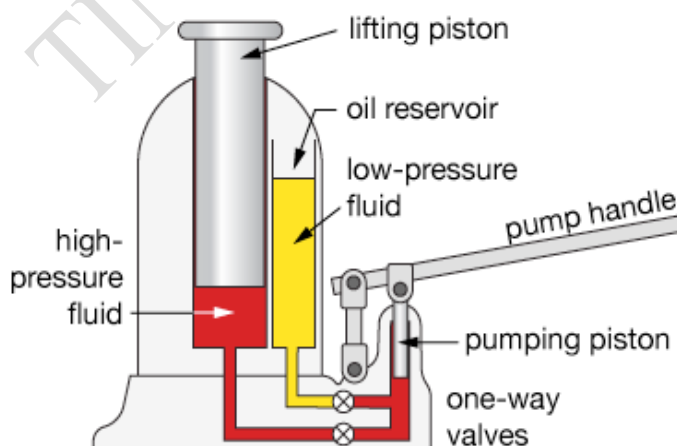


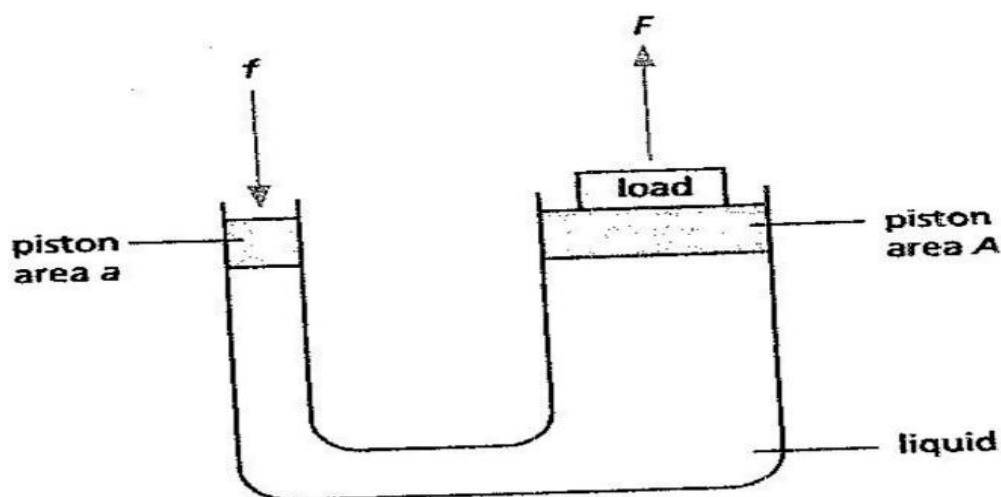
- When the brake pedal is pressed, this creates a pressure in the fluid which is **transmitted equally through the cylinder** and into the brake (slave) cylinders.
- Pressure acts on slave cylinder pistons and force is exerted on brake shoes so that they come into contact with the wheel drum. This causes friction which stops the car.
- By having large slave cylinders than the master cylinder, the driver will exert a small force on the brake pedal than is finally felt by the brakes.

Hydraulic jack

Operating principle of the Hydraulic jack

- Pascal's law states that a pressure change at one point in an enclosed incompressible fluid leads to the exact change in pressure throughout its volume, without any losses.
- This law gave way to the concept of a hydraulic jack, meaning to say that the hydraulic jacks operate based on pascal's law.
- A hydraulic jack lifts heavy objects by utilizing the principle of Pascal's Law: applying a small force on a confined fluid (limited to a particular place), which in turn transmits that pressure to a larger area, resulting in a larger lifting force.
- This is achieved by pumping hydraulic fluid into a cylinder where it pushes a piston, which then elevates the load.





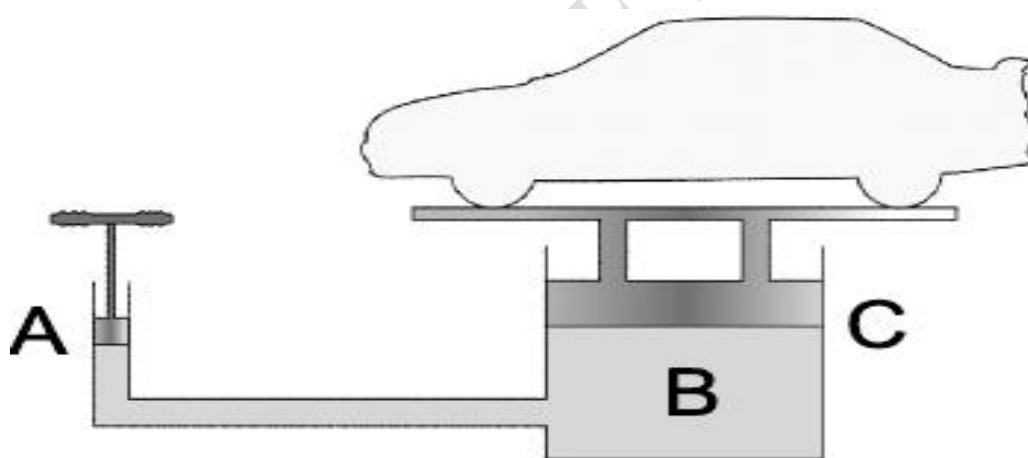
- According to the pascal's law, If a downward **force (f)** acts on a piston of **area (a)**. The pressure transmitted through the liquid is

$$\text{pressure} = \frac{\text{force}(f)}{\text{Area}(a)}$$

- This pressure acts on a second piston of larger area (A), producing an upward Force(F).

$$\text{Force} = \text{pressure} \left(\frac{f}{a} \right) \times \text{Area}$$

- N.B - If pressure is constant and area increases, force also increases.



- The diagram above illustrates Pascal's Law and how hydraulic jacks work.
- A – a small pushing force(effort)
 - B - incompressible, confined fluid
 - C – a large lifting force(load)
- If the plunger piston (labeled A) is pushed downwards, it creates a change in pressure.
- The fluid in the larger hydraulic cylinder (labeled B) equally changes in pressure, applying it to the ram piston (labeled C).
- Since pressure is the ratio of force to area, the ram piston's larger surface area will distribute a larger lifting force.
- The larger force lifts the load above the ram cylinder.
- To raise the load to a reasonable height, the plunger piston has to move a greater distance than ram piston.

Question?

- Suppose **force(f)** on effort piston = 40N; **area** of effort piston(**a**) = 0.2m² and **Area** of load piston(**A**) = 0.5m², calculate force on the load. [2]

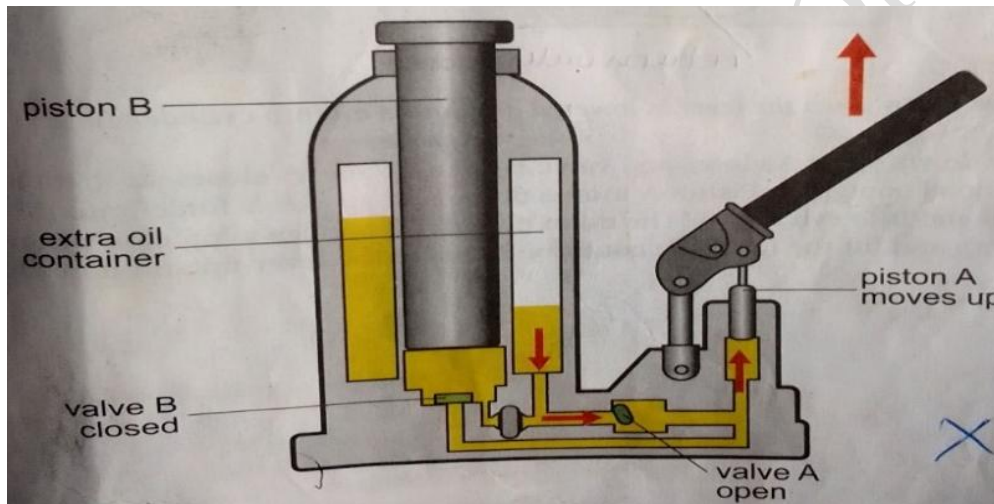
Solution

$$\text{Force} = \frac{f}{a} \times A$$

$$= \frac{40\text{N}}{0.2\text{m}^2} \times 0.5 \text{ m}^2$$

$$= 100 \text{ N}$$

- **N.B** - A force of 1N could lift a load of 50N; the hydraulic system multiplies the force 50 times.



- On diagram above when we pull the lever up valve A opens and valve B closes
- When we push the lever down; valve A closes and valve B opens
- Valve A closes so that the oil does not go back into the extra oil container
- Piston A moves down and applies a force onto the oil
- The oil is forced through valve B and into cylinder B
- The oil in cylinder B applies a force on piston B
- This forces piston B to move up and lift the load
- A hydraulic jack is used to lift a part of the car so that the tyre can be changed if it is damaged.

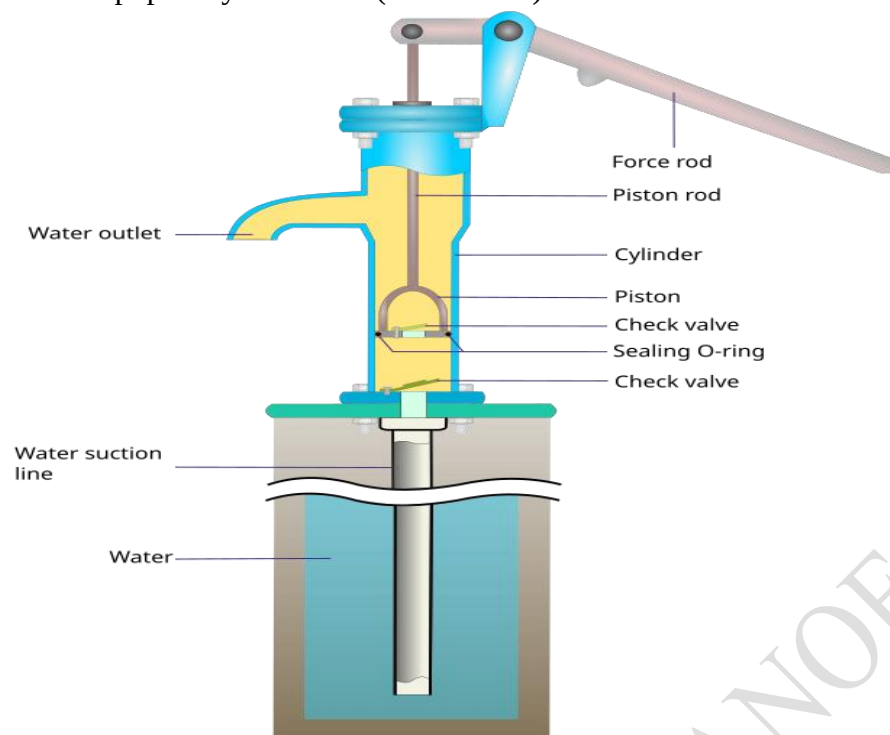
SIMPLE PUMPS

- Simple pumps re used to move fluids from one place to another.

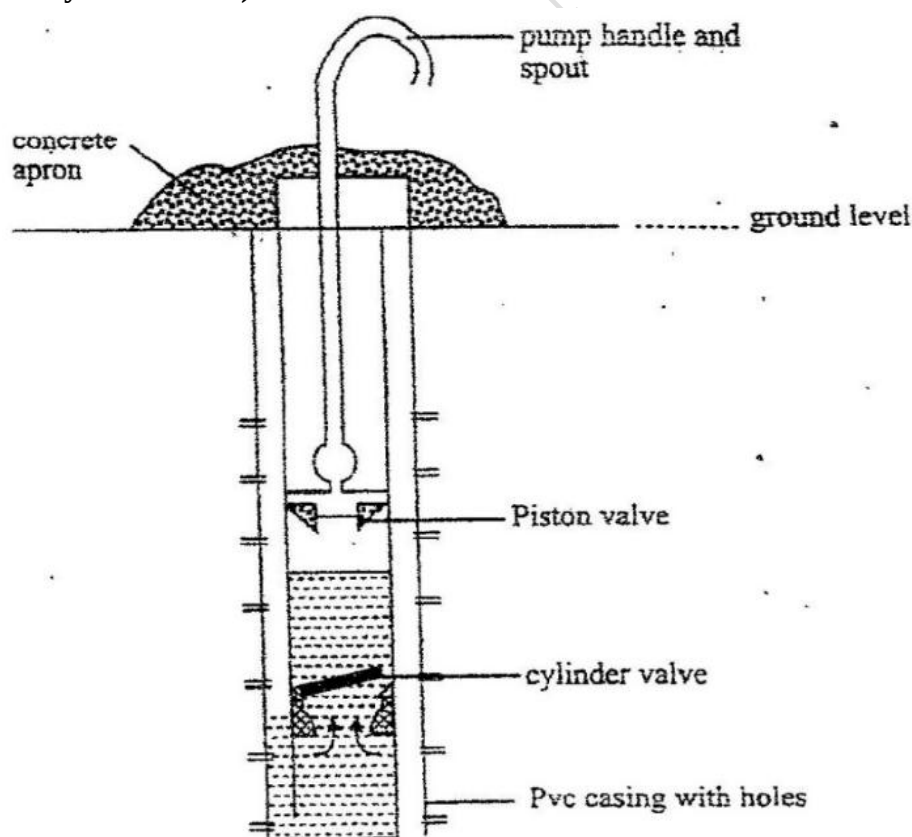
✚ **Lift pump:** Blair pump

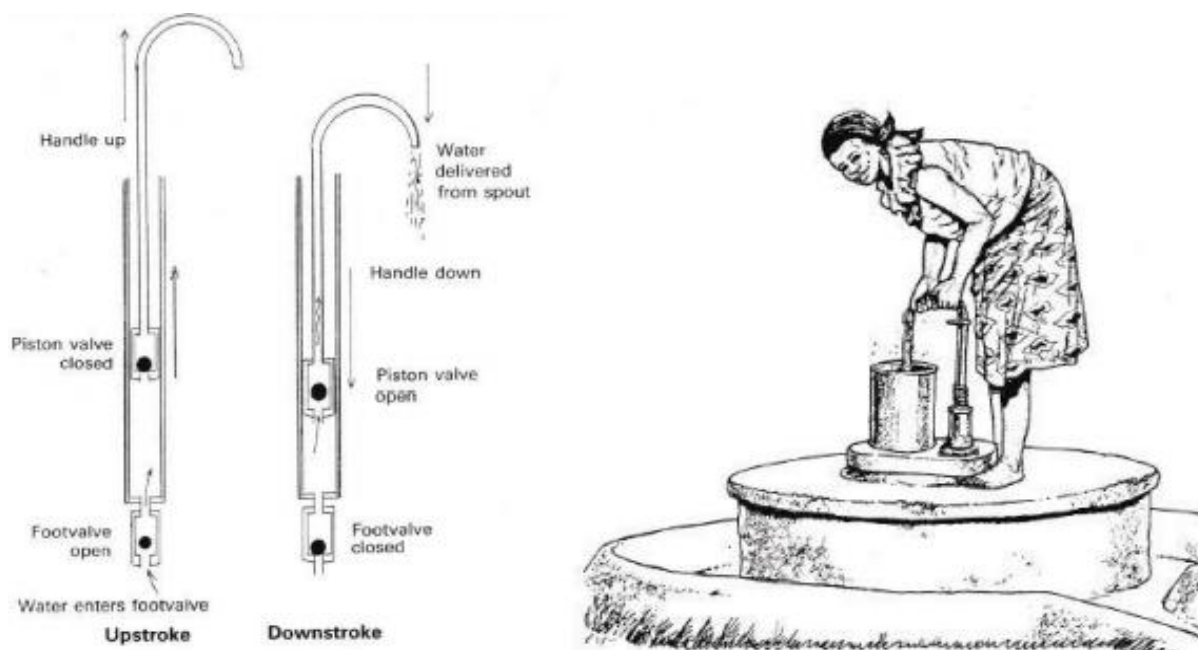
- The Blair pump is a lift pump that pulls water up from covered wells.

- It is popularly known as (**chibhorani**) in rural areas.



- It can lift water up to 12 meters.
- It operates by atmospheric pressure, forcing the fluid into the pump through valves.
- The pump handle is connected to a piston which moves (**the piston valve**).
- In a cylinder that is fixed below the water level in a well, there is another valve (**the cylinder valve**).

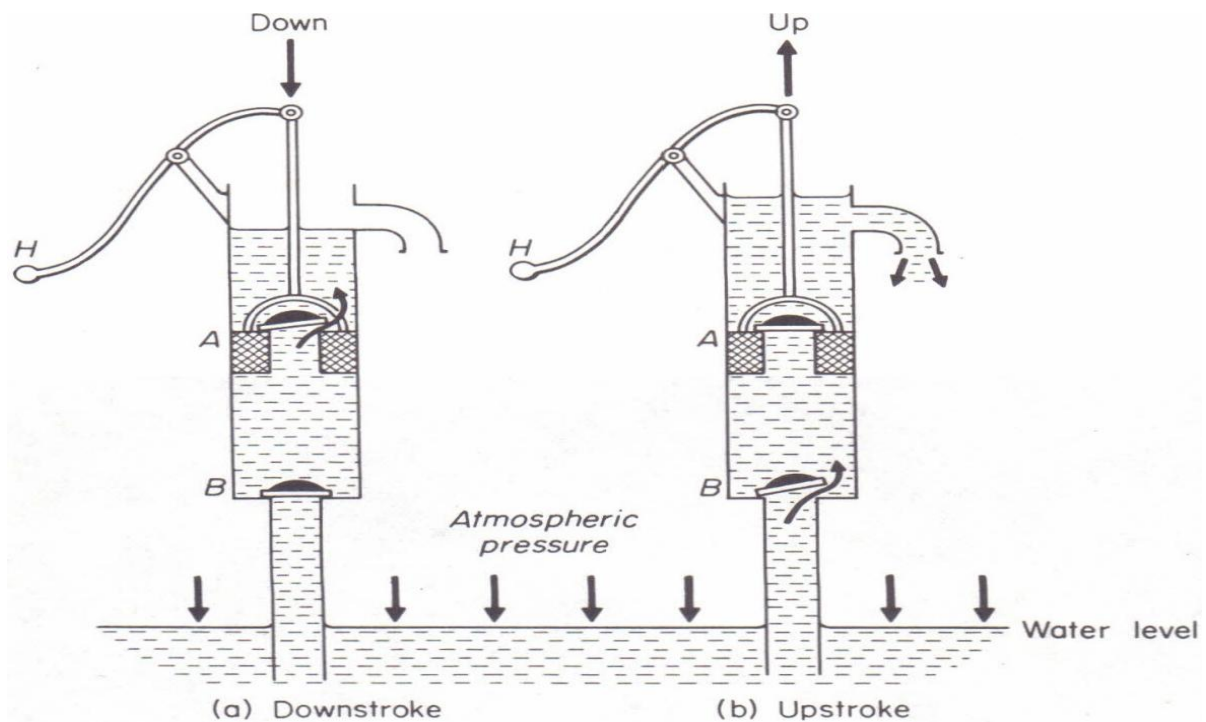




- **During a downstroke (When the plunger(handle) is moved down),** the foot valve (cylinder valve) at the base of the cylinder closes thereby trapping water in the cylinder.
- The piston applies force to the water in the cylinder, and the water moves up past the piston valve into the cylinder above the piston (pump raising main) and out of the handle tube.
- The water flows out of the handle tube through the nozzle into the container e.g. a bucket.
- **During an upstroke (When the handle is pulled up),** the piston valve closes due to the weight of the water above the valve.
- Water moves up past the foot valve (cylinder valve) at the base of the cylinder and fills the cylinder to be collected when the handle moves down again.
- **N.B – water is only obtained during a downstroke.**

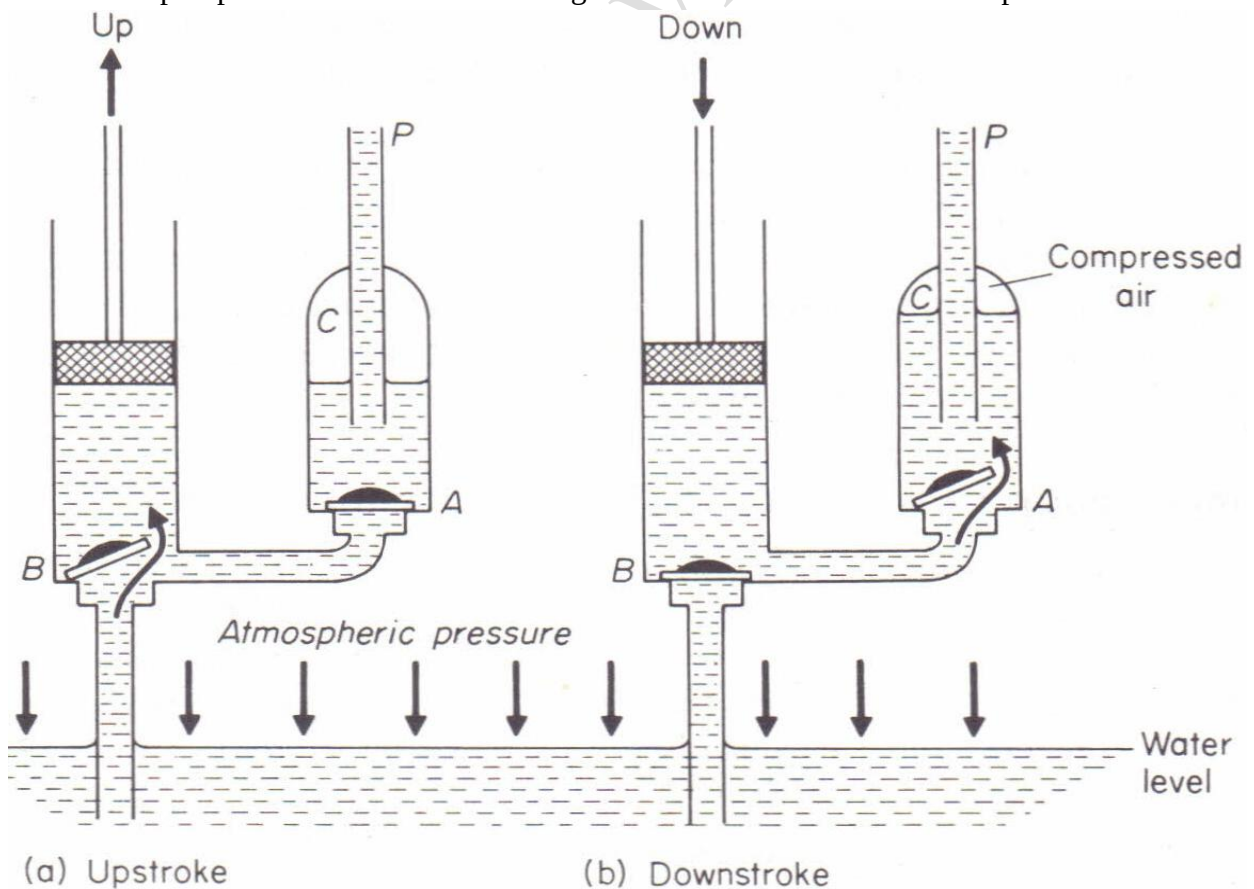
Common Lift Pump

- **During an upstroke,** piston moves up and a partial vacuum is created in the cylinder and atmospheric pressure forces water into the cylinder through the cylinder valve.
- Piston valve remains closed and water passes out through the nozzle.
- **During downstroke,** piston moves down and a partial vacuum is created in the cylinder above the piston.
- Pressure below the piston closes the cylinder valve and forces the piston valve to open.
- Water moves through piston valve and fills the cylinder above the piston.
- Continuous upstroke and downstrokes will result in continuous flow of water.
- Always remember that **water is only collected during upstroke.**
- It should be noted that the common lift pump **cannot pump more than 10 metres high.**
- **N.B – from the diagram given below, the piston valve is labelled A and the cylinder valve is labelled B.**



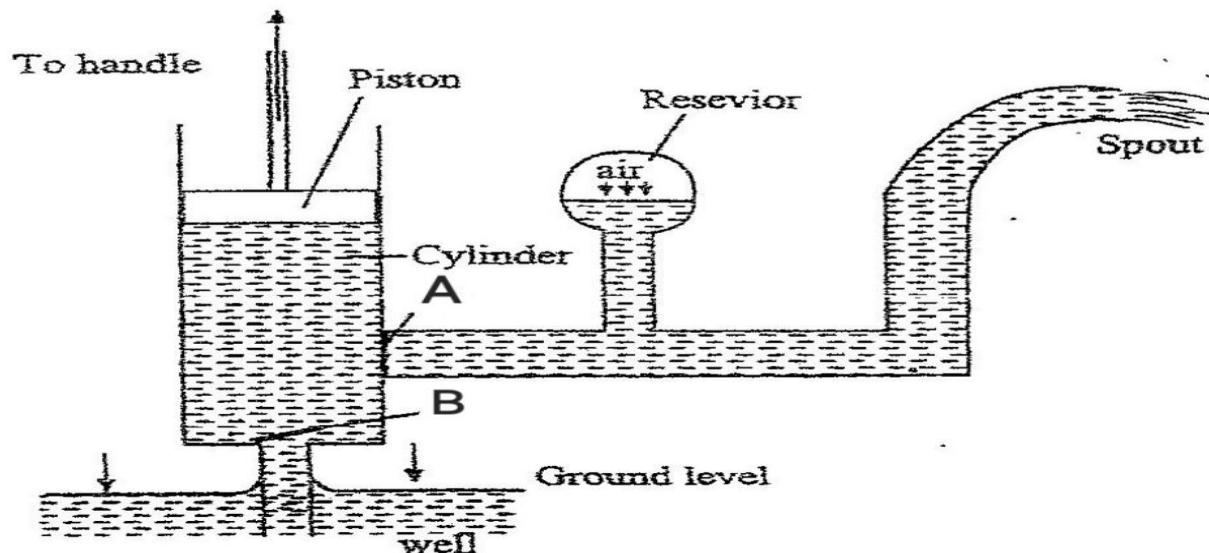
Force Pump

- A force pump uses force to obtain underground water that is situated at depths above 10m.



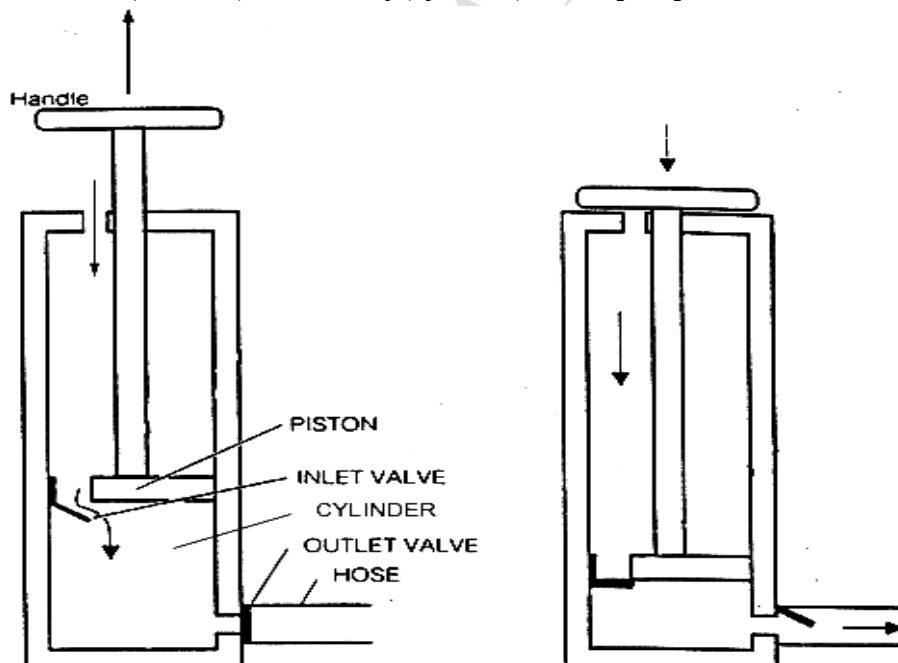
- **During downstroke**, piston moves down, valve A closes and valve B opens.
- Water is filled in the temporary tank with air being highly compressed.

- Water is forced out through the nozzle.
- **On upstroke**, valve A opens as atmospheric pressure forces water into pump.
- Air pressure in the reservoir (temporary tank) is exerted on the water.
- This causes valve B to close, while water is forced to continuously come out through the nozzle.
- Always remember that **water is collected during both upstroke and downstroke**.



Bicycle pump

- A bicycle pump is used to force air into tyres.
- The pump has a handle that is connected to a piston and the piston moves up(outwards) and down(inwards) in the body(cylinder) of the pump.



- **When the piston is moved inwards**, the leather washer presses against the sides of the cylinder forming an air tight seal.
- The air in the cylinder is forced into the tyre through the tyre valve.
- The valve allows air to pass one way only.

- **When the piston is pulled out**, pressure in the cylinder is reduced and the washer collapses allowing air into the cylinder.
- The strength of the washer is vital if a lot of air is to be forced into an already high pressure tyre.
- The force pump used must be able to produce higher pressures than this, otherwise the force air would not be able to enter the tyre.
- **N.B** – the bicycle pump has valves which allow air to flow in one direction only. When one valve is closed the other will be open.

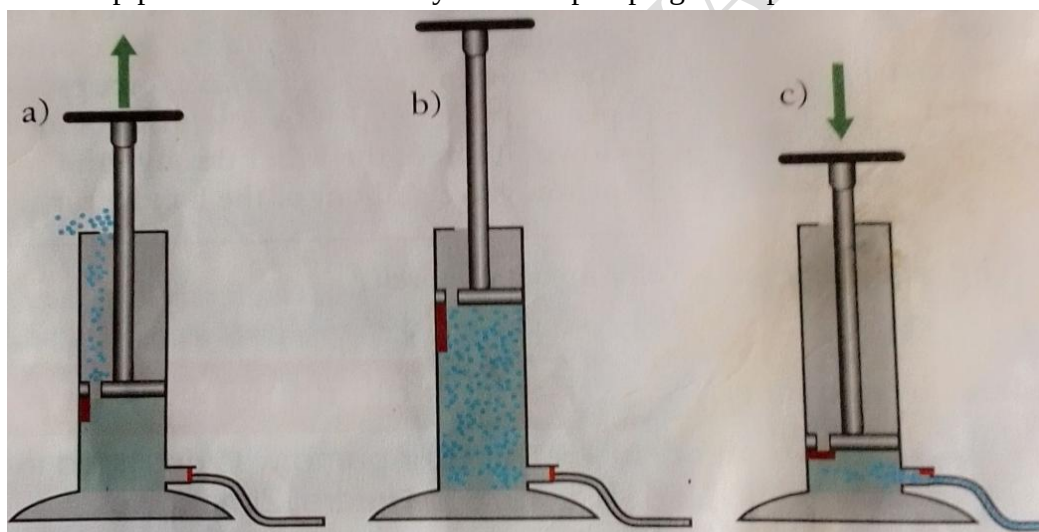
Additional information

Diagram A

- When we pull the handle up the piston moves up; piston valve opens and compression chamber fills with air.

Diagram B & C

- When we push the handle down; the piston moves down and the piston valve is closed
- The piston tries to compress the air and forces the foot valve to open
- The air is forced out of the foot valve and into the pipe
- The pipe is connected to the tyre and so pumping takes place.



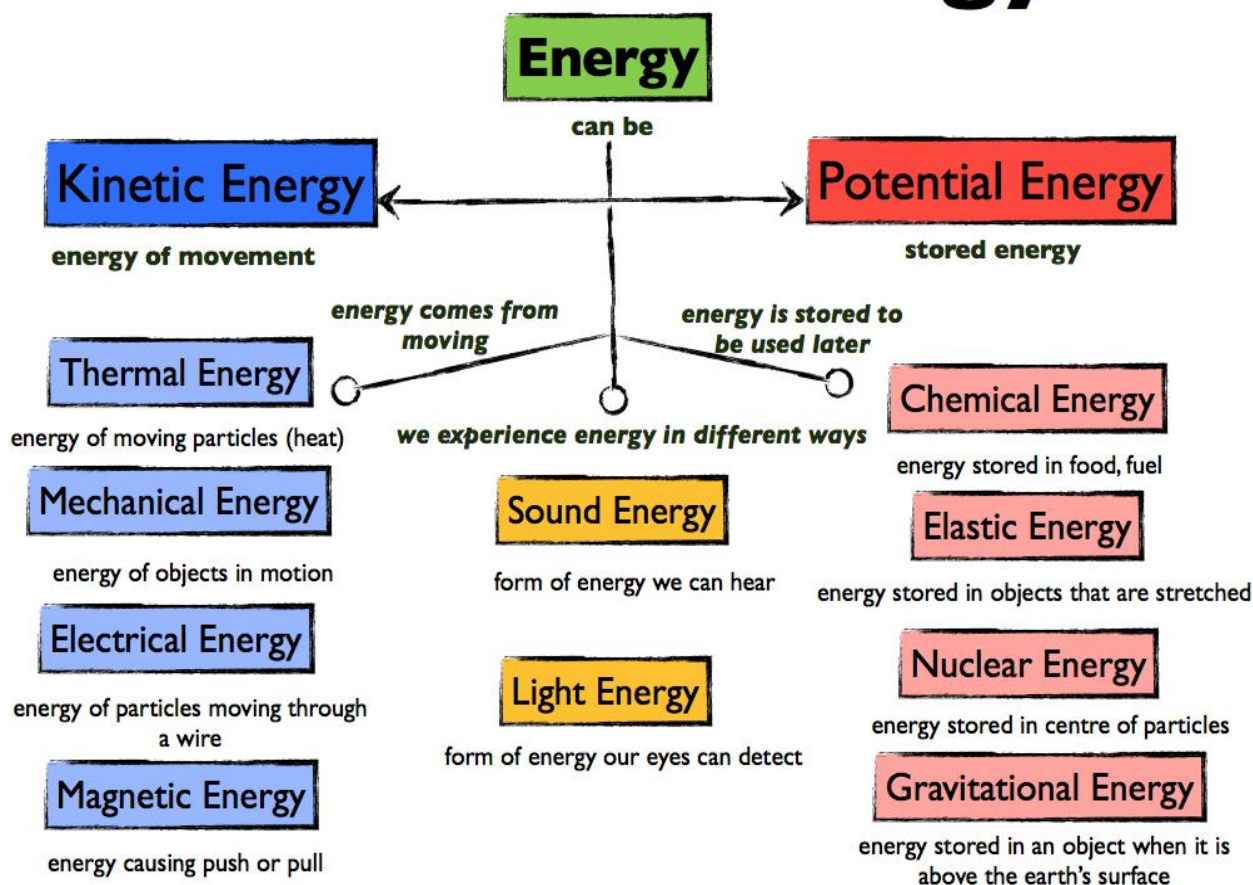
ENERGY

- **Energy**, is the ability to do work.
- It is the ability to exert a force causing displacement of an object.
- It's a fundamental property of matter which **can be transferred or converted between different forms, but not created or destroyed.**

Conservation of energy

- The principle of energy conservation states that **energy can neither be destroyed nor created but can be transformed from one form to another.**

Forms of Energy

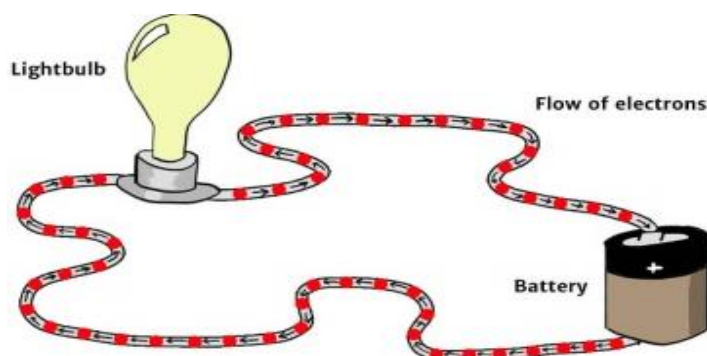


Forms of energy

- Energy is Categorized into two groups, which are **kinetic energy** or **potential energy**.
- **Kinetic energy** is associated with motion, while **potential energy** is stored energy, ready to be converted into other forms.

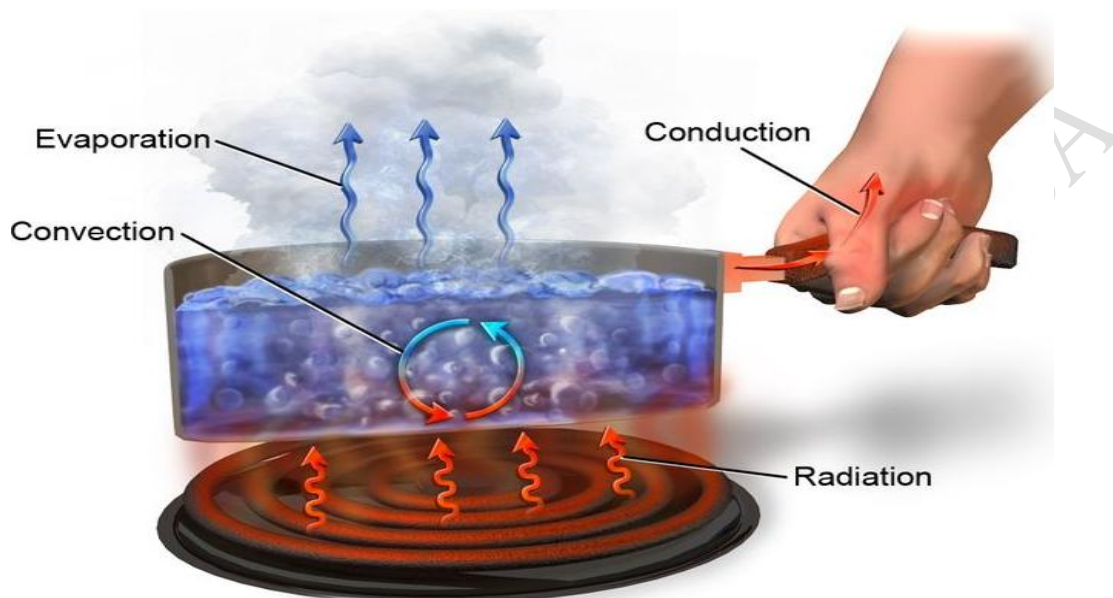
Kinetic Energy

- This type of energy is possessed by an object due to its motion.
- Examples of kinetic energy include:
 - Electrical energy** is form of energy derived from the movement of charged particles, primarily electrons. It is the flow of electrical charge.

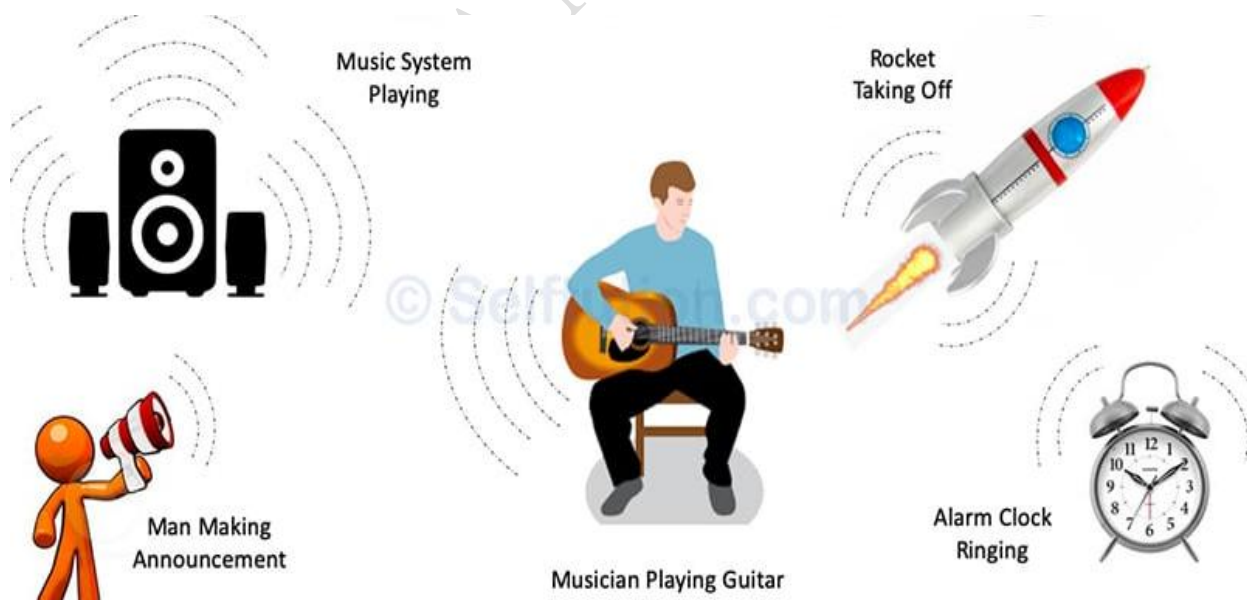


- b) **Thermal energy** is the energy contained within a system that is responsible for its temperature. It can also be defined as energy possessed by an object due to the movement of particles within the object. Thermal energy transfers occur in three ways namely: conduction, convection, and radiation. Heat energy is associated with the movement of molecules.

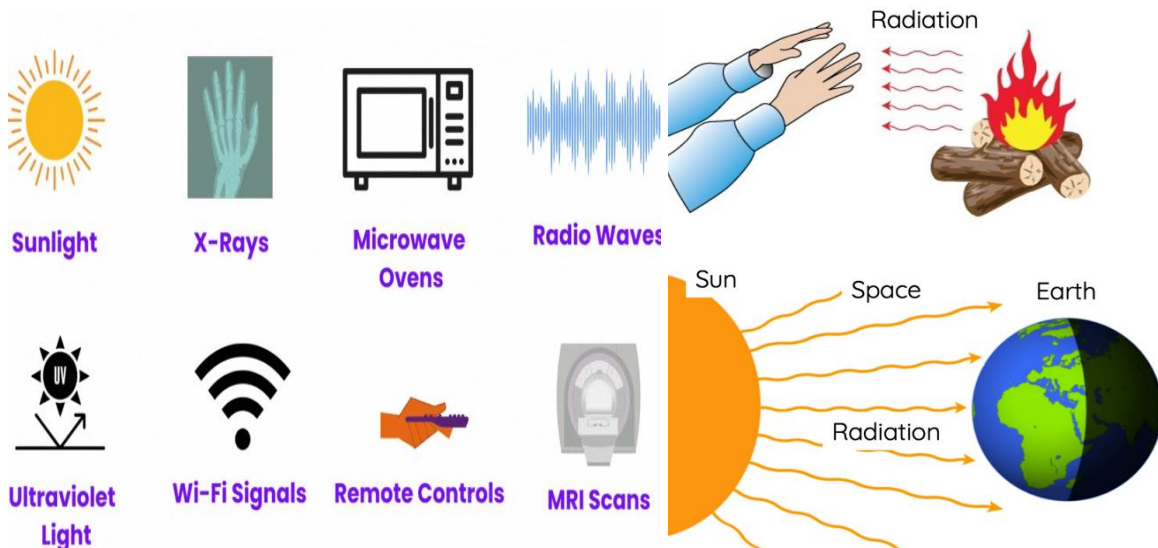
N.B -thermal means relating to or caused by heat or changes in temperature.



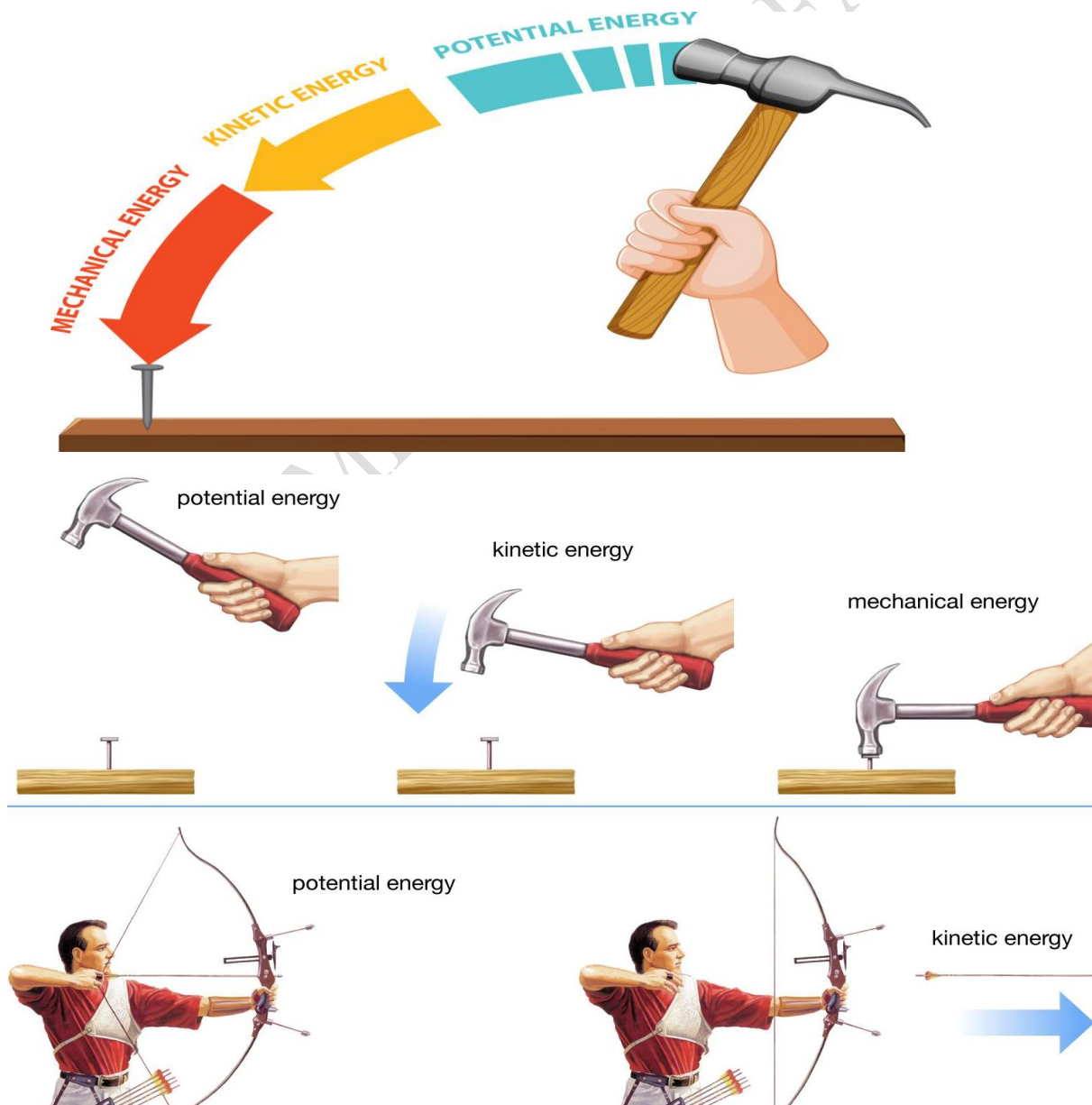
- c) **Sound energy** is the movement(transferring) of energy through substances in longitudinal waves. It can be found in anything that vibrates.



- d) **Radiant energy** is a form of energy that travels through space in form of electromagnetic waves. This includes light and other forms of radiation.



e) **Mechanical energy** is the sum of kinetic and potential energy in a system.



Potential Energy

- This type of energy is stored and can be converted into kinetic energy.
- Examples of potential energy include:
 - a) **Chemical energy** is the Energy stored in bonds of chemical compounds, like atoms and molecules. This energy is released when a chemical reaction takes place. Usually, once chemical energy has been released from a substance, that substance is transformed into a completely new substance.
 - Batteries, biomass, petroleum, natural gas, and Fossil fuels (coal) are examples of stored chemical energy.



Burning Wood



Eating Food



Gasoline in Cars



Photosynthesis



Digestion



Fireworks



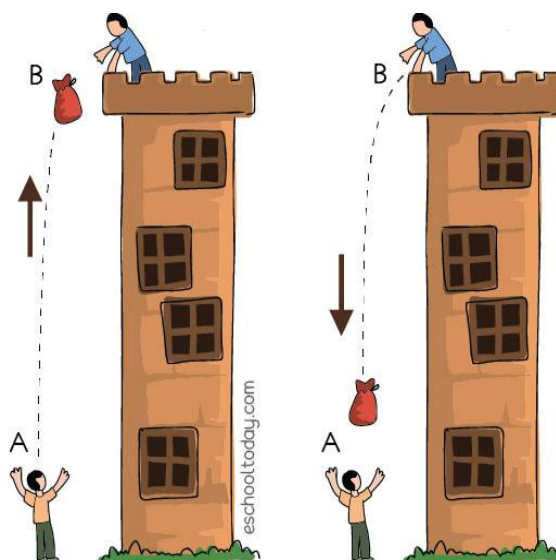
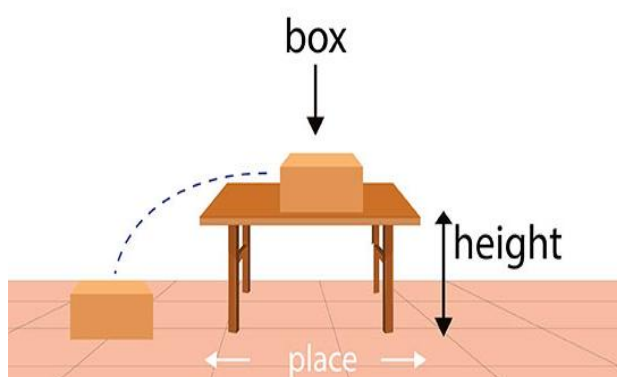
Rusting of Iron



Natural Gas Heating

- b) **Gravitational potential energy** is the Energy stored by an object due to its position relative to a gravitational field.

Gravitational Potential Energy



- c) **Elastic potential energy** is the Energy stored in an object when it is deformed, like a stretched rubber band



Spring in its normal state
(no elastic potential energy)

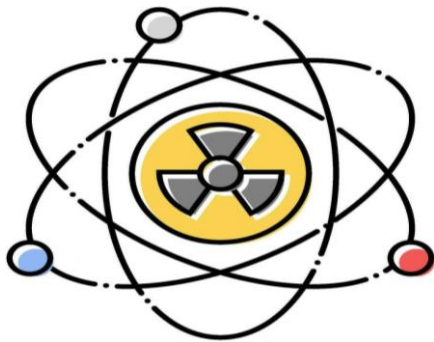


Stretched spring
(has elastic potential energy)



Compressed spring
(has elastic potential energy)

d) **Nuclear energy:** Energy stored in the nucleus of an atom.



- These different forms of energy can be transformed from one to another, for instance, **chemical energy in a battery** can be converted into **electrical energy**, which can then be transformed into **light and heat**.

Chemical Energy

Stored in the battery

is transformed into

Electrical Energy

(as electrons are produced)

is transformed into

**Light Energy
(Radiant)**



Know
this
well!

Clean Sources of Energy

Sun

- Solar energy is the energy we receive from the sun, mainly as visible light and other forms of electromagnetic radiation not seen by the naked eye.
- The term solar energy is often used to describe processes that convert this kind of energy to heat or electricity for human use.
- Solar-powered photovoltaic (PV) panels convert the sun's rays into electricity by using photons of light from the sun to excite electrons in silicon cells. Hence, generation of electricity takes place.



Wind

- It is the term used for the kinetic energy of the wind.
- Using windmills and turbines this energy is harnessed to obtain electrical energy.
- Wind power is the use of air flow through wind turbines or wind mills to mechanically power generators for electricity.



Geothermal (Geysers and hot springs)

- Geothermal energy is the energy that is stored and produced from the earth's core.
- It is clean, sustainable and environmentally friendly. High temperatures are produced continuously inside the earth's molten core.
- Hot rocks present below the earth heats up the water trapped beneath the Earth's surface which in turn produces steam that can be captured and controlled to drive power generators.

**Gyser****Hotspring**

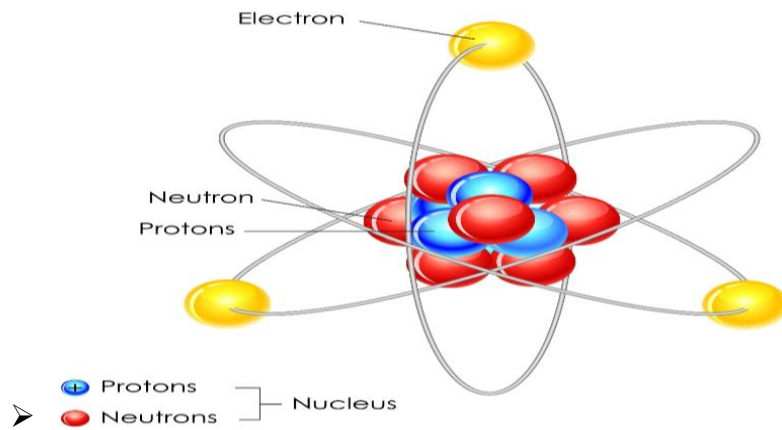
Tides

- Tidal energy is a form of hydropower that converts the kinetic energy of the tides into electricity or other useful forms of energy.
- The tide is created by the gravitational effect of the sun and moon on the earth causing cyclical movement of the oceans and seas.
- The harnessing of tidal energy is mostly prevalent in coastal regions.



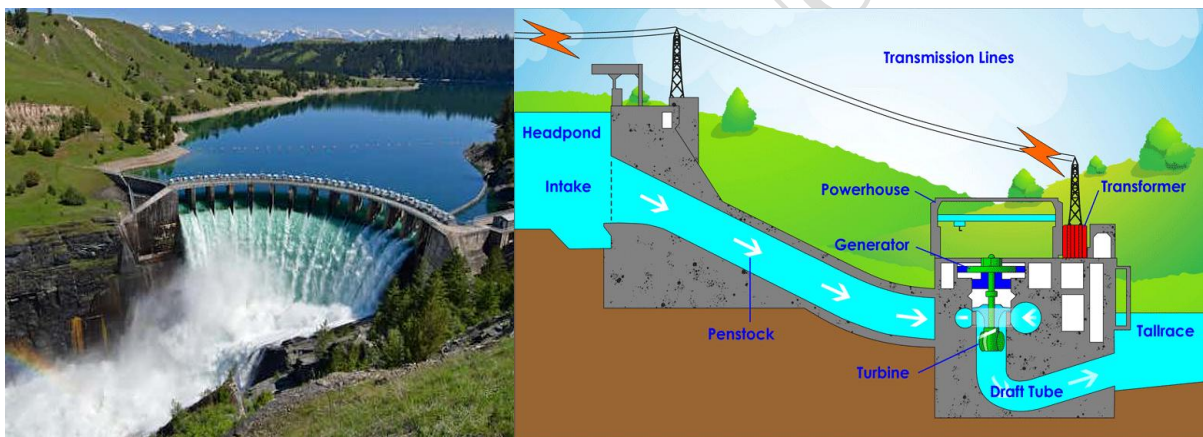
Nuclear

- Nuclear energy is the energy in the nucleus of an atom. At the core of each atom there are two types of particles (neutrons and protons) that are held together by this nuclear energy.
- This energy can be obtained in two ways which is either nuclear fusion or nuclear fission.
- In nuclear fusion, energy is released when atoms are combined together to form a larger atom.
- In nuclear fission, atoms are split into smaller atoms, releasing energy. Actually, nuclear power plants only use nuclear fission to generate electricity.
- When one of these two physical reactions (nuclear fission or nuclear fusion) takes place, atoms experience a slight loss of mass.
- The loss in mass is compensated by the heat energy produced according to Albert Einstein's famous equation $E=mc^2$.



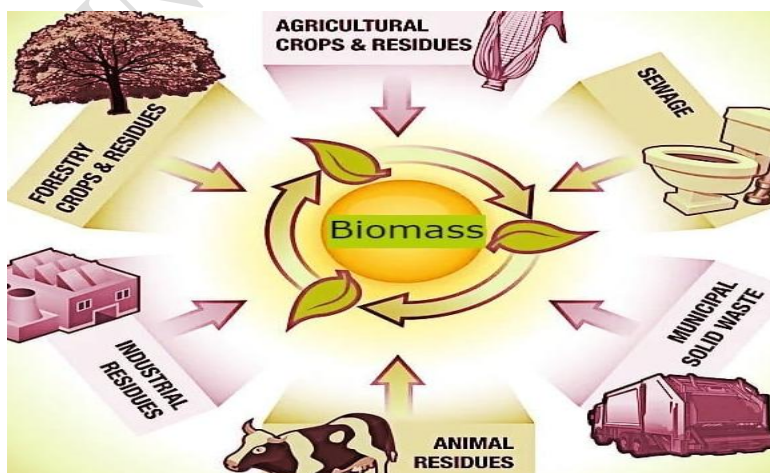
Hydroelectric Energy

- Hydroelectric energy is energy derived from the movement of water.
- The water level is first risen up by a dam wall to increase its potential energy and when released it moves with great kinetic energy due to gravity acting upon its weight and the energy will be used to drive turbines of a power generator.



Biomass

- Biomass is produced from organic materials. When the plants are burned, the carbohydrates within them are converted into water and carbon dioxide in addition to heat released.



➤ Biomass generally includes crops, plants, trees, wood chips and animal waste.

Differences between Renewable and Non-renewable energy sources

Renewable	Non Renewable
1. Can be replaced by natural process in a short amount of time or can be recycled	1. These are natural resource that either cannot be replaced or may take millions of years to replace by natural process like coal and oil.
2. Can be reused or recycled and used multiple times.	2. Cannot be reused or recycled.
3. Some of the examples are: wind energy, solar power, hydroelectricity, geo thermal.	3. Some of the examples are: Petrol, coal, Natural gas, nuclear energy, fossil fuels.
4. No harm done to the environment because of its use.	4. Huge harm done to the environment because of the harmful emissions.

RENEWABLE ENERGY



Wind



Hydropower



Solar



Geothermal



Biomass

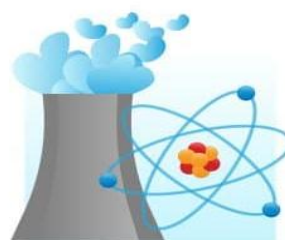
NON-RENEWABLE ENERGY



Oil



Coal



Nuclear



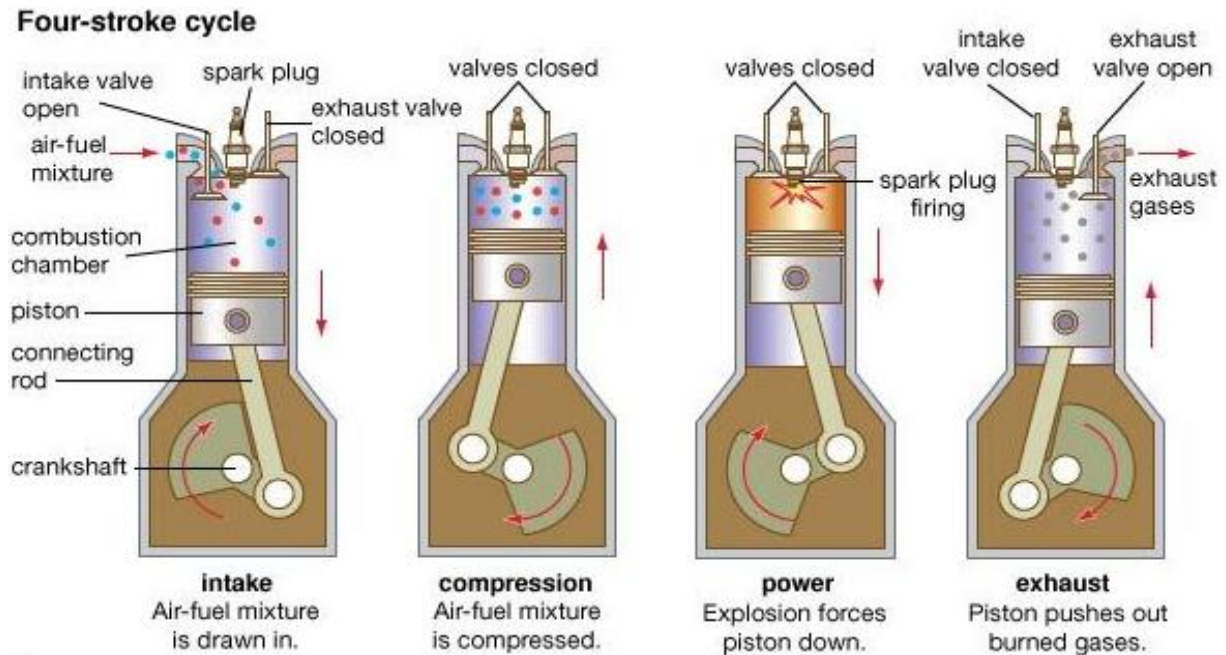
Natural Gas

PETROL AND DIESEL ENGINES

Petrol engine

- A **petrol engine**, also known as a gasoline engine, is an internal combustion engine that runs on petrol (gasoline) and similar volatile fuels.
- It's a spark-ignition engine, meaning it uses a spark plug to ignite the air/fuel mixture inside the cylinder.
- Petrol engines operate on the Otto cycle, which involves four main strokes: intake, compression, power, and exhaust.

Operation of a four-stroke petrol engine



Intake stroke

- Piston moves downwards and pressure in the cylinder decreases.
- Inlet valve opens.
- Petrol air mixture is sucked into the cylinder.
- Exhaust valve is closed.

Compression stroke

- The inlet valve and the exhaust valve are both closed (Both valves are closed)
- Piston moves upwards.
- Petrol air mixture is compressed

Power stroke

- The inlet valve and the exhaust valve are both closed (Both valves are closed).
- A spark from the spark plug ignites the petrol-air mixture.
- The mixture explodes and expands forcing the piston to move downwards.
- Chemical energy is converted to kinetic energy causing motion in the vehicle

Exhaust stroke

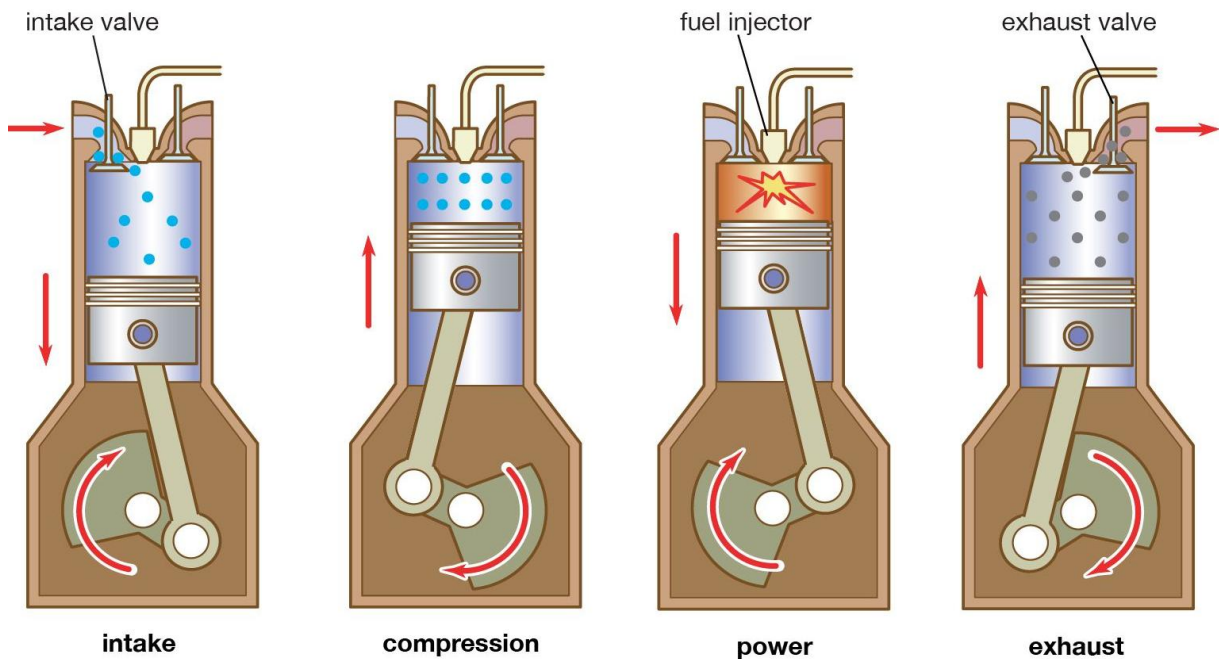
- Inlet valve is closed and exhaust valve opens.

- Piston moves upwards and exhaust gases are expelled.

Diesel engine

- A diesel engine is an internal combustion engine where ignition of the fuel occurs due to the heat created by compressing air in the cylinder.
- This type of engine is known as a compression-ignition engine and is commonly used in vehicles like trucks, buses, and heavy machinery.

Operation of a four-stroke diesel engine



Intake stroke

- Inlet valve opens and exhaust valve closes.
- Piston moves downwards.
- Air is drawn into the cylinder.

Compression stroke

- The inlet valve and the exhaust valve are both closed (Both valves are closed).
- Piston moves upwards, compressing the air so that it heats up.

Power stroke

- Diesel is injected into the hot air through the fuel injector.
- Diesel immediately ignites and drives.
- Pressure caused by the quick expansion of the gases pushes the piston down.

Exhaust stroke

- Inlet valve is closed and Exhaust valve opens.
- Piston moves upwards and exhaust gases are expelled.

Conclusion on the four stroke operations

- For you to easily remember how the four strokes of both diesel and petrol engines, you can just say **I CAN PASS ENGLISH OCCO DUDU**.

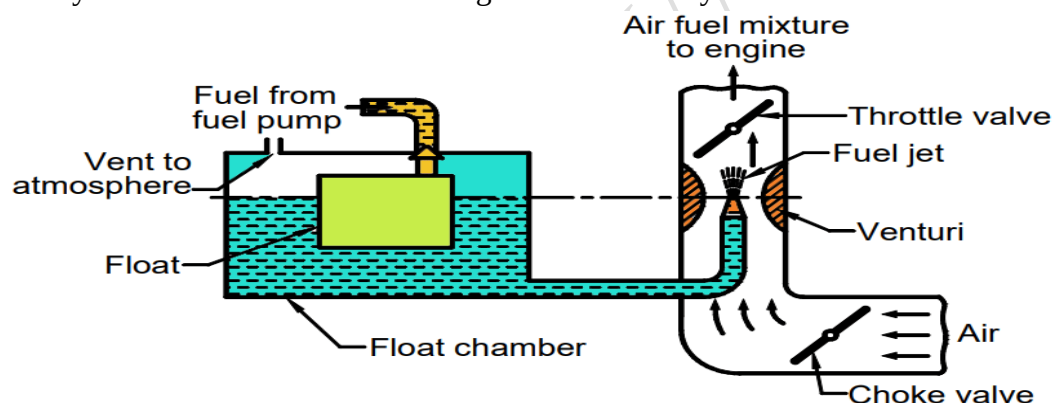
Name of stroke	I (Intake)	Can (Compression)	Pass (Power)	English (Exhaust)
Conditions of the Valves	Open (only the inlet valve is open)	Closed (both valves are closed)	Closed (both valves are closed)	Open (only the exhaust valve is open)
Movement of piston	Downwards	Upwards	Downwards	Upwards

Role of the Fuel injector in a Diesel engine

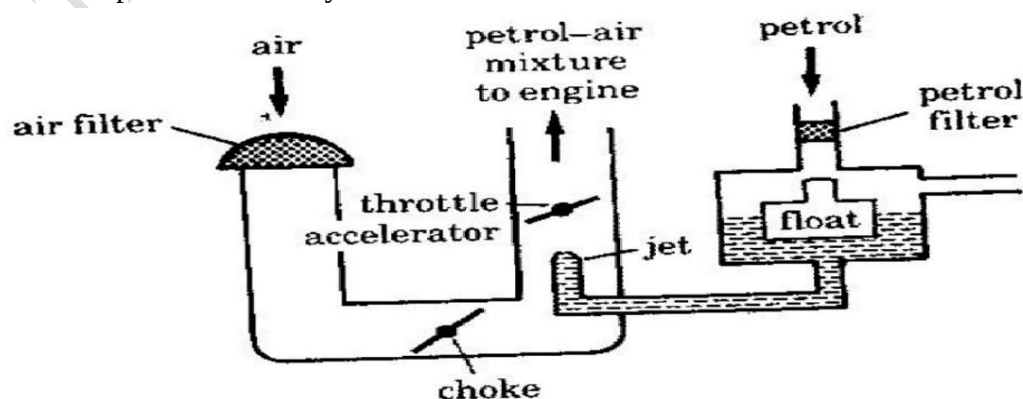
- The **Fuel injector** Pumps fuel into the combustion chamber.
- It makes sure that the correct quantity of fuel is injected at the correct time.

Role of the Carburettor in a Petrol engine

- The **Carburettor** ensures that the correct amount of air and fuel are mixed together in the cylinder to make sure that the engine runs correctly.



- When the engine is running, a stream of air passes above the jet in the venturi.
- A low pressure is created above the jet such that atmospheric pressure pushes petrol out of the jet.
- The petrol comes out as fine sprays that mixes with air forming a petrol-air mixture which then passes into the cylinder.



Parts of the carburetor and their functions

- **Air filter** cleans air by removing dust particles.
- **Petrol filter** cleans the petrol.
- **Choke** controls the air supply
- **Jet** regulates the flow(supply) of fuel into the main petrol-air mixture.
- **Throttle** controls amount of petrol-air mixture that gets to the cylinders.
- **Float** maintains a consistent level of fuel, ensuring a steady supply to the engine's fuel metering system.

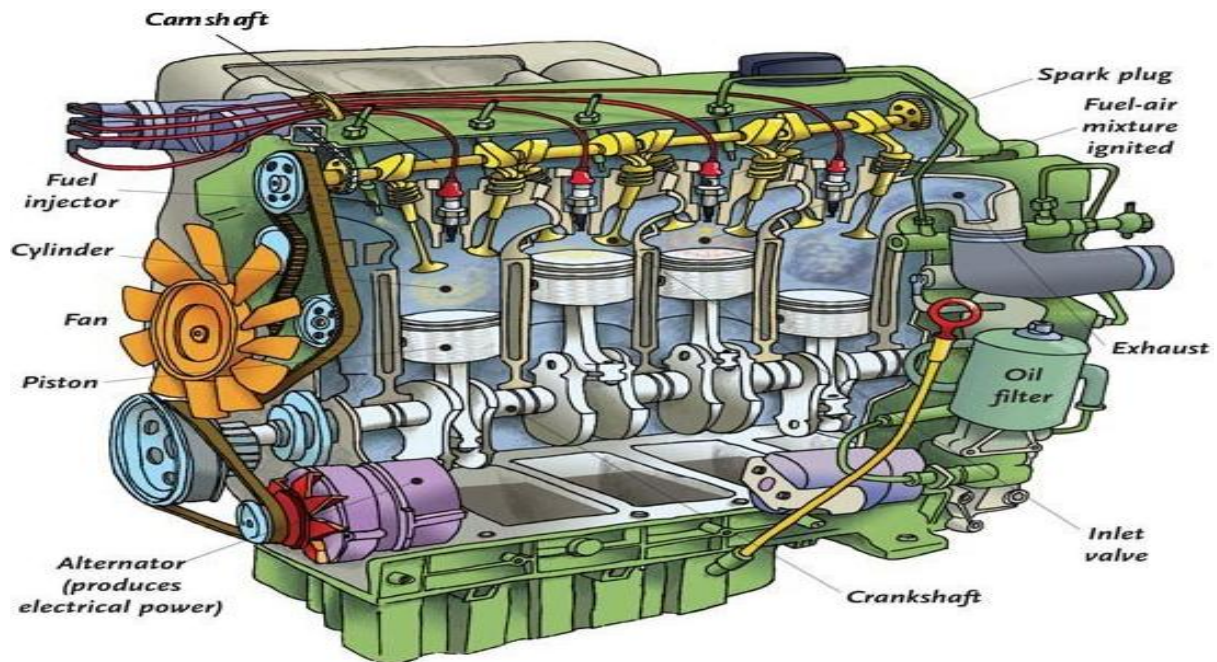
Comparison between the carburetor and fuel injector

FEATURE	CARBURETOR	FUEL INJECTION
DESIGN COMPLEXITY	Simple mechanical design, easy to maintain and repair	Complex system with electronics and sensors, requires specialized tools for repair
THROTTLE RESPONSE	Slower throttle response due to inexact fuel ratio	Faster and more precise throttle response
FUEL EFFICIENCY	Generally lower fuel efficiency	Higher fuel efficiency due to precise fuel-air mixture control
COLD START	Difficult to start in cold conditions	Easy to start in cold conditions
EMISSIONS	Higher emissions due to less efficient combustion	Lower emissions, cleaner combustion
COST	Cheaper initial cost	More expensive due to advanced technology
MAINTENANCE	Easier and cheaper to maintain	Limited repair options, often requires full system replacement if damaged
DURABILITY	Durable and long-lasting	Generally less durable than carburetors
ADAPTABILITY	Requires manual adjustments for environmental changes	Automatically adjusts to environmental changes
PERFORMANCE	Lower performance compared to fuel injection	Higher performance with optimized power output
QUALITY SENSITIVITY	Less sensitive to fuel quality	Requires specific fuel quality, contaminated fuel can cause issues
USER INVOLVEMENT	Requires more user involvement and tuning	Less user involvement, managed by ECU
STARTING METHOD	Can be bump-started easily	Bump-starting is more challenging

Operation of modern petrol and diesel engines

Petrol Engine

- Have more powerful, better fuel efficiency and release few harmful emissions.
- Carburetors have been replaced with an electronically controlled fuel injector to allow for more accurate ratio of fuel and air before the spark plugs ignite the mixture.
- This improves efficiency of the petrol engines.
- Most of the operating systems of new petrol engines are run by computers, which makes the process more accurate.
- It has turbo chargers to increase compression power



Advantages of modern petrol engines over old petrol engines

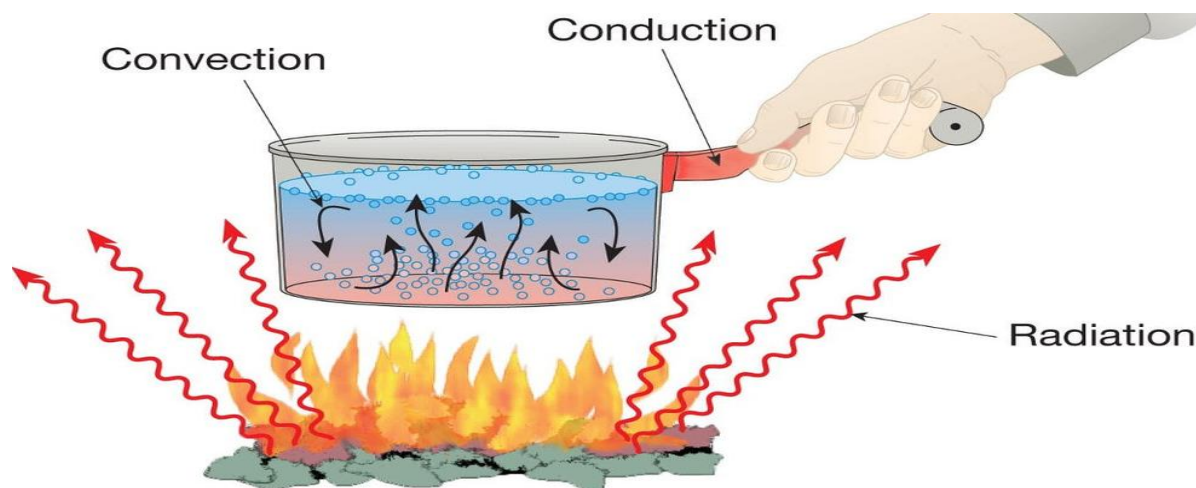
- They are now fast to start.
- Increased efficiency due to use of computers and stronger materials.
- Increased power due to use of more accurate fuel injector and more powerful explosions can occur in the combustion chamber.
- Reduces the number of litres of petrol needed to drive a certain distance thus reducing running costs of the vehicle.

Diesel Engine

- Cylinder made of stronger material to allow the temperature of the air in the cylinder to increase even more.
- Higher air temperature increases the efficiency of the ignition of the diesel fuel. It allows for a more powerful explosion that increases the efficiency and power output of the engine.
- It has turbo-intercooler to improve efficiency.
- Has pre-combustion chambers and has multiple holes on the injector

HEAT TRANSFER

- **Heat transfer** is the process whereby thermal energy moves due to temperature differences.
- Heat moves from a hot body(particle) to a cold body(particle).
- There are 3 methods of heat transfer namely:
 - a) **Conduction**
 - b) **Convection**
 - c) **Radiation**



Conduction Heat transfer

- **Conduction** is the heat transfer through solids by vibrations of particles.
- All solids are made up of atoms or molecules.
- When heat is applied at one end free electrons are given kinetic energy by heat and they move along the metal carrying energy.
- Particles begin to vibrate vigorously and in turn transmit the energy they have to their neighboring particles.
- **Good conductors** of heat are objects that allow heat to flow through them easily.
- Examples include metals, graphite (a crystalline form of carbon), diamond etc....
- **Bad conductors (thermal insulators)** of heat are objects that do not allow heat to flow through them easily.
- Examples include plastics, rubber, wood, air, glass etc...



Ironing Clothes



**Cooking on
a Gas Stove:**



**Ice Melting
in Hand**



**Metal Spoon
in Hot Soup**



**Computer
CPUs**



**Car Engine
Cooling**

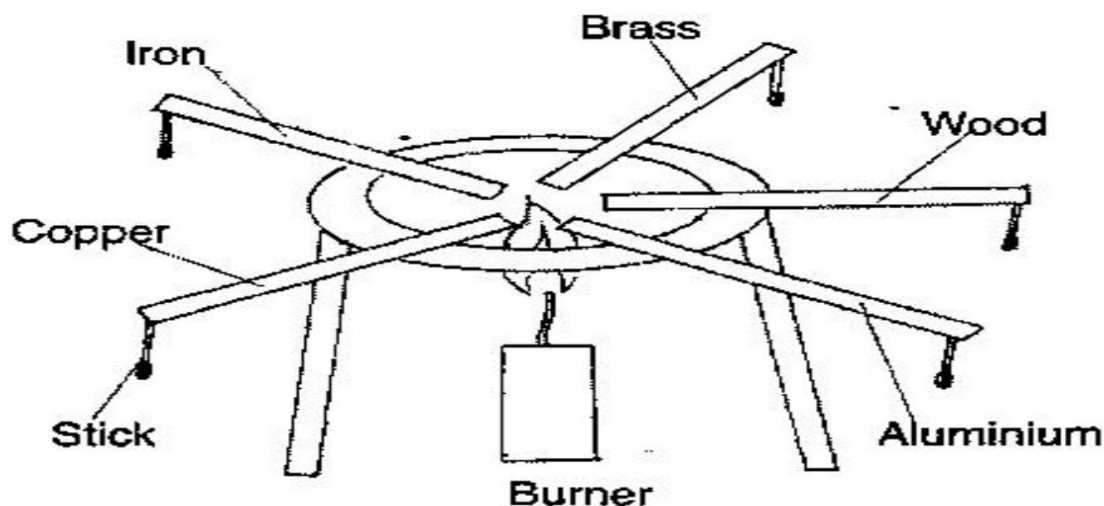


Lighting a Match



**Hot Coffee
in a Mug**

An experiment to determine the conductivity of objects



- Equal sized rods of different materials had match sticks attached on ends by wax and heat is applied.
- Heat flows from a region of higher temperature to a region of lower temperature, therefore the wax on the rods melt.
- The wax melts faster on the copper rod, followed by brass, iron, aluminium and lastly wood.
- Metals are good heat conductors and transfer heat energy rapidly while nonmetals, gases and liquids are poor conductors and transfer heat slowly (They are called thermal insulators).

Radiation Heat transfer

- **Radiation** refers to the heat transfer from one place to another through a vacuum in form of waves.
- No material medium is required, because the heat travels in form of waves i.e. electromagnetic waves.
- Examples include warmth from the sun, a fire, microwave ovens etc...



Sunlight



X-Rays



**Microwave
Ovens**



Radio Waves



**Ultraviolet
Light**



Wi-Fi Signals



Remote Controls



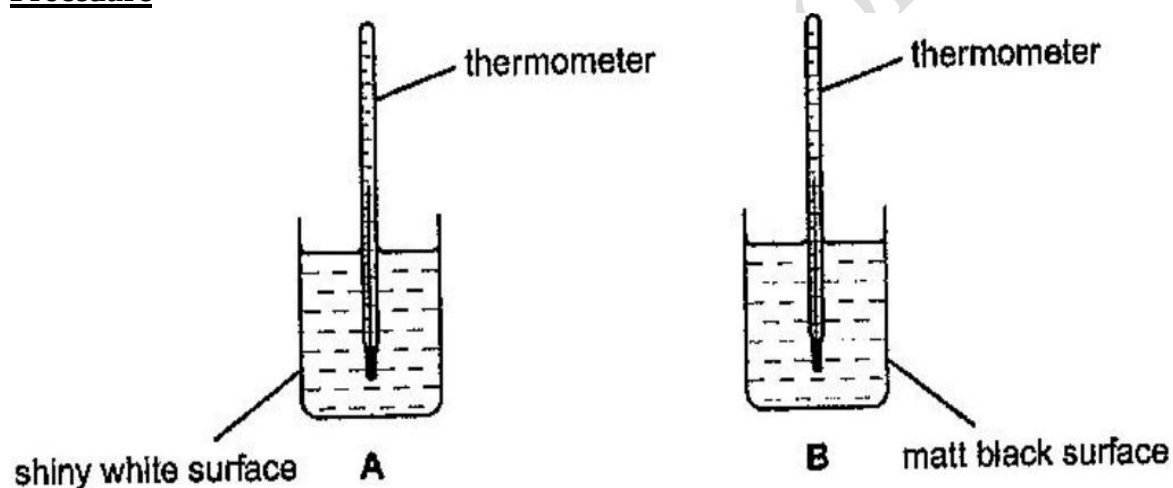
MRI Scans



An experiment to determine the Reflection, Absorption and Emission of heat.

- **Absorption** occurs when an object receives heat energy and increases in temperature.
- **Reflection** occurs when heat energy strikes an object and bounces back off the object.
- **Emission** is the transfer of heat through radiation from hot objects/materials. Emission often causes the temperature of the object to fall.

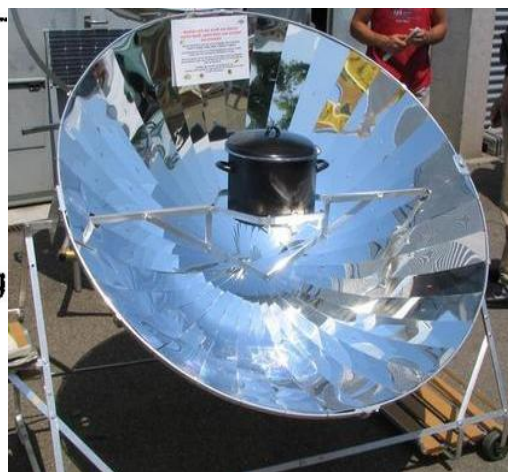
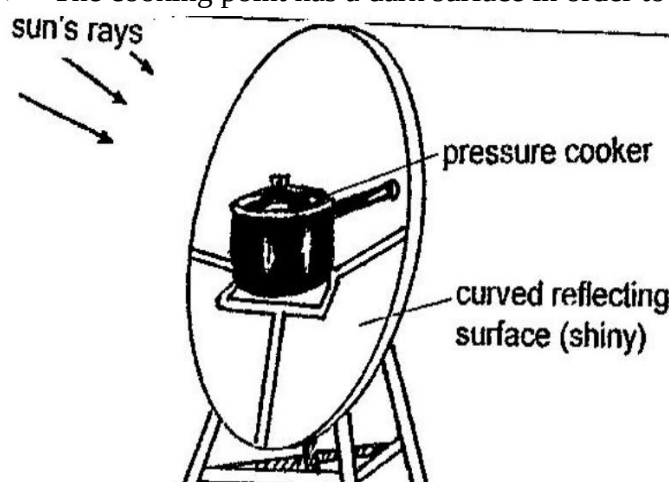
Procedure



- Pour same amount of water in both tins.
- Close the tins and place a thermometer.
- Record initial temperature of water in each tin.
- Place the tins in sun for 30 minutes and record the temperature again.
- **Result - Black or dull tin will have a larger increase in temperature while the white tin will have smaller increase in temperature.**
- Black dull tins absorb all the heat faster while the white tin reflects some of the heat and absorbs less heat.
- Place the tins in a room for 30 mins and record the temperature.
- **Result - Temperature of water in the black tin decreased faster and was lower than that of the white tin after 30 minutes.**
- **Conclusion - Dull, black surfaces are good absorbers and emitters of heat while shiny, white surfaces are good reflectors but bad absorbers and emitter of heat.**

Solar cooker

- A **solar cooker** uses the sun's ultraviolet rays and converts them into infrared light rays.
- This heat is trapped and used to cook food or boil water.
- It is made up of a **large curved reflectors (mirror)** which **focuses solar radiation on to the black pot at the center.**
- Radiant energy falls on mirrors and is reflected.
- The reflected radiation is concentrated and focused at one point (at the black pot).
- This produces intense heat which can reach temperature of up to 200°C.
- The cooking point has a dark surface in order to increase heat absorption.



Advantages of using solar cookers

- It is cheap
- Easy to use
- Environmentally friendly
- It is clean
- It saves time
- It preserves the nutrition value of food because the cooking is done at low temperatures
- It keeps the food hot for a long time



Disadvantages of using solar cookers

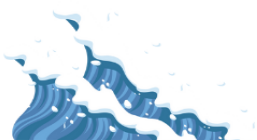
- It cannot be used on a rainy day and when there is cloud cover.
- It is very slow
- Cannot be used during the night

Convection Heat transfer

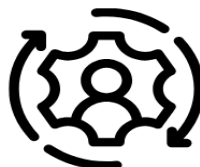
- Convection is heat transfer through fluids (liquids or gas).
- The movement of a liquid or gas allows heat to be transferred by convection currents.
- A convection current is the movement of fluid by the change in densities in various parts of the fluid.



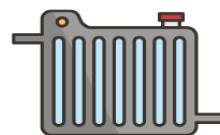
Boiling Water



Ocean Currents



Atmospheric Circulation



Radiators in Homes



Hot Air Balloons



Sea Breezes



Heating Soup

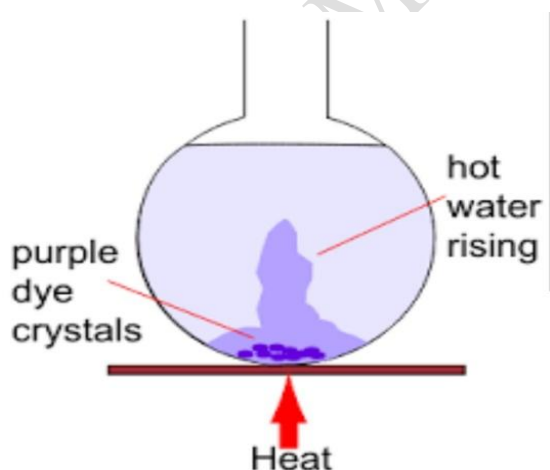


Air Conditioners

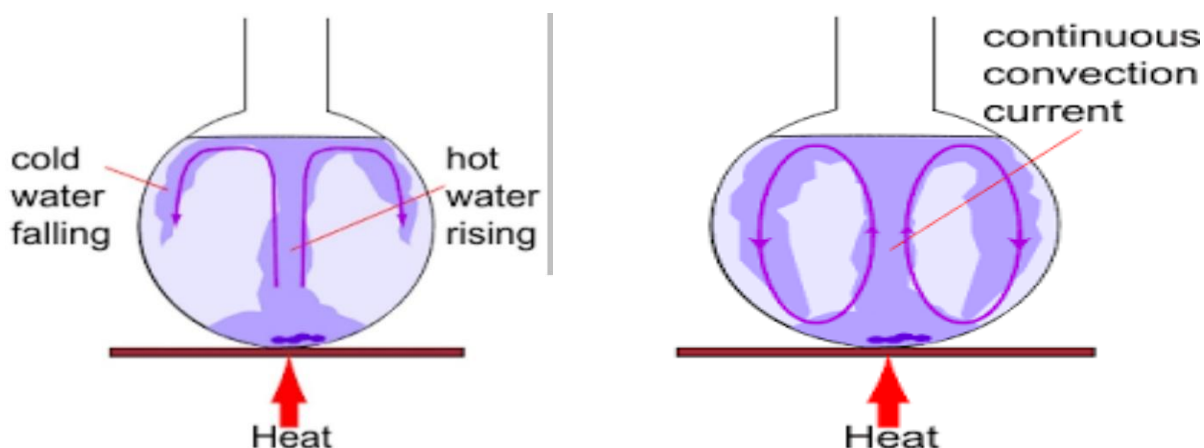
- Convection currents occur only in fluids (liquids and gases) because convection involves the bulk movement of the fluids which carry heat with them. It does not occur in solids.
- Below are two examples of convection, one with a liquid (water) and one with a gas (air).

a) **Showing Convection in liquids using potassium permanganate (convection in liquids)**

- The potassium permanganate is used to colour the moving water so that the convectional movement can be observed.
- A small amount of a purple dye is put at the bottom of the flask (to "colour" the water).
- As the water begins to heat up at the bottom of the flask, the potassium permanganate which is purple in color dissolves in water and rises up above the heat source:



- When the bottom of the flask of water is heated warm less dense water rises.
- The potassium permanganate dye rises with the warmed water.
- When the water cools it becomes denser and sinks down to the bottom of the flask.
- The potassium permanganate dye also sinks with the cooled water.



Explaining of why the above process happens

- As the water at the bottom of the beaker is heated, the water particles/molecules gain energy.
- This extra energy causes them to move faster and further apart; they take up more space.
- So, the heated water has increased in volume, becoming less dense than the colder water around it. (Remember $\text{Density} = \text{Mass}/\text{Volume}$, so if V increases whilst M remains unchanged, then D decreases.)
- Since this water is less dense than the surrounding water, it rises.
- Colder water moves in to take its place and it is heated and rises so there is a column of rising warm water.
- Eventually as the warmed water moves further from the heat source it cools, becomes denser and falls, joining the colder water at the bottom of the beaker, getting reheated, forming a continuous convection current.

b) Showing Convection in gases using a smoldering paper in a chimney (convection in gases)

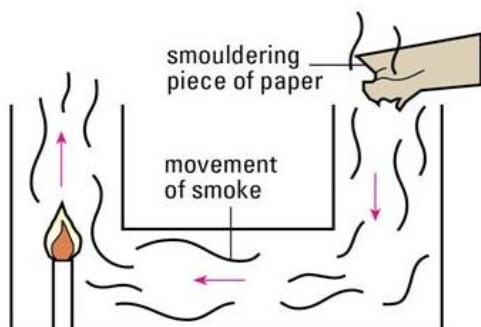
- A smoldering paper is a paper that produces smoke.



- Air above the candle is heated, becomes lighter and rises, creating a partial vacuum which in turn is filled in by cold dense air that flows into the apparatus through one chimney.
- This creates convectional currents and smoke is seen continuously coming out through the other chimney.

Objective

To show convection in air



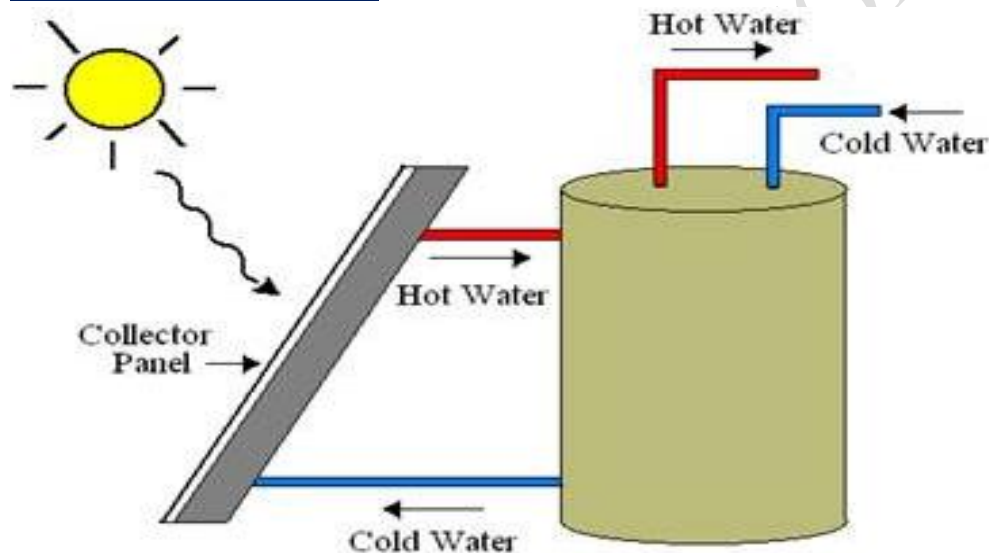
Apparatus

large box with two chimneys on top, a piece of clear glass on one side, candle, matches

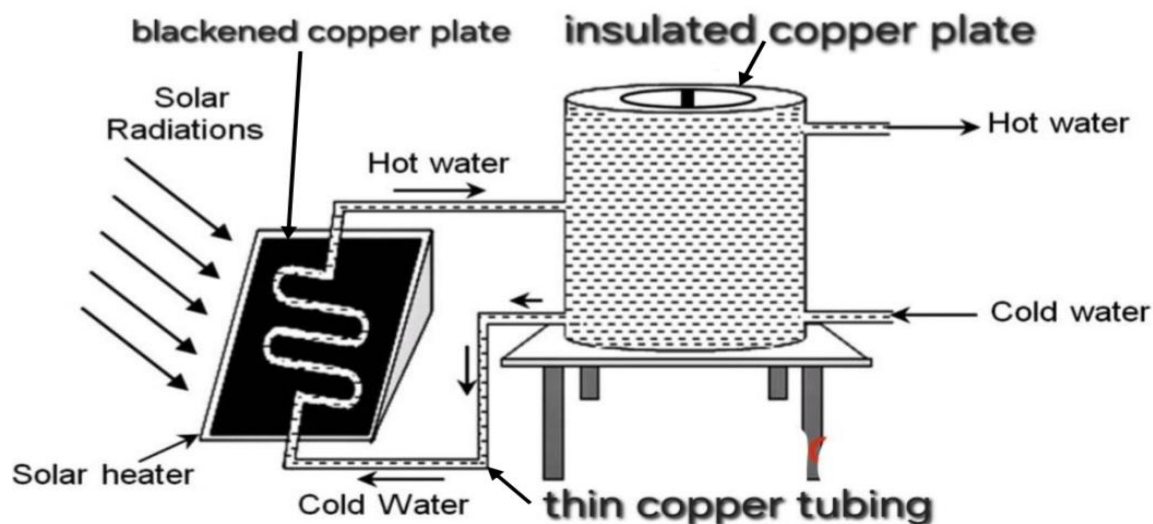
Procedure

1. Place the candle below one of the chimneys. Light the candle.
2. Introduce smoke into the other chimney by placing a piece of smouldering paper over it and observe the movement of the smoke.

The Solar water heater



- The solar water heater is placed on the roof of a house to absorb maximum solar radiation since there are no obstacles.
- The solar panel is heated directly by radiation from the sun.
- The heat is then transferred to water inside the panel by **conduction**.
- Radiation falls on **blackened copper plate**. Black surfaces are good absorbers of heat. The absorbed heat is passed on to thin copper tubes. Copper is a good heat conductor.
- Heat is then passed to the water; the thin copper tubes easily pass heat to the water.
- The thin copper tubing is coiled to increase surface area for heating the water.
- Heated water is stored in an **insulated tank to reduce heat loss**.
- The glass plate traps hot air in the panel by the **greenhouse effect**.
- The expanded polystyrene base **prevents heat loss by convection** while air spaces prevent heat loss by conduction to the roof.

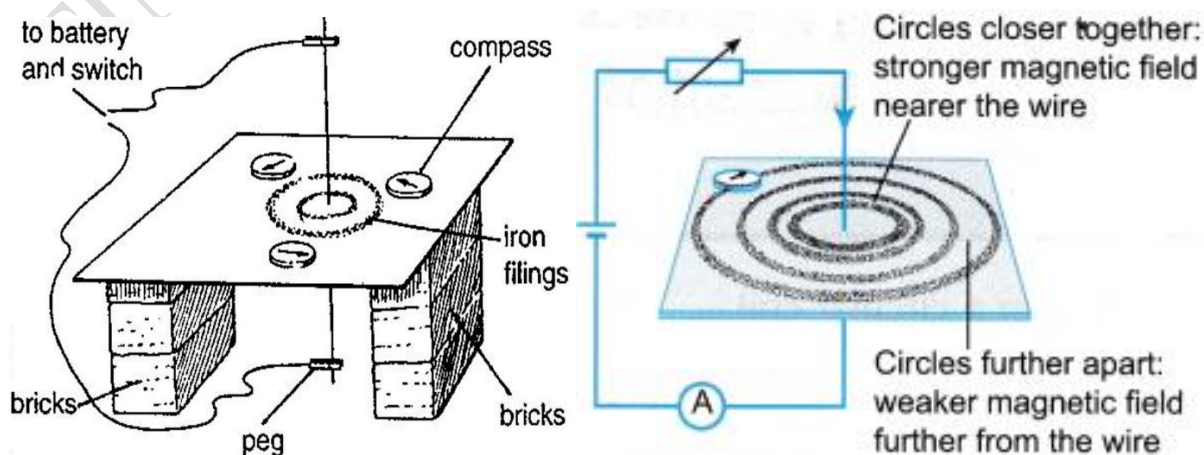


- A solar water heater can work together with a geyser boiler and this allows to save electricity because the solar water heater will keep the water warm.

MAGNETISM

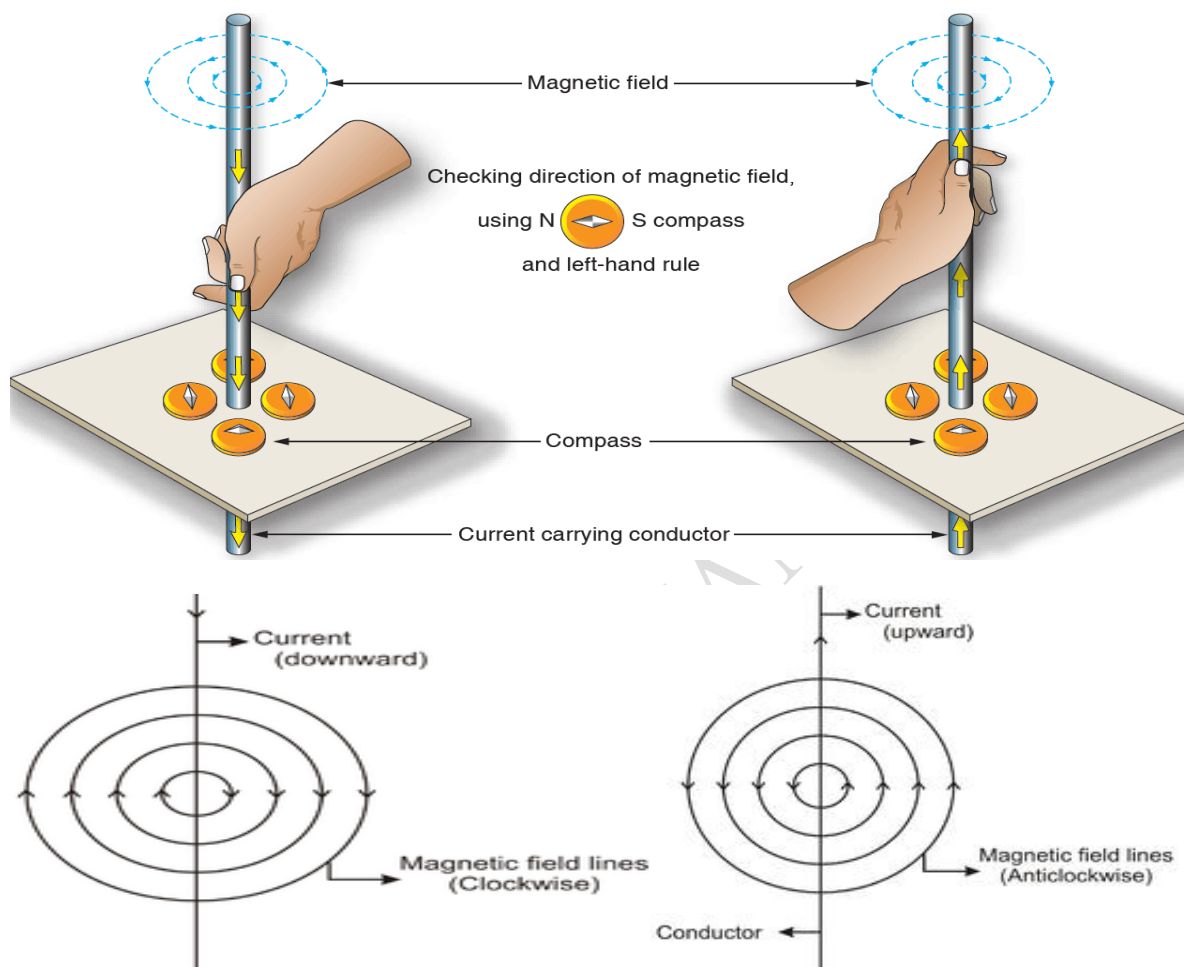
Electromagnetism

- **Electromagnetism** is a branch of physics that studies the interactions between electric charges and magnetic fields.
 - A **magnetic field** is generated around any current-carrying conductor, and this field can be demonstrated using various methods like **iron filings, compasses, or even the movement of a wire in a magnetic field.**
 - The direction of the field can be determined using the Fleming's left-hand rule.
 - Here's a more detailed explanation:
- a. **Demonstrating magnetic field in a conductor current carrying using iron filings**
- A straight copper wire is set up vertically, passing through a hole in the centre of a piece of a cardboard supported horizontally.
 - A fine layer of iron filings is sprinkled evenly on the cardboard.
 - Current is passed through the wire and iron filings will align themselves in a series of circular patterns around the wire, showing the shape of the magnetic field.



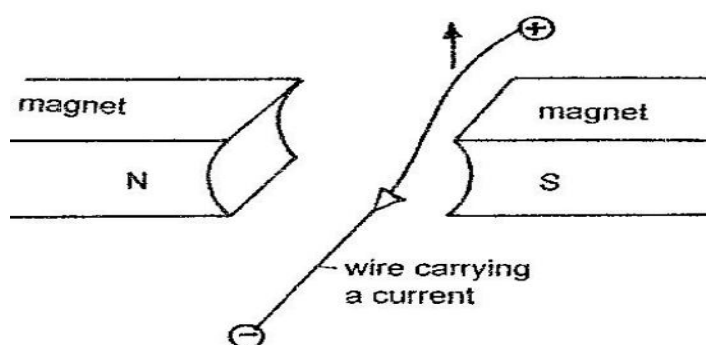
b. Demonstration magnetic field in a conductor current carrying using compasses

- Arrange several plotting compasses around a straight wire.
- Pass a current through the wire.
- The compass needles will deflect and align themselves, forming a circular pattern around the wire, demonstrating the magnetic field's direction.



c. Demonstration magnetic field by the movement of a wire

- Place a wire in a magnetic field (e.g., between the poles of a magnet).
- Pass a current through the wire.
- The wire will experience a force and move, demonstrating the **interaction between the current-carrying wire and the magnetic field**.
- This movement can be used to demonstrate the force on the wire based on Fleming's left-hand rule.

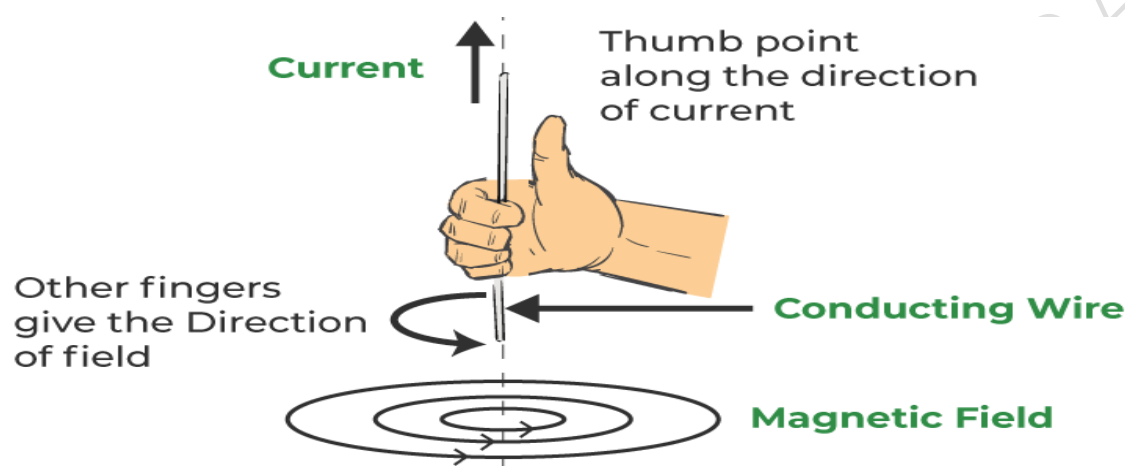


Fleming's hand Rules of determining direction of magnetic field and force on a wire

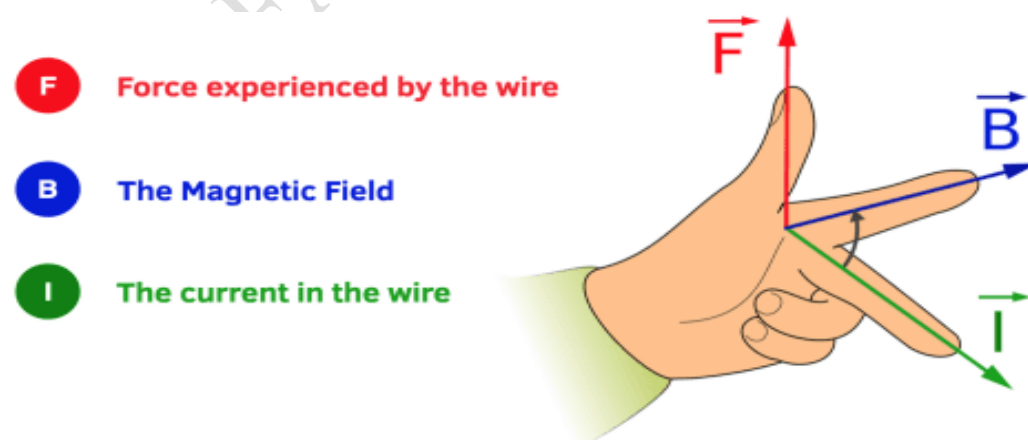
- **Current** always **travels** from the **positive end** of the cell to the **negative end** of the cell
- Just like current, **magnetic field also has a direction.**

✚ Right-hand rule (or Fleming's right-hand rule):

- In order to determine the direction of the magnetic field around a current-carrying wire:
 - Hold the wire in your right hand, your thumb will be pointing in the direction in which the current is flowing.
 - Your fingers will then curl in the direction of the magnetic field.

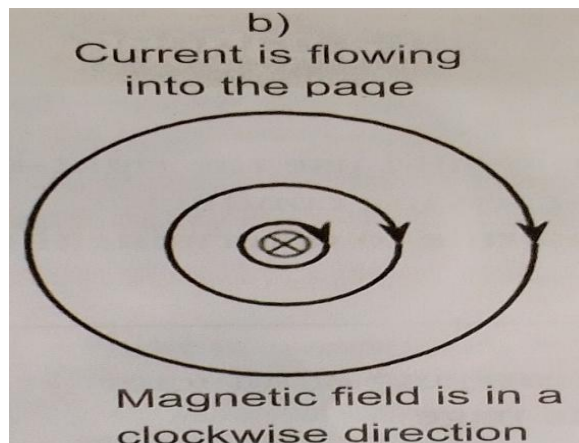
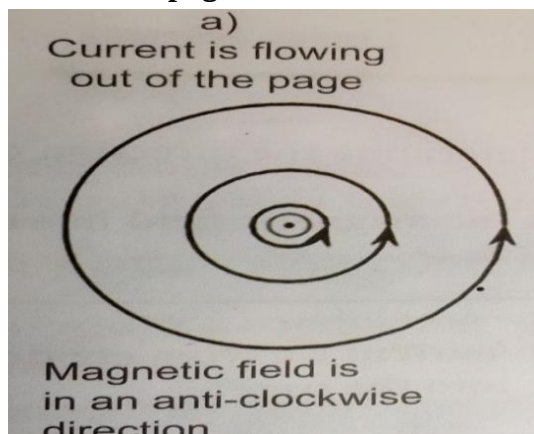
**✚ Fleming's left-hand rule:**

- In order to determine the direction of the force on a current-carrying wire in a magnetic field:
 - Extend your thumb, index finger, and middle finger of your left hand so that they are all mutually perpendicular to each other.
 - Point your **index finger** in the **direction of the magnetic field**.
 - Point your **middle finger** in the **direction of the current**.
 - **Your thumb** will point in the **direction of the force**.

**Direction of current in and out of the page**

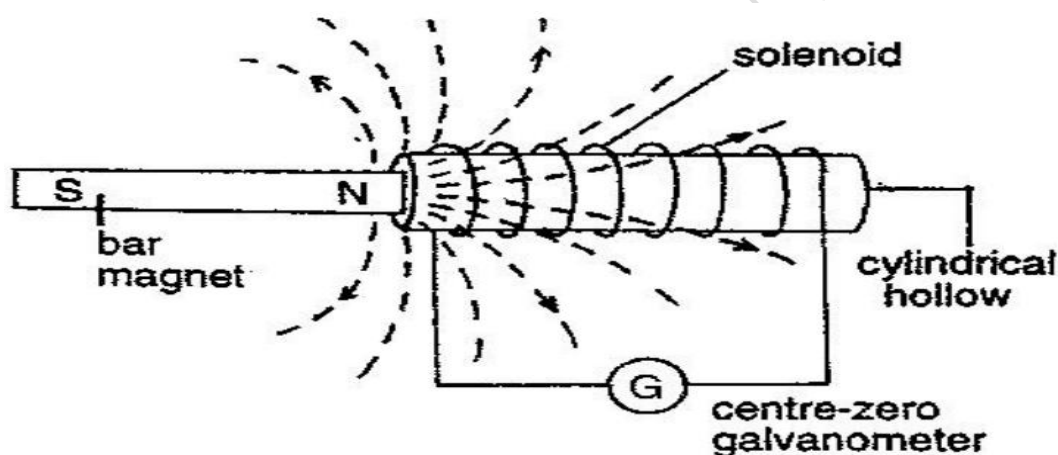
- A dot at the centre of magnetic field diagram indicates that the direction of the current is **out of the page**.

- A cross at the centre of magnetic field diagram indicates that the direction of the current is **into the page**.



Magnetic field around a solenoid (coil)

- When an electric current is passed through a solenoid (coil) the resultant magnetic field is similar to that of a bar magnet.
- It acts as a magnet when current flows through it.



MOTORS

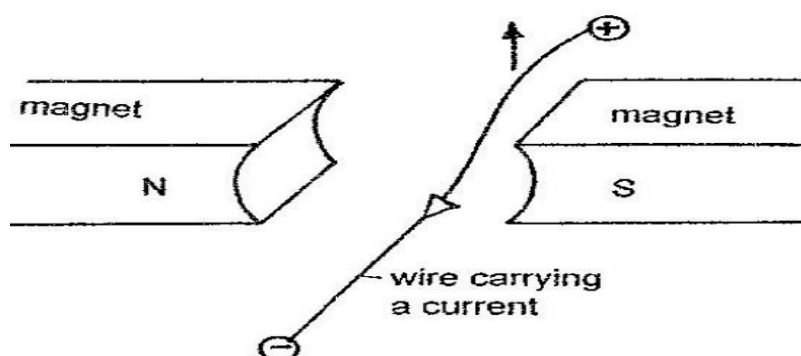
The motor effect

- The **motor effect** is a principle which describes the force experienced by a current-carrying conductor (wire) placed in a magnetic field.
- **N.B** - A current-carrying wire creates its own magnetic field, which interacts with an external magnetic field (like that of a magnet).
- It's the basis for how electric motors **convert electrical energy** into **mechanical energy**.

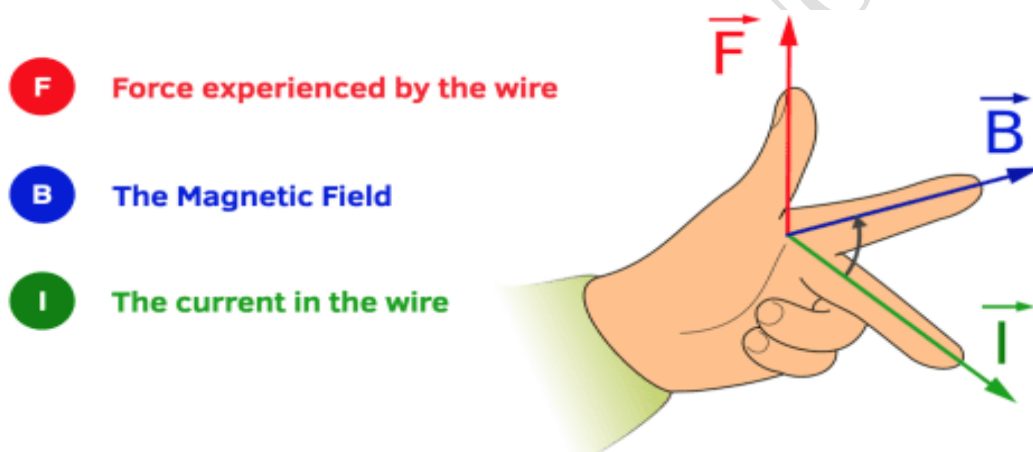
How movement is produced in a current carrying wire in a magnetic field

- A current carrying conductor has a magnetic field around it.
- When the current carrying conductor is placed in another magnetic field that is created by two magnets; the interaction of the two fields creates a force that causes movement of the conductor(wire) and runs the motor.

- The force on the wire is perpendicular to both the direction of the current and the magnetic field.
- The wire moves up when the current is in one direction and moves down when the current is reversed.



- Fleming's Left-Hand Rule helps determine the direction of the force on the wire when given the directions of the magnetic field and the current.
- Stretch the thumb, index and middle finger of the left hand so that they are perpendicular to each other



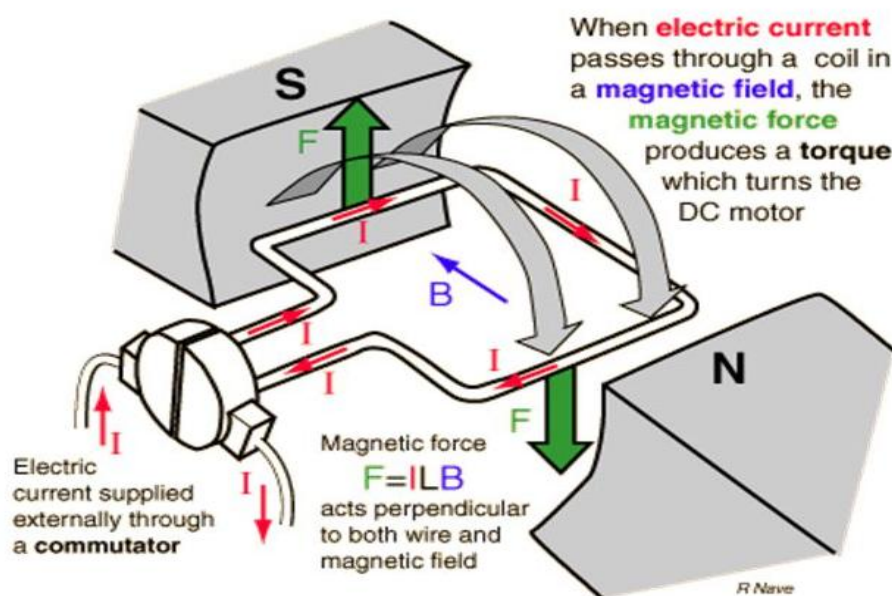
- **Index or first finger** shows direction of the field i.e. north to south.
- **Middle or second finger** shows the direction of current
- **Thumb** finger shows the direction of force or movement.

Application of motor effect: Electric Motors

- The motor effect is the driving force behind how electric motors work, converting electrical energy into rotational motion.
- How it Works:
 - A current-carrying wire generates a magnetic field around itself.
 - When placed in an external magnetic field (e.g., from a magnet), these two fields interact.
 - The interaction creates a force on the wire, pushing it in a specific direction.
 - In a motor, this force on multiple wires in a coil can **cause the coil to rotate**.

Direct current (D.C) motor

- A DC motor converts **electrical energy** from a direct current into **mechanical energy**, typically rotational motion.
- This is achieved through the interaction of magnetic fields and electric current.
- The motor's rotor, containing coils of wire, rotates within the magnetic field of the stator, which is either permanent magnets or electromagnets.
- A commutator and brushes ensure continuous current flow to the rotor, maintaining rotation.



DC Motor

Right hand rule:

The Index finger points in the direction of the current;
 Middle finger points in the direction of the magnetic field (N to S);
 Thumb direction indicates the direction of force.

The operation of a direct current (d.c.) motor

- When a DC current is applied, the rotor coils (armature) become electromagnets, creating a magnetic field.
- The magnetic field due to the current in the coil interacts with that of permanent magnets (stators) creating a rotational force on the rotor.
- The commutator and brushes reverse the current flow through the rotor coils every half-turn, ensuring continuous rotation.
- The magnitude of the rotational force depends on the strength of the magnetic fields and the current flowing through the rotor coils.
- Using Fleming's left-hand rule, the forces are in the clockwise direction.
- The motor converts **electrical energy** to **mechanical energy**.

Components of a DC motor:

- ✓ **Rotor:** The rotating part containing coils of wire.
- ✓ **Stator:** The stationary part with either permanent magnets or electromagnets.
- ✓ **Commutator:** A segmented ring that reverses the current flow in the rotor coils, ensuring continuous rotation in the same direction.

- ✓ **Brushes:** Contact points that transmit current from the stationary power supply to the rotating commutator.

Factors that affect the rotation of the coil

- **Strength of magnetic field.** Increasing the strength of the magnetic field by using strong magnets increases the motor's speed.
- **Number of turns in the coil.** More turns in the coil can increase the magnetic field strength, potentially increasing speed.
- **Amount of current passed.** Increasing the current in a coil within a magnetic field increases the force on the coil, leading to a faster rotation speed.
- **Cross sectional area of the coil.** A larger coil area will experience a greater force, resulting in faster rotation.

The speed of a motor is increased by

- Increasing number of turns in the coil.
- Using stronger magnets to increase density of magnetic field.
- Increasing current.
- Place a soft-iron core within the magnetic field lines

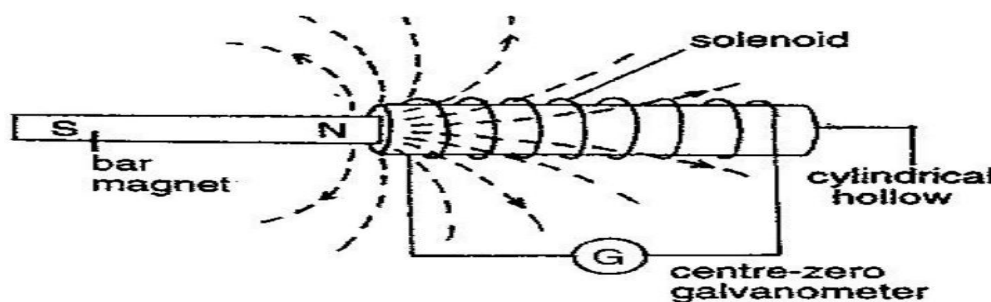
GENERATORS

Generator effect

- The **generator effect** refers to the process of generating a voltage and potentially a current in a conductor when it experiences a changing magnetic field, called a magnetic flux.
- The voltage induced(produced) is the **electromotive force (e.m.f)**
- The rotation of the coil in a magnetic field induces an e.m.f in the coil.

An experiment showing that a change in magnetic field can induce e.m.f in a conductor.

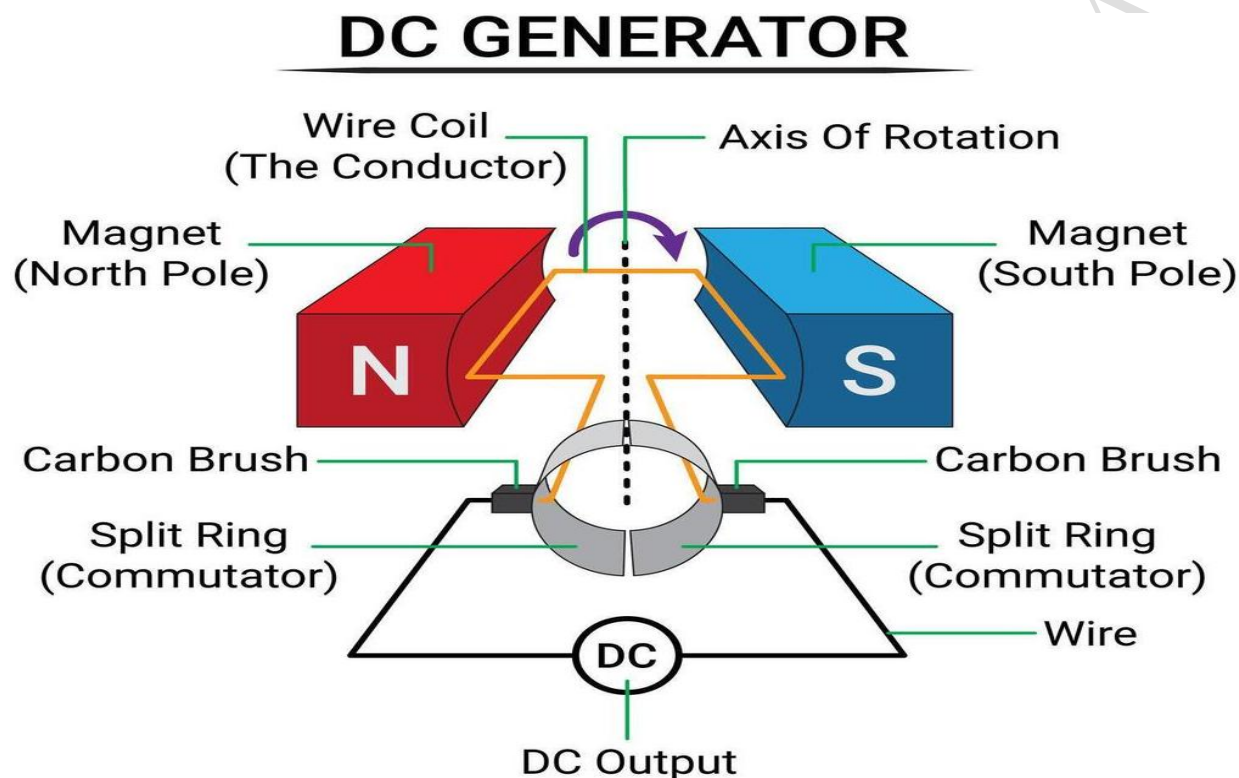
- When the magnet approaches the coil, north-pole first, the induced current should flow in a direction that makes the coil behave like a magnet with its top a north pole.
- The downward motion of the magnet will then be opposed since like poles repel.
- When the magnet is withdrawn, the top of the coil should become a south pole and attract the north pole of the magnet, so hindering its removal.
- The induced current is thus in the opposite direction to that when the magnet approaches.
- When there is a change in the magnetic lines of force passing through the coil, an e.m.f is induced in the conductor.



Factors which affect the magnitude of induced e.m.f.

- **Strength of a magnet** – a stronger magnet induces a stronger e.m.f
- **Relative motion** – moving or rotating the conductor (coil) faster induces a stronger e.m.f
- **Number of turns** – the more the turns are in the coil, the stronger the induced e.m.f
- **Area of coil** – the larger the area that the magnetic field acts over the stronger the induced e.m.f

 Simple D.C. generator

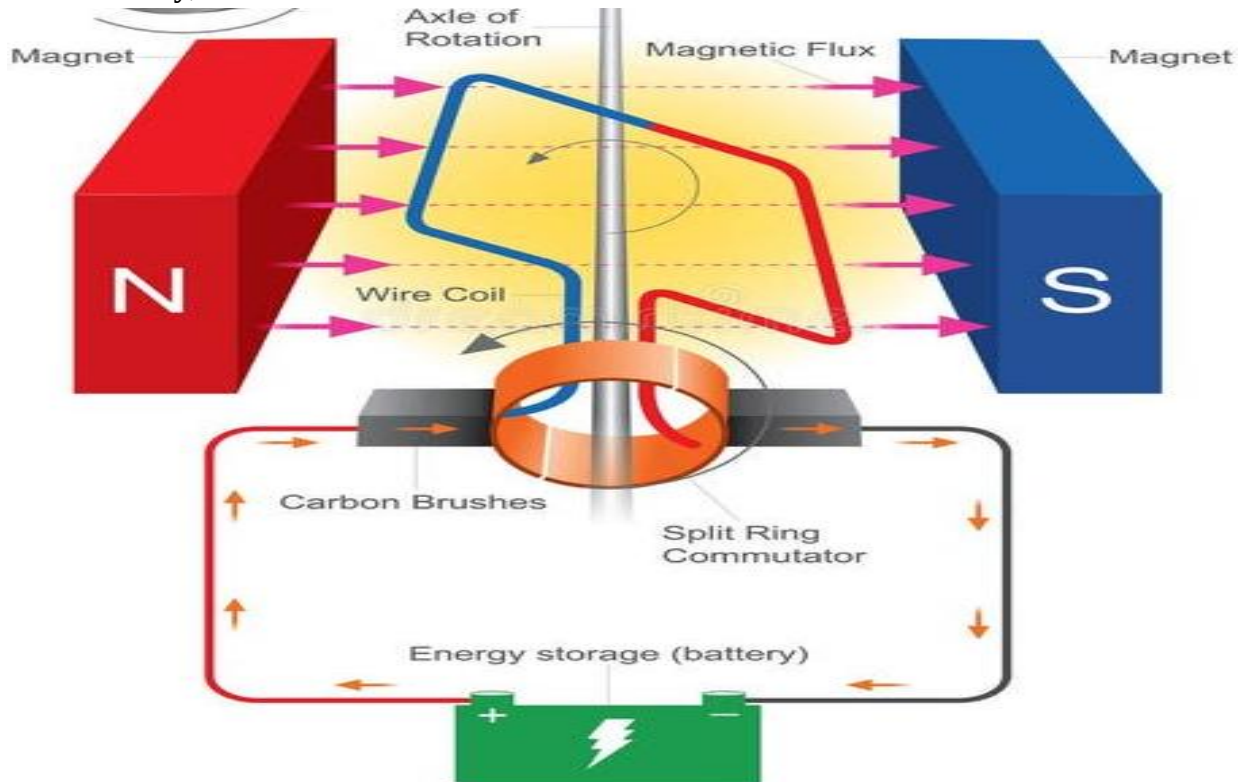


- A simple DC generator works by converting **mechanical energy** into DC **electrical energy** using electromagnetic induction.
- A coil of wire rotates within a magnetic field, causing an electromotive force (emf) to be induced in the coil.
- A split-ring commutator ensures that the current always flows in the same direction in the external circuit.

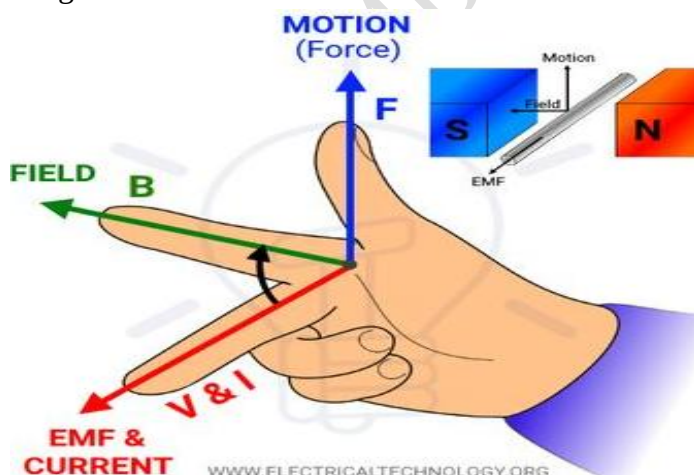
Working Principle of a DC generator:

- In a dc generator; a coil which is attached to the external circuit is placed in a magnetic field.
- The coil is physically rotated within the magnetic field, either by hand or by a mechanical source of power.
- As the coil rotates, it cuts through the magnetic field lines, and a change in magnetic field (magnetic flux) occurs.
- The change in magnetic field induces an electromotive force (emf) through the coil.

- The coil is attached to a split ring commutator which ensures that the direction of the induced current is always in the same direction as it is to the external circuit.
- The brushes on the coil touch the split ring commutator while the coil rotates.
- The dc generator converts the **mechanical energy** of the rotating coil into **direct current electricity**, which is then available for use in an external circuit.



- Fleming's right hand rule is used to determine the direction of flow of current in a dc generator.



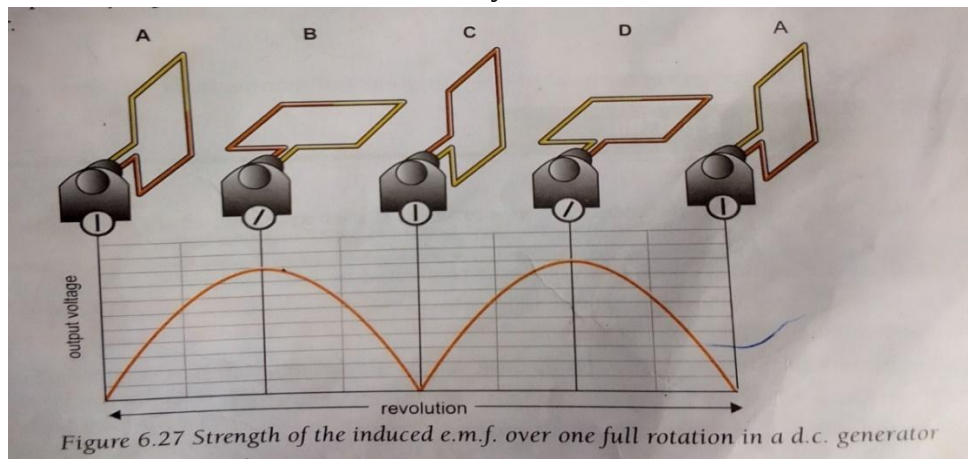
Key Components of a DC generator:

- **Coil:** A coil of wire (usually copper) that rotates within the magnetic field.
- **Magnetic Field:** Created by permanent magnets or electromagnets.
- **Split-Ring Commutator:** A split ring that connects to the coil and rotates along with it.

- **Brushes:** Conductive contacts that contact the split-ring commutator and provide a connection to the external circuit.

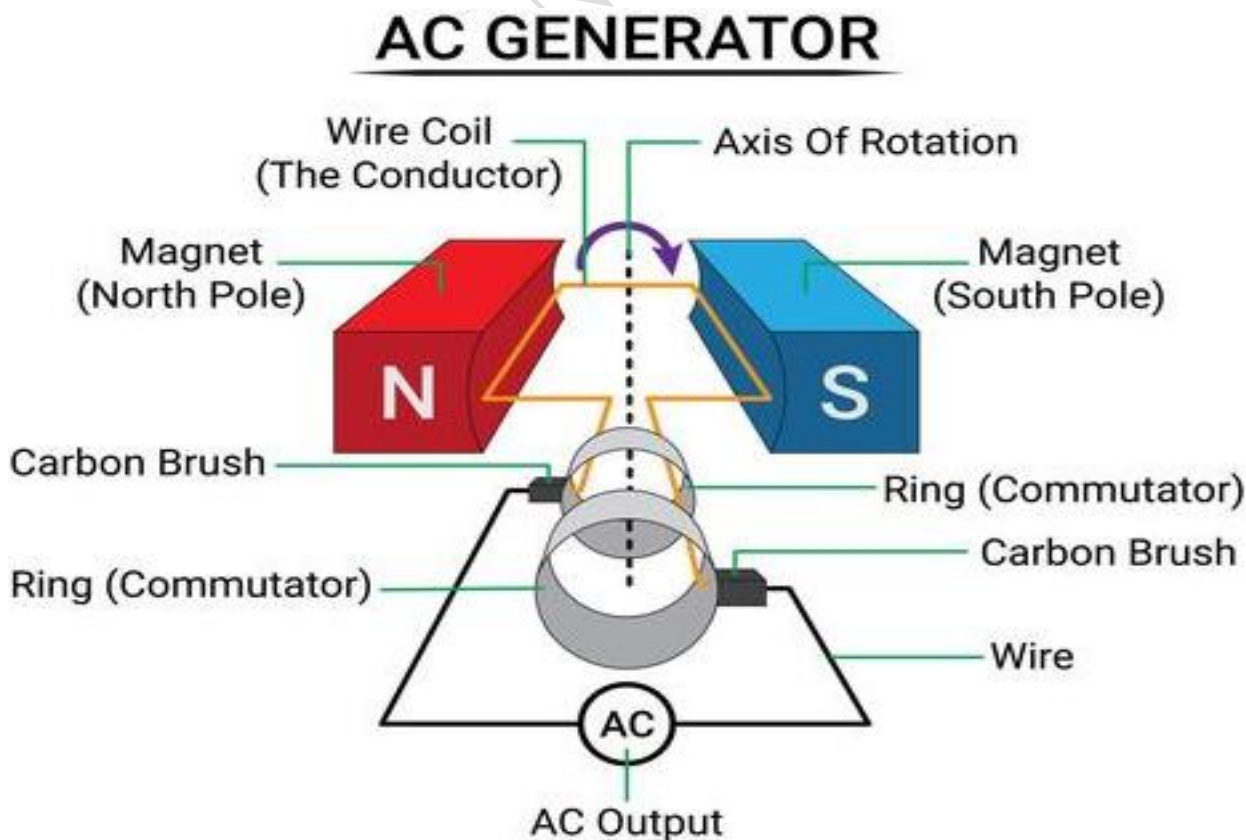
Graphical representation of output voltage in a DC generator

- Current flows in one direction only



- The emf on the graph always has a positive value
- When the coil is vertical, no magnetic field lines are being cut by the coil, hence no current.
- When the coil is horizontal, the coil is cutting the field lines, hence maximum current in the coil.

Simple AC generator



- As the coil rotates the magnetic field through the coil changes and therefore induces an e.m.f. between the ends of the coil.
- The induced current does not flow unless the ends of the coil are connected to an external circuit with an electrical load.
- The slip rings allow the transfer of alternating e.m.f. induced in the rotating coil to the external coil.
- Each ring is connected to one end of the coil wire and electrically connected by a carbon brush to the rest of the circuit.
- The brushes on the coil touch the slip rings while the coil rotates for a full, continuous cycle thus causing the current in the external circuit to change direction every time the coil makes half turns.
- An a.c current is generated
- The current in the external circuit changes direction every half revolution

Graphical representation of output voltage in an AC generator

- The current in the external circuit changes direction every half revolution
- The emf on the graph changes from a positive to negative

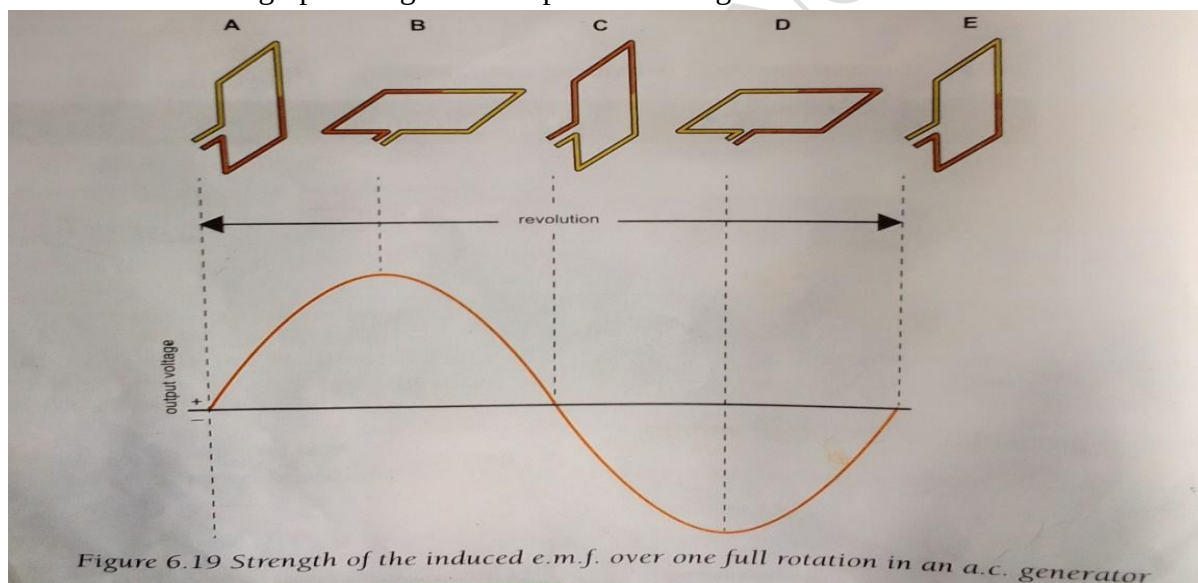


Figure 6.19 Strength of the induced e.m.f. over one full rotation in an a.c. generator

- This shows that there is a change in the direction of the flow of current

POWER GENERATION

Hydropower generation

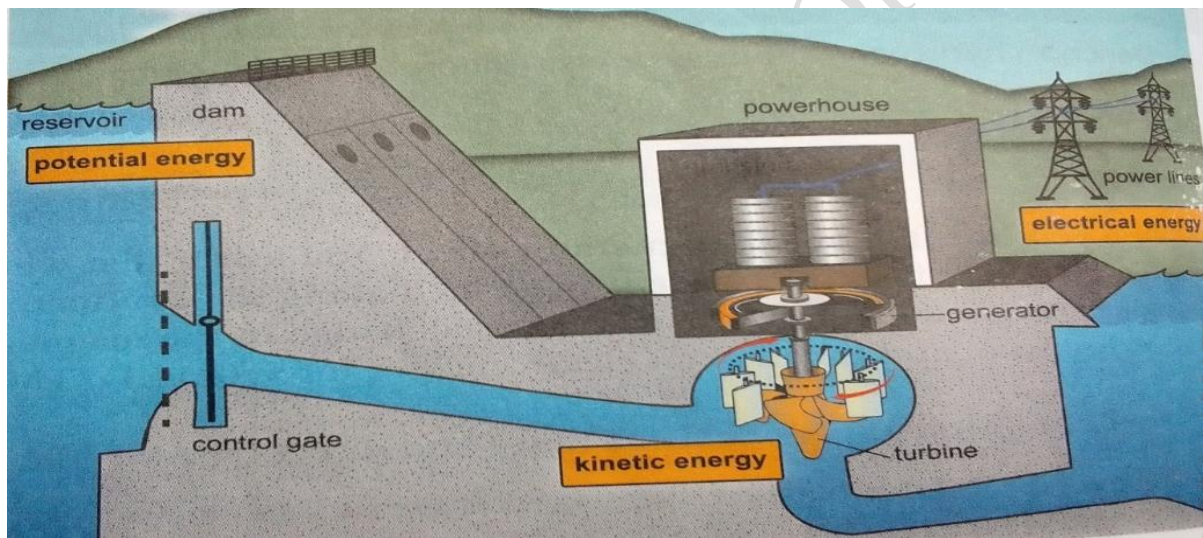
- **Hydroelectric power stations** generate electricity by using the force of moving water to turn turbines, which in turn drive generators.
- Hydropower is economical; renewable and environmentally friendly.
- Hydropower is a renewable energy source

How Hydroelectric power station operates

- Dams are built to create reservoirs that store large volumes of water.

- This stored water has **potential energy** due to its height above the turbine.
- Water is released from the reservoir through a narrow tunnel called a penstock, which directs the flow to the turbine.
- The water, now with **kinetic energy**, is channeled to a turbine.
- The force of the water causes the turbine's blades to rotate.
- This rotation converts the water's kinetic energy into **mechanical energy**.
- The turbine's rotating shaft is connected to a generator.
- As the shaft spins, the generator converts the mechanical energy into **electrical energy**.
- This electrical energy is then transmitted through power lines to homes and businesses.
- After passing through the turbine, the water is returned to the river downstream of the power station.
- In essence, a hydroelectric power station acts as a water-powered electricity generator, using the natural flow of water to produce a renewable energy source.
- Energy conversion in hydroelectric power generation:

Potential energy → kinetic energy → mechanical energy → electrical energy



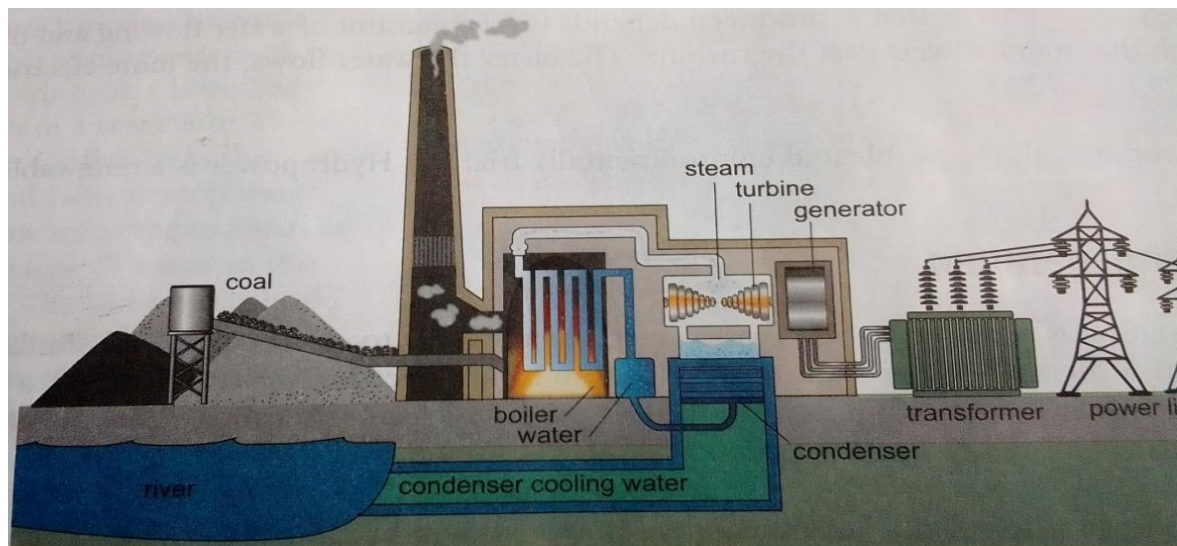
Thermal power generation

- **Thermal power stations** generate electricity by converting **heat energy** into **mechanical energy**, which then drives a generator.
- This process typically involves burning fuel to produce heat, which is used to generate high-pressure steam.
- The steam then powers a turbine, and the turbine's rotation drives an electrical generator, producing electricity.

How Thermal power station operates

- Fossil fuels like coal, oil, or natural gas are burned in a boiler.
- The heat from combustion is used to heat water, turning it into high-pressure, high-temperature steam.
- The high-pressure steam is directed to a turbine, causing its blades to rotate.

- The rotating turbine is connected to an electrical generator which converts **mechanical energy** of the turbine into **electrical energy**.
- The steam is then condensed back into water and recycled in a closed loop, with any excess heat dissipated, often through cooling towers.



- Energy conversion in thermal power generation:

Chemical potential energy → heat energy → kinetic energy → mechanical energy → electrical energy

ELECTRICITY

Uses of electricity in the home

- Powering electrical devices e.g. motors
- Lighting
- Heating and cooling.

Static electricity

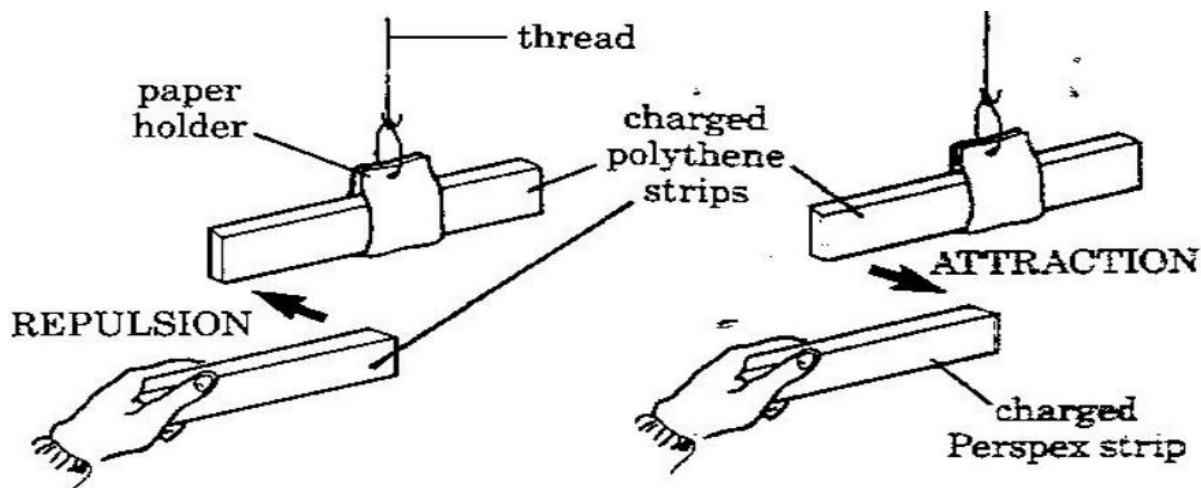
Electrostatic charging

- Polythene and Perspex can be charged through friction creating static electricity.
- Rubbing these materials together, such as with a cloth, causes electrons to transfer from one material to the other, resulting in one material becoming negatively charged (gaining electrons) and the other positively charged (losing electrons).
- This phenomenon is known as charging by friction or triboelectric effect.

How electrostatic occurs

- When two different insulating materials are rubbed together, electrons can be transferred from one material to the other.
- Materials have different affinities for electrons. The triboelectric series ranks materials based on their tendency to gain or lose electrons.

- When two materials are rubbed together, the one higher on the series tends to gain electrons (become negative), and the other tends to lose electrons (become positive).
- **Polythene** typically becomes **negatively charged** when rubbed with a cloth, while **perspex** tends to become **positively charged**. The cloth, in turn, becomes oppositely charged.
- The charged materials can then exert forces on other objects, causing them to be attracted or repelled. This is the basis of many static electricity demonstrations and experiments.



- For example, if you rub a polythene rod with a woolen cloth, the rod will become negatively charged, and the cloth will become positively charged. Then suspend the charged rod by a cotton thread from a retort stand.
- Charge a second polythene rod and bring it towards the suspended rod and repulsion occurs, thus like charges move away from each other (repel).
- Charge a perspex rod and move it towards the suspended polythene rod, and attraction occurs, thus unlike charges move towards each other (attract).
- Moreover, if you then bring the charged polythene rod near a stream of water, you will observe the water bending towards the rod due to the electrostatic attraction between the opposite charges.
- Similarly, if you bring the charged polythene rod near small pieces of paper, the paper will be attracted to the rod.

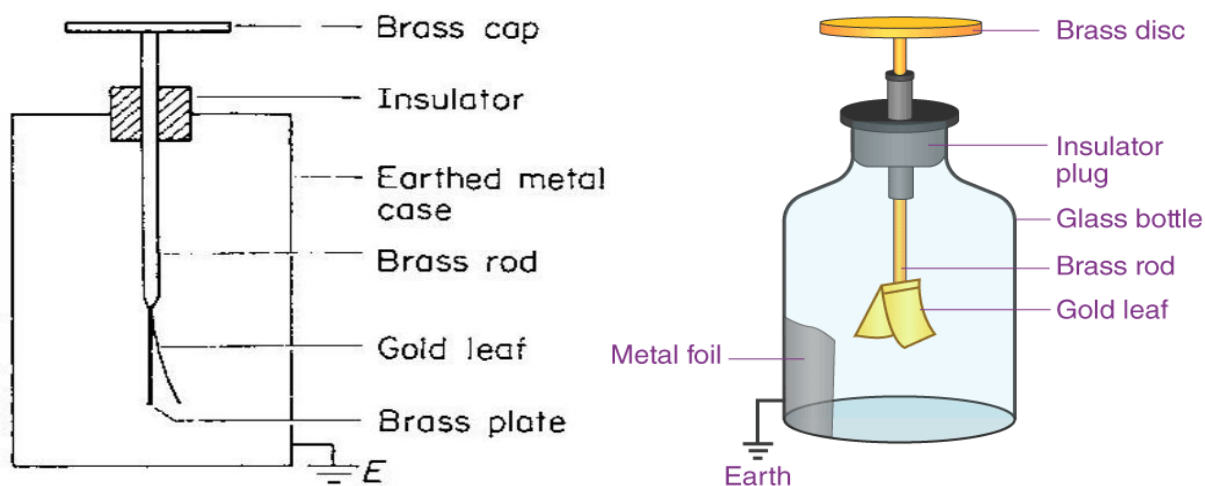
Gold leaf electroscope

- A **gold-leaf electroscope** is a scientific instrument used to detect the presence and relative amount of electric charge in a body.
- It consists of a vertical metal rod with thin, flexible gold leaves hanging from its end, typically enclosed within a glass case.
- When charged, the leaves repel each other, demonstrating the presence of charge.

Key Components of the gold-leaf electroscope:

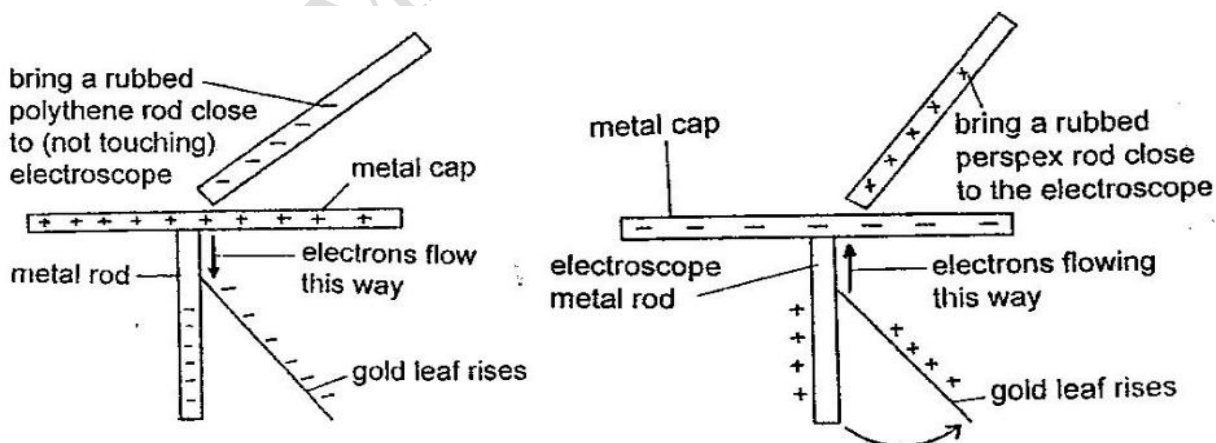
- A gold-leaf electroscope consists of a brass rod, with a brass (metal) cap and a brass plate at the bottom.
- A thin leaf of gold or aluminium foil is attached to the brass plate. The brass rod is mounted in a glass case, supported by a plug of insulating material.

- **Brass Rod (Electrode):** This is a conductive rod, usually brass, that's insulated from the surrounding case.
- **Gold Leaves:** Two thin, flexible strips of gold foil, hanging from the end of the rod.
- **Glass Case (Shield):** Encloses the gold leaves to protect them from air currents, which can interfere with their movement.
- **Brass Plate:** May be present inside the glass case to act as safety measures and improve sensitivity.



How the gold leaf electroscope Detects a charge

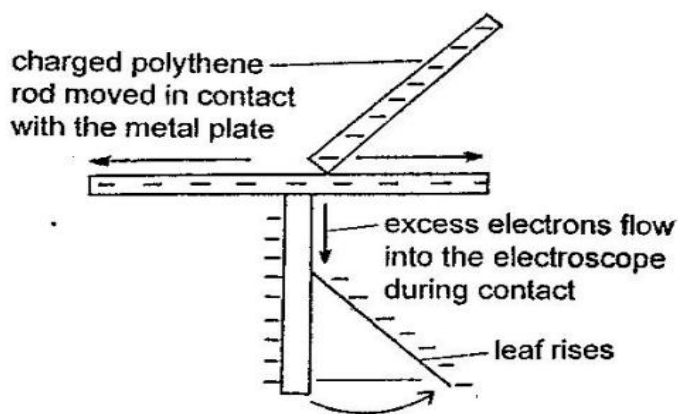
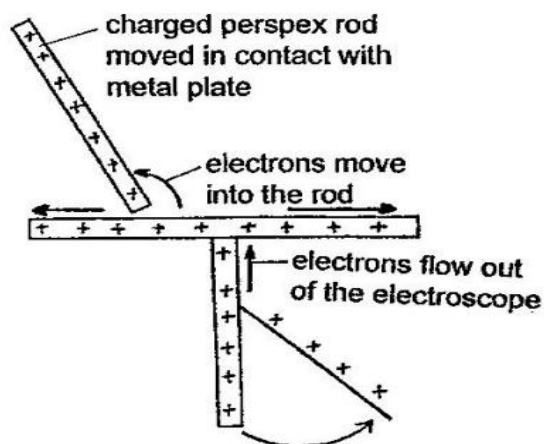
- When the gold leaf is close to the brass plate in a collapse state then the brass cap, brass rod, brass plate and the gold leaf are electrically neutral. There is no excess charge.
- When a charged insulator (such as a positively-charged glass rod) is brought near to the brass cap, the free electrons from the brass and gold parts of the electroscope are attracted to the brass cap leaving the brass plate and gold leaf positively charged.
- This causes the gold leaf to diverge due to the repulsion between like positive charges.



- **N.B - If the charge on the leaves is of the same polarity (both positive or both negative), the leaves will repel each other, causing them to spread apart.**
- The degree of separation of the leaves indicates the relative amount of charge.
- The gold-leaf electroscope can also indicate the polarity of a charge (positive or negative) by observing how the leaves behave when charged objects are brought near.

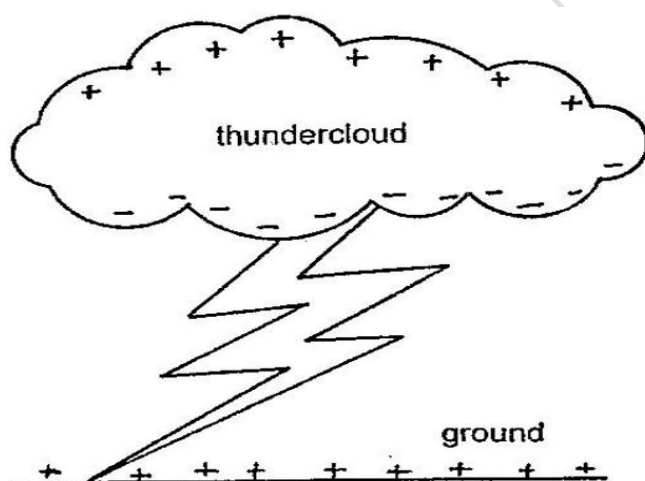
Charging an electroscope

- To charge the electroscope negatively, slide a negatively charged rod on the brass cap of the electroscope so that some electrons from the charged rod can be transferred to the cap causing the gold leaf to deflect or diverge.
- Charge detection uses only the phenomenon of repulsion between like charges is used.



Lightning

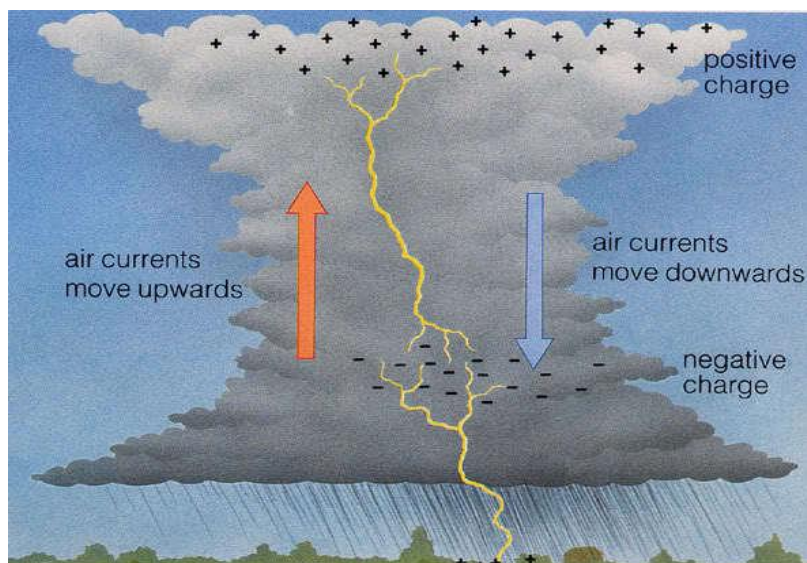
- **Lightning** is a giant spark of electricity created by a buildup of electrical charges in the atmosphere.
- These charges accumulate within a cloud, between clouds, or between a cloud and the ground, and when the electrical potential becomes too great, a discharge occurs, resulting in lightning.



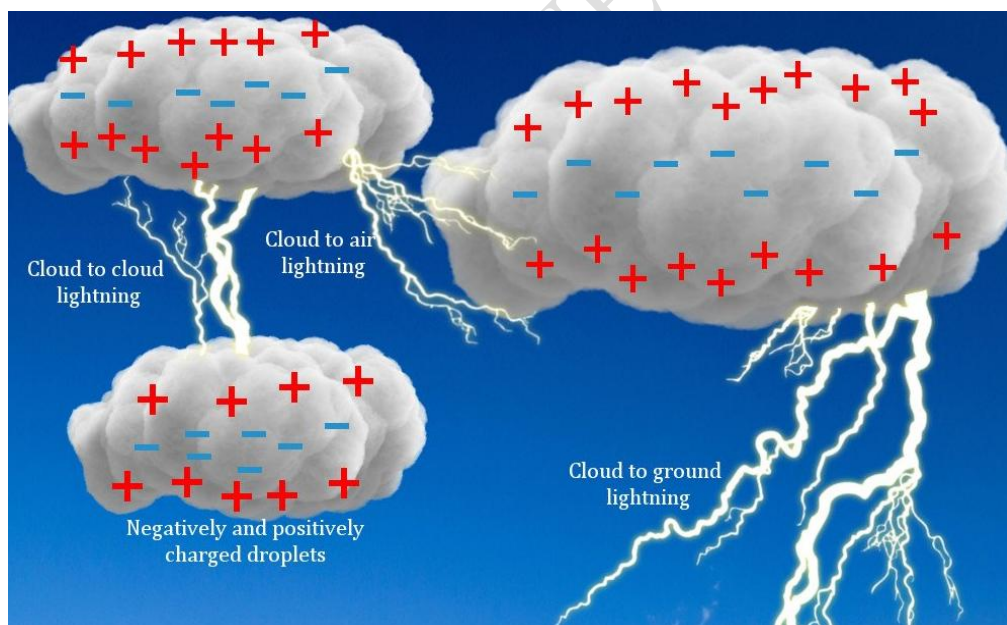
How a Lightning Forms:

- In a thunderstorm cloud, collisions between ice crystals and water droplets cause a separation of electrical charges.
- Typically, the top of the cloud becomes positively charged, while the bottom becomes negatively charged.
- These opposite charges build up, creating a strong electrical potential difference within the cloud or between the cloud and the ground.

- When the electrical potential becomes too high, the air, which normally acts as an insulator, can no longer withstand the voltage. This causes a rapid discharge of electricity, which is seen as lightning.



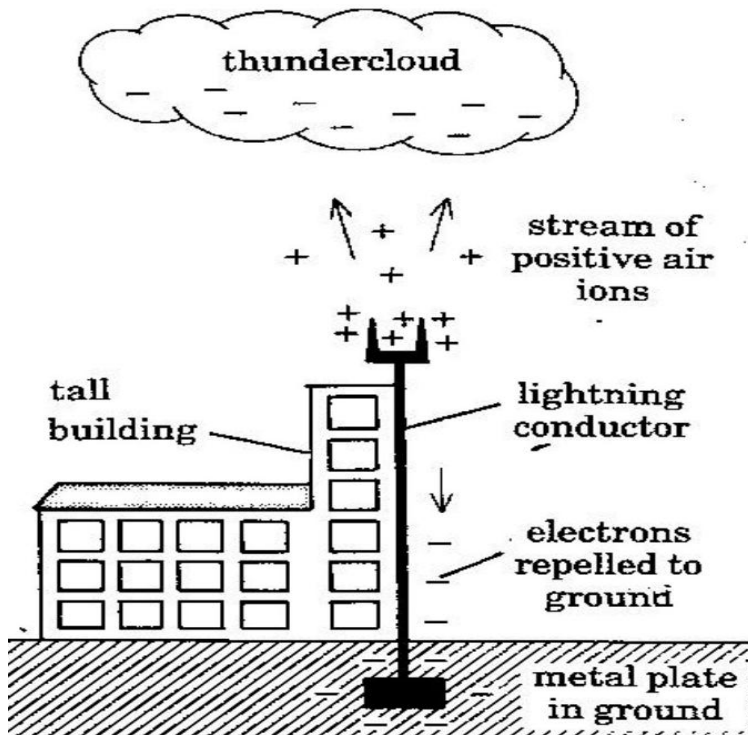
- There are 3 types of Lightning:
 - **Intra-cloud lightning** – is the lightning that can occurs within a cloud.
 - **Cloud-to-cloud lightning** – is the lightning that occurs between clouds.
 - **Cloud-to-ground** – is the lightning that occurs between a cloud and the ground.



Cloud-to-Ground Lightning:

- Cloud-to-ground lightning is the lightning that we typically see.
- A negative charge moves from the cloud towards the ground, and when it gets close, a positive charge from the ground to meet it, creating a conductive channel.
- The connection between the stepped leader and the streamer forms a channel for the main electrical discharge, called the return stroke, which is the bright flash of lightning.

- A typical cloud-to-ground flash can have multiple return strokes, one after the other, along the same channel.
- The lightning channel heats the surrounding air to extremely high temperatures, causing it to expand rapidly and create a shockwave, which we hear as thunder.



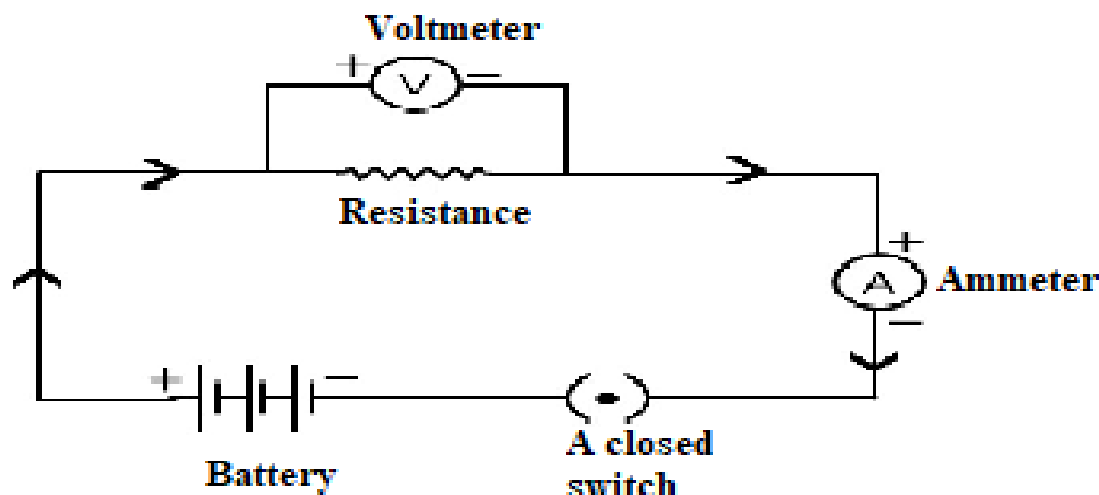
- A **lightning conductor** is a metal rod made of copper, that protects structures from lightning strikes by providing a safe, low-resistance path for the electrical current to travel to the ground.
- It intercepts the lightning strike and directs the current away from the building, minimizing potential damage.
- **N.B** - The lightning conductor, being the highest point on a structure, provides an alternative steady path for the flow of charge from the top of the building to the earth.

The principle of a lightning conductor

- The conductor is placed above the building to collect charges and pass them harmlessly down the side of the building into the ground, providing a low-resistance path for the electrical current.
- The conductor is made of thick copper cables(strip) to conduct electrons to the ground since copper is a good conductor of electricity.
- When a negatively charged thundercloud passes overhead, it acts inductively on the conductor, charging the **spikes positively** and the **earth plate negatively**.
- Negative electrons are attracted to the spikes and are discharged.
- The electrons are then passed down the conductor and are dissipated (earthed) to the ground from the copper plate, preventing them from passing through the building and causing damage.
- **N.B** - The copper strip is attached to a copper plate in the ground to earth excess electrons.

Electricity in circuits

- An **electrical circuit** is a closed path that allows electricity to flow.
- It consists of components like power source, wires, and a load (like a bulb or a resistor), all connected to enable a continuous flow of electric current.



Components of a circuit

- **Power source:** provides the electrical energy such as a battery.
- **Conductors:** are wires that allow electricity to flow easily.
- **Load:** a device that uses the electrical energy, like a bulb, motor or resistor.
- **Switch:** controls the flow of electricity by opening or closing the circuit.

Current

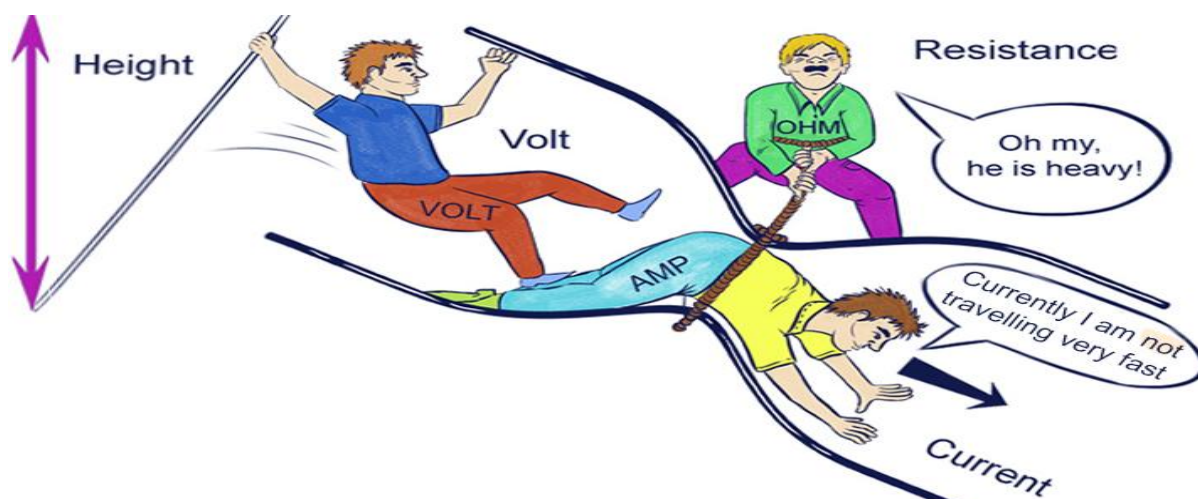
- **Electric current** refers to the flow of (electrons) electricity in an electronic circuit.
- It is measured in **amperes (A)**, which are also called amps.
- It is measured using an instrument called **ammeter**.
- The larger the value in amperes, the more electricity is flowing in the circuit.

Voltage

- **Voltage** (also known as potential difference), is the difference in potential energy between two different points in a circuit.
- It is the pressure/force from an electrical circuit's power source that pushes charged electrons (current) through a circuit, enabling them to do work. In brief, **voltage = pressure**.
- It is measured in **volts (V)** using a voltmeter.

Resistance

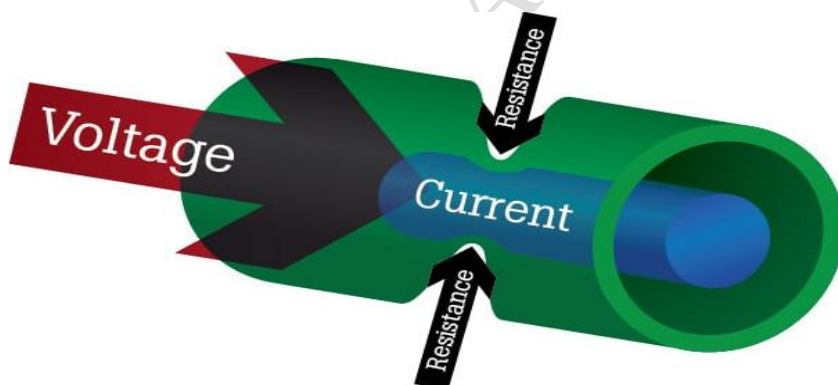
- **Electrical resistance** is the opposition to the flow of electric current in a material or circuit.
- It is also defined as a measure of how much a material opposes the flow of electric charge.
- It's a property that determines how much the current is reduced when a voltage is applied.
- The higher the resistance, the lower the current flow for a given voltage.



- Resistance is measured in **ohms (Ω)**.
- Resistance arises from collisions between electrons and the atoms within a material.
- These collisions impede the flow of electrons, thus increasing resistance.
- **N.B** - the following is an image of resistors:



Factors Affecting Resistance:



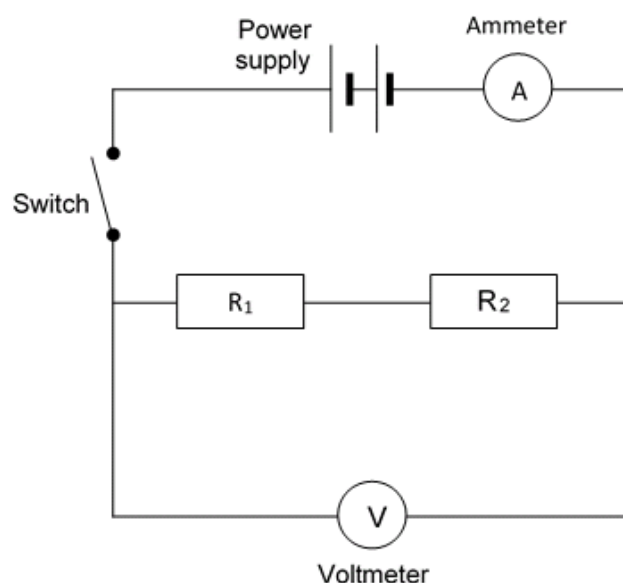
- **Material:** Different materials have different inherent resistances. Conductors (like copper) have low resistance, while insulators (like rubber) have high resistance.
- **Length of conductor (wire):** Longer wires have higher resistance because electrons have more atoms to collide with. Therefore, resistance increases with an increase in length of the material. The longer the wire, the greater the resistance it has.
- **Thickness of the conductor (cross sectional area):** Thicker wires have lower resistance because electrons have more space to flow through. Therefore, resistance increases with an increase in cross sectional area.
- **Temperature:** In general, higher temperatures lead to higher resistance due to increased atomic vibrations and more collisions.

Connection of resistors

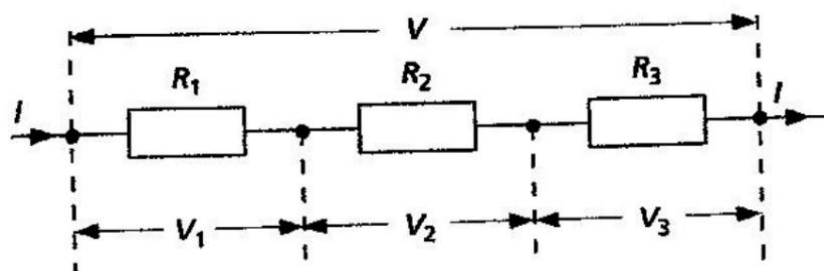
- Resistors can be connected in two primary ways in an electrical circuit: **series and parallel**.
- The way resistors are connected significantly impacts the overall resistance of the circuit and how current and voltage are distributed.

Series Connection:

- In a **series connection**, resistors are placed one after the other, forming a single path for current flow.



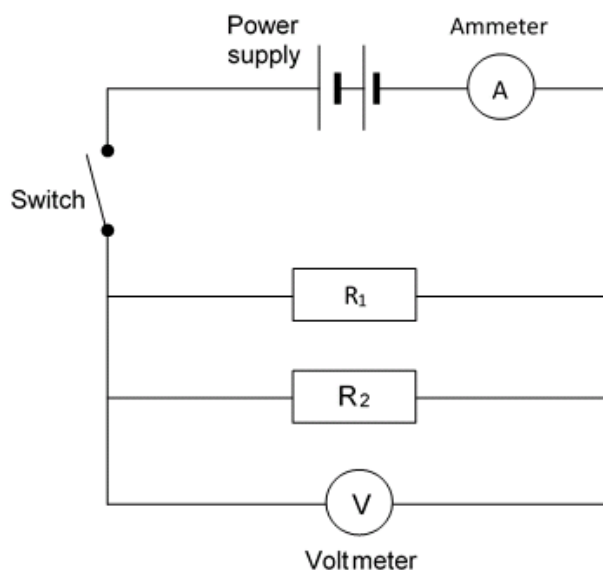
- The current that flows through all resistors is the same.
- The total voltage across the series resistors is the sum of the individual voltage drops across each resistor.
- The total resistance (equivalent resistance) is the sum of the individual resistances.
- Formula: add all the resistances.



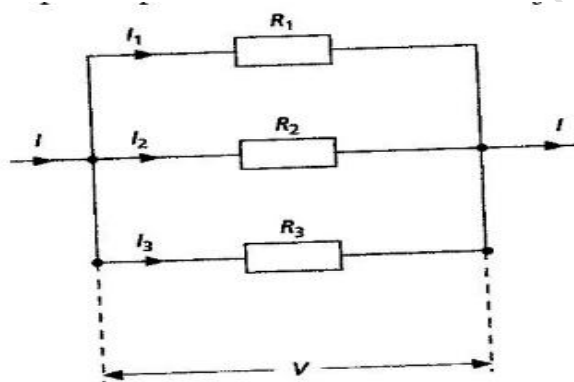
- Given the above diagram, Total Resistance = $R_1 + R_2 + R_3$

Parallel Connection:

- In a **parallel connection**, resistors are connected side-by-side, providing multiple paths for current to travel.



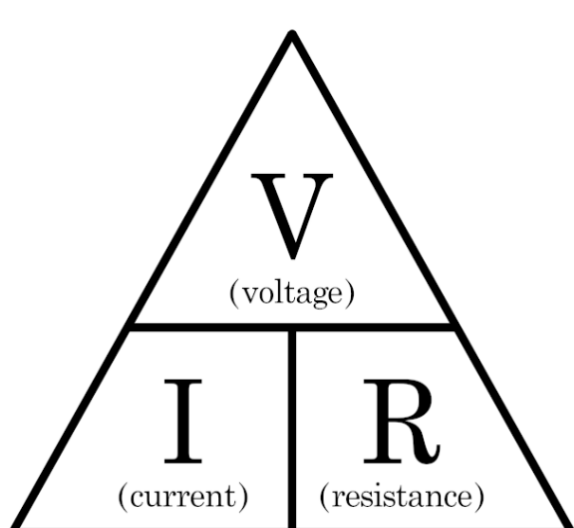
- The voltage across each resistor in a parallel connection is the same and equal to the source voltage.
- The total current flowing into the parallel combination is the sum of the currents through each individual resistor.
- The total resistance is less than the smallest individual resistance. Calculating the total resistance requires using the reciprocal formula.
- Formula: **Total Resistance** = $\frac{\text{Product of Resistances}}{\text{Sum of Resistances}}$



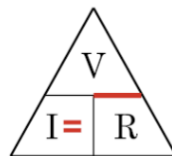
- Given the above diagram, **Total Resistance** = $\frac{R1 \times R2 \times R3}{R1 + R2 + R3}$

The Ohm's law

- **Ohm's law** describes the relationship between voltage, current, and resistance in a circuit.
- It states that the current flowing through a conductor is directly proportional to the voltage across it, and inversely proportional to its resistance.
- This relationship can be expressed as: **V = I R**
 - **V** = Voltage: Measured in volts (V).
 - **I** = Current: Measured in amperes (A).
 - **R** = Resistance: Measured in ohms (Ω).



$$V = I \times R$$



$$I = V \div R$$



$$R = V \div I$$

Key points about Ohm's law:

- **Direct proportionality:** If the voltage increases, the current also increases, assuming the resistance remains constant.
- **Inverse proportionality:** If the resistance increases, the current decreases, assuming the voltage remains constant.
- **Constant Resistance:** Ohm's law is most accurate when the resistance of the conductor remains constant, regardless of the voltage or current.

Limitations of the Ohm's law

- If temperature increases, the resistance also increases and if it decreases resistance also decreases.
- This will cause variations in the answers when using Ohm's law.
- Ohm's law is not obeyed when:
 - Temperature of conductor changes e.g. bulbs and heater elements.
 - The conductor is cooled.
 - The conductor does not have a uniform cross-sectional area.

Question 1

- A potential difference measured across a light bulb is 12V. The current measured is 5A. Calculate the resistance.

Question 2

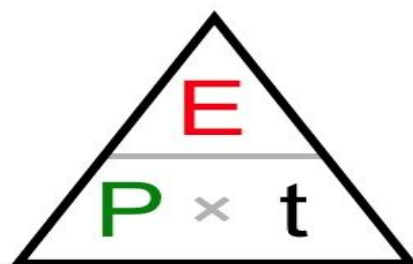
- A potential difference of 240V applied across the heating coil of an electric kettle drives a current of 8A through the coil. Calculate:
 - (a) The resistance of the coil.
 - (b) The new current flowing through the coil if the p.d. applied is changed to 220V.

Electrical Energy

- **Electrical energy** is the energy associated with the movement of electric charges, like electrons.

- A potential difference (voltage) creates a force that pushes these charges, causing them to move from a point of higher potential to a point of lower potential.
- The flow of electric charge is known as electric current, which is the basis for electrical energy being used in various applications.
- As charges move, they can do work, transferring energy to other components in a circuit or system.
- Electrical energy is often generated by converting other forms of energy (mechanical, chemical, etc.) into electrical energy, such as in power plants or batteries.
- Electrical energy can be transmitted through conductors (like wires) over long distances, allowing it to be used in various locations.
- SI unit for energy is the joule (J)
- The formula for electrical energy is typically expressed as:

$$\begin{array}{lcl} \text{Energy} & = & \text{Power} \times \text{Time} \\ \text{(Joules, J)} & & \text{(Watts, W)} \quad \text{(Seconds, s)} \\ \text{Or } E & = & P \times t \end{array}$$



Question 1

- An electric iron has a heating element of resistance 60Ω . If the operating current flowing through it is 4A, Calculate
 - the supply voltage
 - the electrical power produced
 - the heat energy produced in 5 minutes

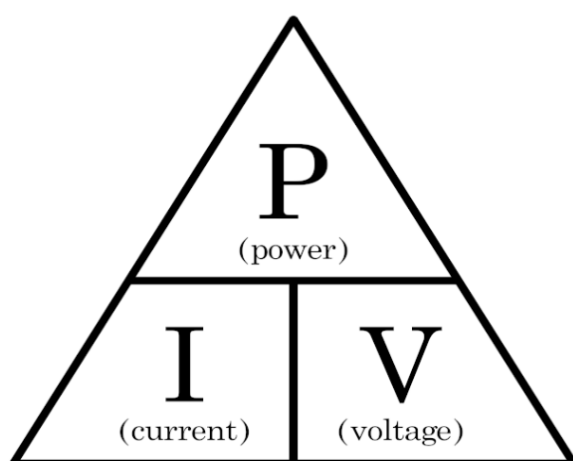
Question 2

- A filament lamp is rated 60W and 240V. Find
 - The current flowing through the lamp;
 - The resistance of the filament
 - The energy produced by the lamp in one hour

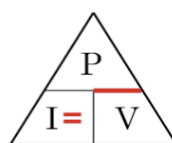
Electrical Power

- **Electrical energy** is the rate at which electrical energy is used in an electrical circuit.
- It is also defined as the rate at which electrical energy is converted from one form to another.
- This could be from potential energy (stored energy) to kinetic energy (moving energy) or even other forms like heat or light.

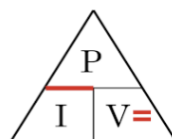
- The SI unit of power is the **watt (W)**, which is equivalent to one joule per second (J/s).
- *N.B – for example, a light bulb with a higher wattage (100W) consumes more power and produces more light (and heat) than a lower wattage bulb (e.g. 25W).*
- *An electric heater with a higher power rating will heat a room faster than a lower powered one.*
- If a bulb is marked 100W, it is converting electrical energy to heat and light at rate of **100J/s**
- It can be calculated using different formulas, such as : **$P = IV$** or **$P = I^2R$**



$$P = I \times V$$



$$I = P \div V$$



$$V = P \div I$$

S.I Units:

- Energy is measured in joules (J).
- Power is measured in watts (W).
- Voltage is measured in volts (V).
- Current is measured in amperes (A).

Relationship between Power and Energy

- Power is closely related to electrical energy(E).
- Energy is the capacity to do work, and electrical power is the rate at which that work is done.
- The relationship is **Energy(E) = Power(P) × Time(T)**

Question 1

- Find the power of an electric iron operating on a 240V supply using a current of 0.5A

Question 2

- Find the voltage of an electric motor if its power is 3kW and the current passing through it is 12,5A.

Question 3

- What is power of an electric light bulb if it is drawing a current of 0.25A from the mains electricity supply of 240V?

Question 4

- What current is drawn by a 1.5kW heater which operates on a 240V mains supply?

Cost of electricity

- The cost of electricity consumption is based on the number of kWh of electrical energy used
- The kWh is the electrical energy used by a 1kW appliance in 1 hour
- Meters are installed to measure the amount of energy used by each house.
- One unit of electricity = 1kWh

$$\text{Energy (number of units)} = \text{power(kW)} \times \text{time(h)}$$

- The cost of electricity each month is determined by reading the electricity meter and multiplying the units by the cost per unit.

$$\text{Cost} = \text{number of units} \times \text{cost per unit}$$

Question 1

- If a company charges 14 cents for each kWh of electrical energy used, calculate the total cost of using a 3kW electric kettle for 20 mins and a 100W filament bulb for 5 hours.

Question 2

- If a unit of electricity costs 11,2c, calculate the cost of using an electric stove that consumes energy at a rate of 8000w for 3 hours.

Question 3

- Suppose you do some ironing for 3 hours using a 750W electric iron.
 - a) How many units of electrical energy are consumed?
 - b) How much did it cost to do the iron in January 2015, if cost per unit was \$0.1 per kWh?

Question 4

A 12V accumulator delivers a current of 3A through a car headlamp for 4 hours. Calculate;

- a) The power supplied in watts.
- b) Energy supplied in kilowatt-hour.
- c) Cost of 300 units of electricity at 14 cents per unit.

Methods of saving electricity

- Use of energy saving bulbs
- Turning off lights during the day or night when no one is in the room
- Wearing warm clothing in winter instead of using electric heating systems
- Use of alternative sources of electricity e.g. solar panels, biogas
- Switch off appliances not in use
- Use of low power rating appliances
- Using appliances with heating elements sparingly as they consume the most energy

Electrical hazards and safety precautions

Electrical hazards

- **Damaged insulation** – fire risks and electric shocks
- **Overloaded circuits cause** - overheating of cables – fire risks because there is maximum current in a circuit
- **Damp conditions** – increase the severity of an electric shock because water lowers the resistance of the path to the earth.

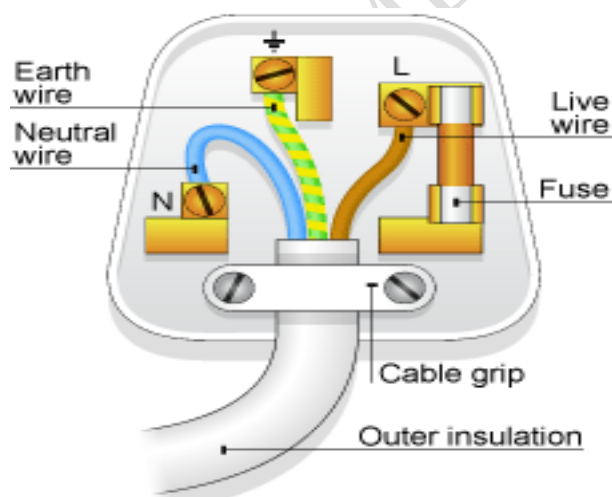
Safety precautions

- Earthing all appliances-the earth wire creates a safe path for current to flow and prevents the current from flowing through you and shocking you
- Avoid overloading circuits
- Use insulated cables
- Do not handle appliances with wet hands
- Put on rubber foot wear when using appliances
- Use Switches and fuses

Electrical plugs

Three pin plug

- This is an electrical plug with three pins or metal projections to fit into a power socket.
- The three pins are connected to three wires namely:
 - **Earth wire**
 - **Live wire**
 - **Neutral wire**



- The **neutral (N)** wire (blue) completes the circuit by forming a path for the current back to the supply. It is usually at zero volt
- The **earth (E)** wire (yellow with green) is a low resistance wire usually connected to the metal casing of the appliance.
- The **live (L)** wire (brown) delivers the energy at high alternating voltage to the appliance

- Earthing protects the user of the appliance from an electric shock, if the metal casing should accidentally become live.
- The earth wire carries excess current to the ground thus protecting the user from electric shocks.

Fuse

- A fuse is a component made from a metal with low melting point.
- The fuse melts if too large a current flow through it thus breaking the circuit. This prevent overheating and damage of appliance and use from electric shocks.
- The fuse should be connected to the live wire so that the appliance will not become live after the fuse has blown.
- Fuses are normally rated 1A, 2A, 5A, 10A and 13A



Switches

- All switches are designed to break or complete an electrical circuit.
- The switch must be fitted onto the live wire so that switching off disconnects the high voltage from an appliance

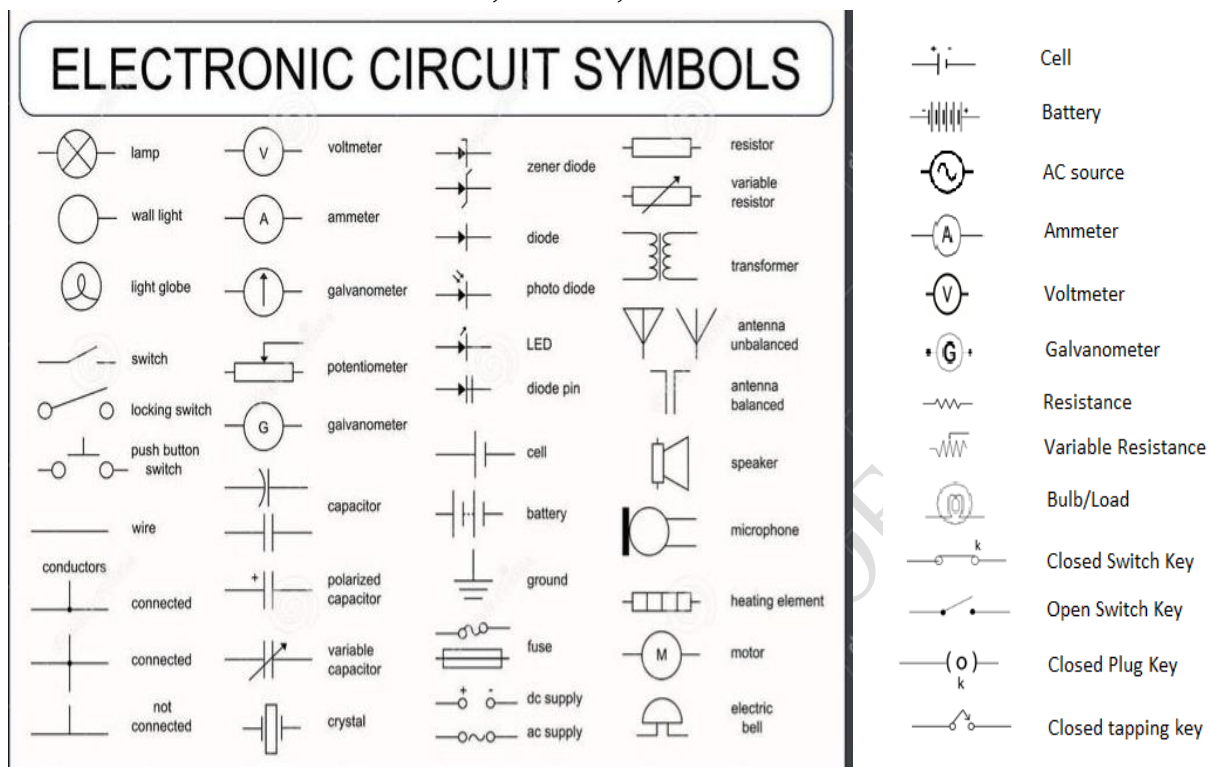
Two pin plug

- Double insulation is a safety feature in an electrical appliance that can substitute for an earth wire. Only the live and neutral wires are required for the appliance.
- This safety feature provides two levels of insulation i.e. the electric cable is insulated from the internal components of the appliances and the metal parts which could become live if a fault developed are also insulated from the external casing.
- Appliances with this feature normally have nonmetallic casing such as plastic or wood e.g. radio, television.

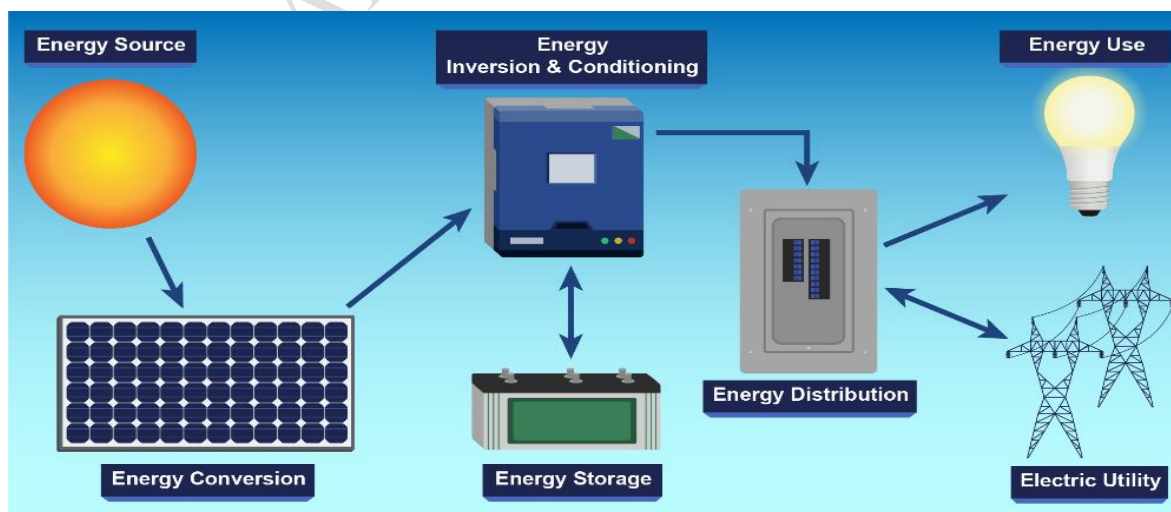


ELECTRIC SYMBOLS

- **Electronic symbol** is a pictogram used to represent various electrical and electronic devices or functions such as wires, batteries, resistors etc.

**SOLAR PHOTOVOLTAIC SYSTEMS**

- **Photovoltaic** means using light to produce voltage.
- Photovoltaic cells convert solar energy to electrical energy. It is made up of semi-conducting material i.e. silicon.
- The electrons in the silicon gain energy from the sun and create a voltage which can produce current in a closed circuit.



- A certain amount of light is required to cause a significant current, thus making solar power inefficient during times with low levels of sunlight. In bright sunlight, each cell produces 0.5V and a current of 0.03A.

- Greater voltages and current are obtained by connecting cells in series and in parallel on a panel. Cells are connected in series to give a higher voltage and in parallel to give a higher current.
- Cells in series supply steady current which last long.

TELECOMMUNICATION

- Telecommunication is the transmission of signals, signs, messages or information of any nature by wire or radio
- Communication is the sending and receiving of information
- The sender is the person who sends a message
- The receiver is the person who receives the message

The transmission of signal in different media transmitters

- The transmission of signal or information follows this process
- **Sender → encoding → medium of transmission → decoding → receiver**
- Sender first has to encode the information
- Encoding is the translation of the information into a message that can be understood by others
- The message is transmitted through a transmission medium
- The transmission medium could be a wire or cable or it could be radio waves
- Message travels through the transmission medium to the receiver
- The receiver decodes the message when they receive it
- Decoding a message means reading listening to or seeing the message

Transmitter and receiver

- A **transmitter** is a device that converts useful information such as sounds or images so that the information can travel through a medium
- A **receiver** converts the information back to sounds or images so that we can understand it
- Transmission media is the physical pathway that sends information from the transmitter to the receiver
- There are two types of media transmitters; guided transmission media and unguided transmission media

Guided transmission media

- It is also known as **wired media**.
- Transmitter and receiver are connected by a wire or a cable
- The transmitter converts the information (sound; voice; picture; text) into an electric current (Digital signal)
- Electric current travels along the wire to the receiver
- The receiver converts the electric current back into information
- E.g. of devices that use guided transmission media are telegraph and telephone
- Energy conversion in a guided transmission media
- Sound energy → electrical energy → sound energy

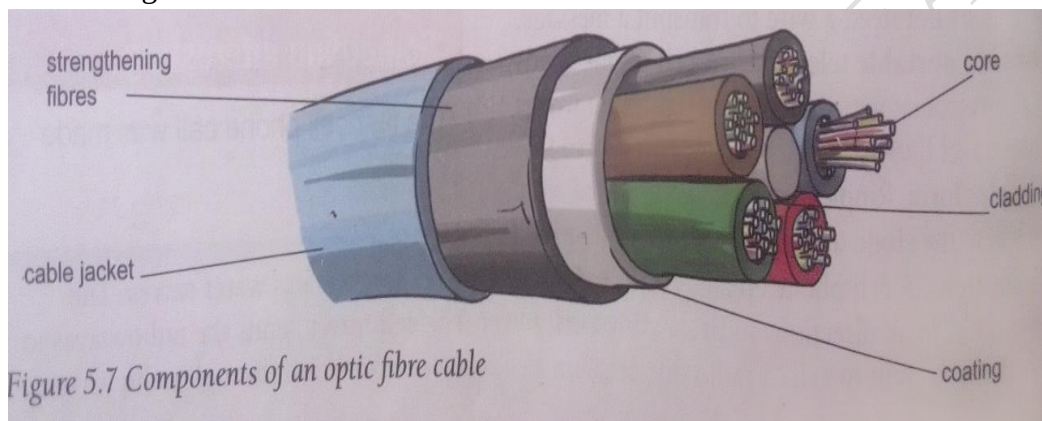
- The difference between a telegraph and a telephone is that a telephone can transmit your voice; telephone transmitter changes your voice into electric current; receiver then changes the electric current back into a voice.

Signal transmission cables

- A signal transmitter cable is a guided transmission medium
- There are three different kinds of guided transmission media namely:
- Optic fibre cables; coaxial cables and sheathed pair cables

Optic fibre

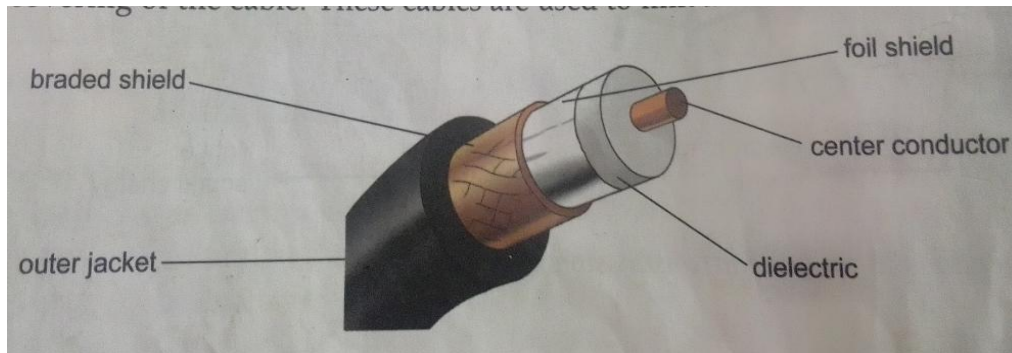
- Has 3 main sections; the core; the cladding and the jacket
- Core is made of thin strands of glass or plastic
- Each strand covered by cladding made from plastic or glass
- Coating surround the cladded strands



- Strengthening fibres surround the coating to prevent the cable from bending too much and damaging strands
- Cable jacket is the outer covering of the optic fibre cable
- Optic fibres carry messages using pulses of light
- Optic fibre sends information coded in a beam of light down a glass pipe
- The light that carries the signal cannot leave the cable but continue to totally internally reflect off the edges of the glass until it reaches its destination
- Can transmit information more efficiently and over a longer distance when compared to other transmitting cables
- Information only travel in one direction; thus for two-way communication a second connection must be made between the sender and the receiver

Coaxial cables

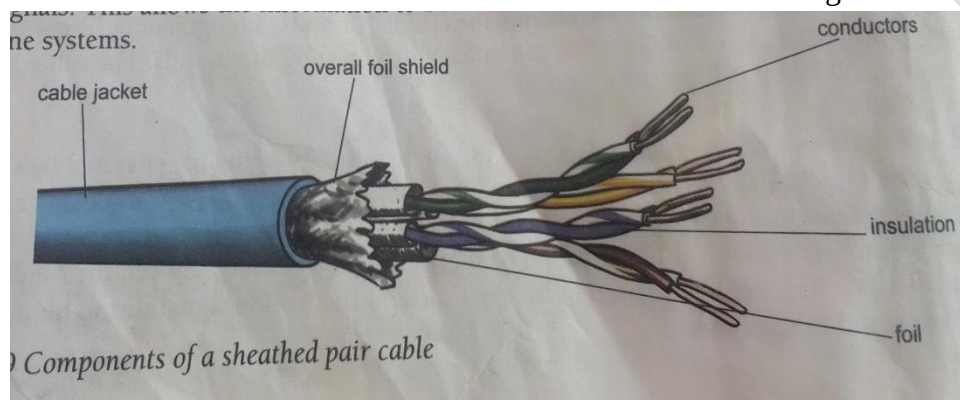
- Is a copper wire that is covered with an insulation casing called a dielectric.
- Information travels through the copper wire as an electric current.
- Bradded shield prevents interference signals from entering the inner cable
- An interference signal is a signal that can disrupt or change a signal that is travelling from the transmitter to the receiver



- Cables are used to link an antenna to a cable television; radio or computer

Sheathed pair cables

- Foil sheath reduces the amount of interference from other signals



- Is used mainly in cable telephone systems

Telephones

- Transmits vibrations created by a person's vocal cords through a wire to a receiver.
- The signal is analogue from sender to receiver.
- When speaking in the mouth piece, the voice makes the diaphragm in the microphone inside vibrate converting sound energy into electrical energy.
- The electrical energy travels from the phone through exchanges to another phone where the diaphragm in the loud speaker inside the ear piece of the receiving phone converting the incoming electrical energy back to sound energy.



Unguided transmission media; wireless signal transmission

- It's also known as **wireless media**
- The transmitter and the receiver are not connected by a wire or a cable
- The transmitter uses an antenna that converts the information into radio waves
- An antenna is a long metal rod that is used to send or receives radio waves
- Electric current forces electrons in the antenna to vibrate
- Vibration produces radio waves (Analogue signals)
- Radio waves travel through the air to the antenna of the receiver
- Radio waves cause the electrons in the receiving antenna to vibrate
- This vibration produces an electric current
- The receiver converts the electric current back into information
- Examples of devices that use unguided transmission media are cell phones; radios and televisions

Energy conversions in unguided transmission media

Light or sound energy → electrical energy → electromagnetic energy → electrical energy → light or sound energy

Cell phones

- A microchip inside a cell phone takes the vibrations of your voice into a tiny microphone which turns it to strings of numbers (encoding) that are transmitted in form of radio waves by means of the phone's antenna to the nearest cell phone mast.
- The mast works in the same way as the tiny antenna in the phone, transmitting the radio waves across to the next phone mast and the next, until it reaches the one closest to the person you are trying to phone.
- From that mast it is transmitted to the other person's phone where the encoded digital message is decoded back into electrical pulses that get the speaker of the receiver vibrating to produce sound waves.
- The signal is analogue at first and the cell phone digitalizes it before sending it off to a mast. The signal quality is very good i.e. there is very little loss of energy and sound quality
- The signal is converted back to analogue by the other cell phone and into sound waves for the other person to hear.



Internet

- The internet works similar to cell phones and regular phones, converting information into a string of numbers and transmitting these through a combination of radio waves through antennas, electrical pulse through phone lines and pulses of lights through fibre optics
- . In order to make this possible, the computer chips in the computer break up information into smaller, addressed packets. It allows exchange of information between two or more computers on a network.

Wi-Fi

- Is the technology that allows our computers; laptops and cell phones to connect to the internet without using a cable
- Is an example of an unguided transmission medium
- In order to have wi- fi one needs to have a small device called a wireless transmitter
- The transmitter receives information from the internet and then converts this information into radio waves
- The wireless transmitter emits these radio waves in a small area around itself called the wireless local area network (WLAN)
- Cell phone or laptop converts the radio waves into useful information than you can hear; see or read.



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Why You Should Consider Us:

- @affordable fees
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